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# Ego-METAS: an Egocentric online Multimodal Energy-efficient Temporal Action Segmentation benchmark

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## Abstract

To operate in the physical world, embodied agents must perceive their environment in an “always-on” fashion, selectively accessing the most informative sensors to balance energy constraints and task accuracy. Despite its importance for resource-constrained devices, energy-aware perception remains under-explored, with most prior work assuming unlimited compute. To address this, we introduce **Ego-METAS**: the first **Egocentric online Multimodal Energy-efficient Temporal Action Segmentation benchmark**. Ego-METAS provides a unified testbed of more than 100 hours of untrimmed egocentric video from EgoExo4D, CMU-MMAC, and CaptainCook4D, spanning 5 modalities (RGB, audio, gaze, IMU, and monochrome camera). We formulate an online temporal action segmentation task where models must dynamically select which sensors to activate at each timestep while strictly adhering to hardware-representative energy budgets. Alongside the benchmark, we release unified splits, cleaned annotations, pre-extracted features, and a diverse suite of baseline routing policies. Our evaluations show that optimal routing is highly scenario-dependent, and that existing policy-learning methods—designed primarily for trimmed clips—struggle to adapt to continuous, untrimmed environments. However, even simple dynamic fusion of complementary modalities (e.g., via random routing) proves critical for balancing predictive accuracy against strict energy budgets. Ultimately, Ego-METAS provides a standardized foundation to develop robust, cost-aware policies for autonomous, always-on embodied AI.

## 1 Introduction

Continuous perception is essential for intelligent agents operating in the physical world, such as robots or assistants deployed through smart glasses [36]. Yet, modern computer vision systems often treat perception as exhaustive, processing all available modalities under implicit assumptions of abundant energy, computation, and memory. This contrasts with biological perception, where multimodal sensing is used selectively to cope with physical constraints: for instance, audio may suffice during a quiet conversation while washing dishes, whereas in a crowded bar we naturally rely more on vision, such as lip reading, to communicate effectively.

We argue that practical embodied AI should reconsider this scaling assumption: rather than treating physical constraints as a nuisance to be overcome by better hardware, we should design systems that embrace resource budgeting and selective modality routing as core algorithmic features. As illustrated in Fig. 1, wearable devices are inherently multi-modal. In these settings, an assistive system could efficiently recognize actions by keeping its energy-intensive RGB camera active while a user puts a meal in the microwave, but power it down during a prolonged, repetitive background task like wiping

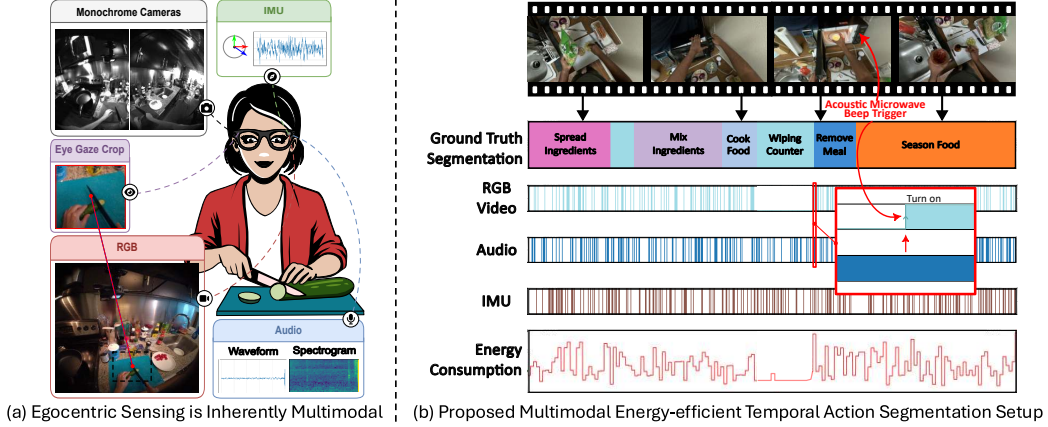


Figure 1: (a) Wearable devices naturally capture rich streams including multiple modalities. (b) In the proposed setup, models dynamically select which sensory modalities to activate for the next timestep, minimizing energy consumption while maintaining accurate, continuous online action segmentation.

the counter. During this time, the system can rely entirely on low-power audio and IMU sensors until an acoustic trigger—such as the microwave beep—dynamically reactivates the visual stream. Current approaches have largely overlooked this dynamic, online setting due to the absence of unified, standardized benchmarks tailored for energy-aware, continuous processing. Indeed, most prior efforts in Temporal Action Segmentation (TAS) focused on static uni-modal settings [13], or exhaustive multi-modal processing [39], with limited works focusing on energy-aware scenarios [21, 34, 51], often scattered and carried out under non-standardized benchmarks.

To address this gap, we introduce EgoMETAS: an Egocentric online Multimodal Energy-efficient Temporal Action Segmentation benchmark. Specifically, Ego-METAS introduces a unified evaluation pipeline designed specifically to test systems under strict operational constraints. We source egocentric videos from three distinct egocentric datasets, EgoExo4D [21], CaptainCook4D [35], and CMU-MMAC [12], which are selected to be representative of real-world operational settings, having been collected natively with diverse hardware such as Aria Glasses, Microsoft HoloLens, and custom multimodal rigs. Unlike prior efforts that restrict evaluation to episodic recognition on single devices [21], Ego-METAS is the first benchmark to target untrimmed, continuous action segmentation across diverse wearable platforms. To resolve historical inconsistencies in prior datasets, we rigorously revise existing annotations and establish a unified evaluation protocol that moves beyond standard task accuracy, allowing us to quantitatively assess model performance against explicit energy consumption limits and strict operational budgets. Furthermore, to accurately reflect the physical realities of wearable systems, we curate a principled set of modalities, ranging from full RGB streams to energy-efficient grayscale images and gaze-centered crops, and provide concrete estimates for their respective acquisition, memory, and processing costs. Alongside this benchmark, we systematically survey the literature on resource-constrained perception and compare the most viable approaches. We open-source unified splits, cleaned annotations, pre-extracted modality features, and evaluation scripts, providing a robust foundation for the community to build upon. Ultimately, Ego-METAS shifts the focus toward designing systems that embrace hardware and energy constraints as core features rather than limitations, laying the groundwork for truly viable, always-on embodied AI.

## 2 Related Work

**Energy Efficient Methods:** Deploying vision systems on low-power devices [5, 46] typically relies on static network optimizations [5, 19, 50]. Standard approaches include quantization [8, 14, 19, 49], pruning [1, 55], and compact architectures [17, 25]. Additional overhead reductions are achieved through Neural Architecture Search [46, 57], efficient recurrent networks like Mamba [22] and xLSTM [2], and knowledge distillation [23]—applied specifically to egocentric perception by EgoDistill [47] to compress video and IMU semantics. However, static model optimization is inherently insufficient for continuous, in-the-wild sensing. To achieve true always-on processing for resource-constrained wearables, systems must move beyond fixed computation and learn dynamic routing policies that dynamically adapt the active sensory budget on the fly.

**Information Selection:** Beyond rigid uniform sampling [52], efficient video understanding relies on extracting salient temporal context. While query-based frame selectors [24] require offline video access, online methods successfully reduce temporal redundancy via lightweight clip sampling [28] or early-exit inference gates [18]. Ego-METAS elevates this paradigm: rather than merely selecting unimodal temporal frames, we formulate the problem as dynamic multimodal sensor routing, evaluating policies tailored for continuous, always-on temporal action segmentation.

**Multimodal Learning:** Multisensory integration significantly enhances representation learning [37, 38] but inherently multiplies computational costs. To mitigate this, AdaMML [34] introduced differentiable modality routing via the Gumbel-Softmax trick [26]. This foundational mechanism has been widely adapted [9, 53] for active feature prioritization [45], task-specific sensor gating [54], and cross-modal policy distillation [6]. Concurrent efforts address joint modality training [29], modality selection under domain shifts [31], and test-time missing modalities [40]. However, these approaches overwhelmingly evaluate on offline, pre-trimmed clips. Real-world continuous perception cannot rely on predefined action boundaries to dictate modality switching. To bridge this gap, Ego-METAS introduces a systematic framework specifically designed to evaluate Energy-Efficient Multimodal Online Temporal Action Segmentation in an always-on, untrimmed setting.

**Temporal Action Segmentation (TAS):** Offline TAS assumes full video access for frame-wise classification [13, 16, 30, 44]. Conversely, embodied wearables require causal, online TAS [42, 56], a setting that remains under-explored [13] despite foundational [11] and recent advances based on causal convolutions and memory banks [4, 42, 56]. We adopt online TAS as the foundation for Ego-METAS, challenging models to perform continuous segmentation under strict energy budgets via dynamic modality selection. While prior energy-efficient benchmarks like Ego-Exo4D [21] evaluate streaming keystep recognition on isolated episodes—explicitly filtering out long background sequences—Ego-METAS forces models to persistently navigate long, unsegmented background transitions in the wild. Furthermore, we expand beyond single-device constraints to establish a comprehensive multi-device testbed unifying Ego-Exo4D [21], CMU-MMAC [12], and CaptainCook4D [35].

### 3 Benchmark

We source egocentric videos and annotations from three existing datasets, namely CMU-MMAC [12], Ego-Exo4D [21], and CaptainCook4D [35], which have been collected using diverse hardware representative of the variety of multimodal observations which can be obtained with wearable devices. We provide a formal definition of METAS and a principled set of evaluation metrics aimed to assess both segmentation accuracy, energy efficiency, and their trade-offs.

#### 3.1 Task definition

Let  $\mathcal{X}_{:t} = \{\mathbf{X}_{:t}^{(m)}\}_{m=1}^M$  denote the continuous input stream up to time  $t$  from  $M$  sensors (e.g., RGB, audio, IMU). Despite varying native acquisition frequencies, we define a common discrete clock  $t$  for system predictions. Operating under a strict hardware power budget  $B$  (in mW), the online inference pipeline at step  $t$  proceeds in three stages: 1) A **routing policy**  $\pi$ , conditioned on historically realized features  $\{\tilde{\mathbf{x}}_{:t-1}^{(m)}\}_m$ , outputs a binary decision vector  $\mathbf{a}_t \in \{0, 1\}^M$ . If  $a_t^{(m)} = 1$ , sensor  $m$  actively collects observation  $\mathbf{X}_t^{(m)}$ , incurring capture cost  $E_{\text{cap}}^{(m)}$ . 2) Active sensors utilize feature extractors  $\phi_m$  to generate dense features  $\mathbf{x}_t^{(m)} = \phi_m(\mathbf{X}_t^{(m)})$ . Inactive sensors bypass extraction, relying on a placeholder  $\mathbf{z}_t^{(m)}$  (we set this to the previous feature  $\tilde{\mathbf{x}}_{t-1}^{(m)}$ ). This yields a strictly defined effective representation:  $\tilde{\mathbf{x}}_t^{(m)} = a_t^{(m)}\mathbf{x}_t^{(m)} + (1 - a_t^{(m)})\mathbf{z}_t^{(m)}$ . 3) The effective history  $\{\tilde{\mathbf{x}}_{:t}^{(m)}\}_m$  is processed by a **TAS model**  $\Psi$  to predict the current action class  $\hat{y}_t$ . Alongside capture costs, feature extraction and TAS computation incur specific processing ( $E_{\text{comp}}^{(m)}$ ) and memory access ( $E_{\text{mem}}^{(m)}$ ) costs. The objective of Ego-METAS is to learn a joint system  $(\pi, \Psi)$  that maximizes prediction accuracy ( $\hat{y}_t = y_t, \forall t$ ) subject to the constraint that total aggregated energy ( $E_{\text{cap}} + E_{\text{comp}} + E_{\text{mem}}$ ) across all active components never exceeds budget  $B$  at any timestep  $t$ .

#### 3.2 Energy computation

For video  $\mathcal{V}$  of duration  $T$ , total energy (mJ) is the sum of capture, computation and memory energy:

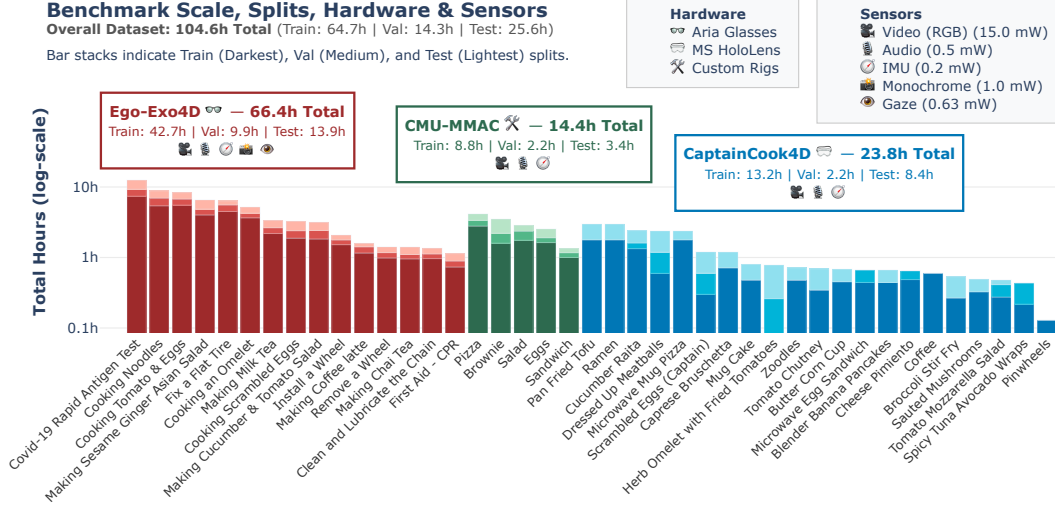


Figure 2: Ego-METAS comprises 104.6 hours of video and 41 scenarios collected with three different wearable devices, encompassing 5 modalities captured with varying energy consumption profiles.

$$E_{V_{\text{total}}} = \sum_{t=1}^T \left( \underbrace{\sum_{m=1}^M a_t^{(m)} P_m \Delta t}_{E_{V_{\text{cap}}}} + \underbrace{\alpha \left( f_{\Psi}^{\text{MAC}} + f_{\pi}^{\text{MAC}} + \sum_{m=1}^M a_t^{(m)} f_{\phi_m}^{\text{MAC}} \right)}_{E_{V_{\text{comp}}}} + \underbrace{\beta \left( f_{\Psi}^{\text{MEM}} + f_{\pi}^{\text{MEM}} + \sum_{m=1}^M a_t^{(m)} f_{\phi_m}^{\text{MEM}} \right)}_{E_{V_{\text{mem}}}} \right) \quad (1)$$

where  $P_m$  is the power consumption of sensor  $m$ ,  $\Delta t$  is the temporal duration between consecutive timesteps. The terms  $f_{\Psi}^{\text{MAC}}$ ,  $f_{\pi}^{\text{MAC}}$ ,  $f_{\Psi}^{\text{MEM}}$ ,  $f_{\pi}^{\text{MEM}}$  represent the MAC operations and CUDA memory events of the core TAS model  $\Psi$  and policy  $\pi$  (we set these to zero for non-learned policies), while  $f_{\phi_m}^{\text{MAC}}$  and  $f_{\phi_m}^{\text{MEM}}$  denote the corresponding costs for the feature extractor  $\phi_m$ . The constants  $\alpha$  and  $\beta$  convert these operations to energy (mJ) [21]. The above formulas allow to compute the total energy for a whole video. In practice, we also compute the power  $P_V = \frac{E_{V_{\text{total}}}}{T}$  as the total energy divided by the length of the video in seconds  $T$  (in mW). In our evaluations, we constrain our models to operate within fixed energy budgets  $B$ , which is achieved by modifying the model’s configuration so that  $P_V < B$ . For more details of the estimation of the energy please refer to Sec. A.4.

### 3.3 Datasets, Modalities and Splits

Ego-METAS comprises 104.6 hours of egocentric video across 41 scenarios. Sourced from three established multimodal datasets described below, the benchmark covers diverse hardware profiles and five sensory modalities with distinct energy signatures, including standardized splits (see Figure 2).

**Ego-Exo4D [21]:** Captured via Aria glasses, this dataset provides procedural activities across five modalities (RGB, audio, IMU, monochrome, gaze). Because prior benchmarks evaluated isolated episodes, the original keystep annotations exhibit inconsistent granularity and overlapping classes that preclude continuous perception (Sec. A.1). To resolve this, we introduce a rigorous re-annotation of this subset, enabling robust always-on evaluation (Sec. A.2). This dataset introduces the most diverse array of scenarios outside of the cooking domain and constitutes 66 h of the total benchmark.

**CMU-MMAC [12]:** Recorded using custom multimodal rigs, captures subjects performing kitchen tasks in a lab setting. It spans five different recipes, with multiple modalities, from which we select video, audio, and the accelerometer and gyroscopes from the IMUs, accounting to 16 hours.

**CaptainCook4D [35]:** Captured via Microsoft HoloLens, this dataset focuses on step-by-step procedural execution and mistake detection. Original annotations provided 325 different classes

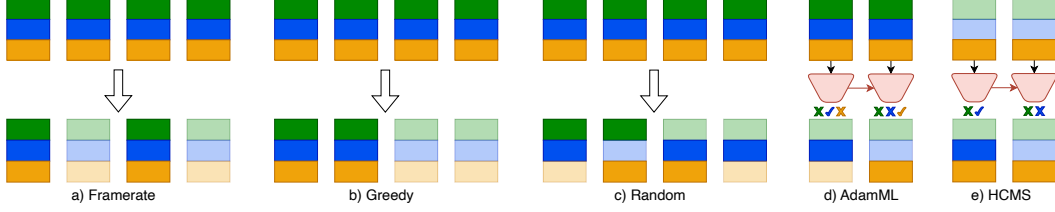


Figure 3: **Policies.** a) Framerate drops entire frames at fixed intervals; b) Greedy uses all modalities until the budget is consumed; c) Random drops individual modalities with a given probability; d-e) Learned policies predict which sensors should be active at each step.

which was too granular for the task we are proposing. Thus, we re-annotated as described in Sec. A.2. The dataset involves 21 different cooking recipes, with synchronized video, audio, and IMU streams.

### 3.4 Pre-Extracted Features

As part of the dataset, we provide the extracted features from each modality. Particularly, a feature vector  $\mathbf{x}_t^{(m)}$  is pre-extracted per modality  $m$  at every timestep  $t$ . We choose a temporal discretization of 30 fps following video. Therefore, all modalities are synchronized with a common time frame. For each modality, we use frozen feature extractors from the literature. Details are reported in Sec. A.7.

### 3.5 Evaluation protocol and metrics

Models are required to produce predictions at 30 FPS (possibly replicating old predictions when idle), even when trained at lower temporal resolutions. This ensures a consistent evaluation setting and prevents models from simplifying the task by skipping frames or actions. Furthermore, following the energy constraints proposed in [21], we define two operating regimes per second for the METAS task: a low-power budget of 20 mW and a high-performance budget of 2.8 W. These budgets are real-device oriented. Tentatively, low-budget settings targets smaller wearable devices (such as Ray-Ban Meta smart glasses [32]), which require low power consumption in order to last for a full working day. In contrast, the Therefore, these budgets are designed to mimic real-world scenarios and encourage models to be not only accurate but also energy-efficient for deployment on real devices.

Following standard TAS literature [13], we evaluate segmentation accuracy using Frame Accuracy (Acc), segmental Edit Distance (Edit) to measure the sequential ordering of predicted actions, and Segmental F1 Scores at varying temporal Intersection-over-Union thresholds (F1@10, 25, 50) to penalize over-segmentation. To evaluate efficiency, we report these metrics alongside the average Energy Consumption per second (mW), and percentage of times a given modality is sensed across the video. While previous works tackling episodic keystep detection [21] used ranking-based metrics like mean Average Precision (mAP), we argue that these are structurally inadequate for continuous Temporal Action Segmentation, as detailed in Sec. A.1.

## 4 Baseline Policies

To evaluate the proposed benchmark, we use ProTAS [41] as the state-of-the-art model to perform the temporal action segmentation of the videos. Using it as a backbone TAS architecture, we consider two families of policies. Fixed policies use heuristics to operate under a fixed energy budget, whereas learned policies involve learning an auxiliary network to decide which modalities to activate at each timestep, often based on a low-power modality input. Specifically, we evaluate three fixed policies and two learned ones, adapted from the literature (Figure 3).

**Frame Rate** [21]: This policy (Fig. 3 (a)) reduces the video sampling frequency to save energy. For skipped timesteps, the model outputs the most recent prediction. While decreasing the framerate significantly reduces the energy consumption, it may result in the loss of fine-grained temporal resolution, and short-duration actions may be entirely missed due to temporal aliasing.

**Greedy** [21]: This policy (Fig. 3 (b)) activates all modalities until reaching the predetermined energy budget per second. When the next second starts, it reactivates all modalities again, repeating the cycle.

Table 1: Performance on Ego-Exo4D [21] across energy budgets and policies. For learned policies, split cells report the modality usage of the TAS model (left) and the policy model (right).

| Budget    | #  | Policy     | Parameters                               | Video (RGB) | Audio      | IMU        | Monochrome | Gaze | Energy   | Acc   | mAP   | Edit  | F1@10 | F1@25 | F1@50 |
|-----------|----|------------|------------------------------------------|-------------|------------|------------|------------|------|----------|-------|-------|-------|-------|-------|-------|
| No Budget | 1  | Frame Rate | 30.00 FPS                                | 100         | 100        | X          | 100        | X    | 11.53 W  | 45.08 | 33.06 | 23.86 | 21.74 | 17.85 | 10.46 |
|           | 2  | Frame Rate | 06.00 FPS                                | 20.0        | X          | X          | X          | X    | 01.86 W  | 36.47 | 22.76 | 19.04 | 19.98 | 15.81 | 8.10  |
|           | 3  | Frame Rate | 10.00 FPS                                | X           | 33.3       | X          | X          | X    | 00.14 W  | 25.08 | 14.95 | 13.06 | 12.77 | 7.28  | 2.97  |
|           | 4  | Frame Rate | 30.00 FPS                                | X           | X          | 100        | X          | X    | 00.00 W  | 24.80 | 13.92 | 13.17 | 12.56 | 8.44  | 2.92  |
|           | 5  | Frame Rate | 30.00 FPS                                | X           | X          | X          | 100        | X    | 01.82 W  | 34.58 | 21.79 | 14.29 | 13.52 | 10.58 | 5.89  |
|           | 6  | Frame Rate | 15.00 FPS                                | X           | X          | X          | X          | 50.0 | 01.54 W  | 24.48 | 14.94 | 10.96 | 10.69 | 7.58  | 3.87  |
|           | 7  | Frame Rate | 06.00 FPS                                | 20.0        | 20.0       | X          | X          | X    | 01.94 W  | 26.02 | 15.71 | 11.54 | 12.39 | 8.07  | 3.74  |
|           | 8  | Frame Rate | 30.00 FPS                                | X           | 100        | 100        | X          | X    | 00.42 W  | 26.67 | 15.21 | 12.59 | 12.70 | 9.09  | 4.07  |
|           | 9  | Frame Rate | 06.00 FPS                                | X           | X          | X          | 20.0       | 20.0 | 00.98 W  | 43.71 | 26.82 | 25.34 | 24.62 | 19.05 | 12.02 |
|           | 10 | Frame Rate | 05.00 FPS                                | 16.7        | 16.7       | 16.7       | X          | X    | 01.62 W  | 38.03 | 23.95 | 25.13 | 24.06 | 18.55 | 10.90 |
|           | 11 | Frame Rate | 06.00 FPS                                | 20.0        | 20.0       | X          | 20.0       | X    | 02.31 W  | 45.99 | 33.56 | 26.58 | 25.22 | 20.96 | 10.61 |
|           | 12 | Greedy     |                                          | 13.4        | 13.4       | 13.4       | 13.4       | 13.4 | 02.37 W  | 43.75 | 31.65 | 25.32 | 25.83 | 20.96 | 12.30 |
|           | 13 | Random     | $\tau_r$ 0.70, $\tau_{inf}$ 0.900, $c=0$ | 10.1        | 10.0       | 10.0       | 10.0       | 10.0 | 01.78 W  | 47.02 | 30.55 | 26.37 | 25.61 | 20.07 | 12.93 |
|           | 14 | Random     | $\tau_r$ 0.70, $\tau_{inf}$ 0.900, $c=1$ | 5.1         | 11.6       | 20.5       | 8.5        | 7.4  | 01.14 W  | 46.80 | 30.47 | 26.51 | 25.91 | 20.18 | 12.80 |
|           | 15 | Random     | $\tau_r$ 0.90, $\tau_{inf}$ 0.900, $c=1$ | 5.1         | 11.5       | 20.5       | 8.5        | 7.4  | 01.13 W  | 45.30 | 29.75 | 26.70 | 25.53 | 20.57 | 11.90 |
|           | 16 | Random     | $\tau_r$ 0.99, $\tau_{inf}$ 0.900, $c=1$ | 5.1         | 11.5       | 20.5       | 8.5        | 7.4  | 01.14 W  | 38.26 | 26.44 | 22.24 | 21.73 | 17.77 | 10.19 |
| 2.8 W     | 17 | AdaMML     |                                          | 0.4         | 0.5 / 100  | 0.6 / 100  | 100        | X    | 02.25 W  | 22.10 | 12.11 | 12.73 | 10.81 | 6.29  | 2.60  |
|           | 18 | AdaMML     |                                          | 11.2        | 13.2 / 100 | 23.5 / 100 | 100        | X    | 03.26 W  | 27.67 | 16.44 | 17.37 | 15.88 | 11.58 | 5.51  |
|           | 19 | HCMS       |                                          | 15.0        | 37.7       | 100        | 27.8       | 19.4 | 03.35 W  | 40.91 | 27.67 | 22.83 | 21.74 | 17.17 | 11.24 |
|           | 20 | HCMS       |                                          | 0           | 0.0        | 100        | 0.0        | 0    | 00.01 W  | 17.82 | 13.89 | 15.57 | 9.84  | 6.26  | 3.03  |
|           | 21 | Frame Rate | 00.05 FPS                                | 0.2         | X          | X          | X          | X    | 16.29 mW | 26.05 | 15.59 | 10.39 | 12.29 | 8.46  | 4.24  |
|           | 22 | Frame Rate | 00.10 FPS                                | X           | X          | X          | 0.3        | 0.3  | 16.70 mW | 28.41 | 18.36 | 11.28 | 13.91 | 9.77  | 4.66  |
|           | 23 | Frame Rate | 00.10 FPS                                | X           | 0.3        | 0.3        | 0.3        | 0.3  | 18.13 mW | 27.02 | 18.36 | 10.06 | 12.60 | 9.06  | 4.16  |
|           | 24 | Greedy     |                                          | 0           | 0          | 0          | 0          | 0    | 00.47 mW | 22.44 | 12.42 | 7.06  | 9.78  | 6.54  | 2.17  |
|           | 25 | Random     | $\tau_r$ 0.90, $\tau_{inf}$ 0.997, $c=1$ | 0.2         | 0.3        | 0.6        | 0.3        | 0.2  | 34.93 mW | 33.09 | 20.06 | 26.73 | 21.24 | 15.05 | 6.77  |
|           | 26 | Random     | $\tau_r$ 0.90, $\tau_{inf}$ 1.000, $c=1$ | 0           | 0          | 0          | 0          | 0    | 00.47 mW | 12.00 | 11.60 | 2.25  | 0.53  | 0.18  | 0.15  |
|           | 27 | AdaMML     |                                          | 0           | 0.0 / 100  | 0.0 / 100  | 100        | X    | 2212 mW  | 18.13 | 12.29 | 5.43  | 5.83  | 2.92  | 0.95  |
|           | 28 | HCMS       |                                          | 0           | 0.0        | 100        | 0.0        | 0    | 05.00 mW | 17.82 | 13.89 | 15.57 | 9.84  | 6.26  | 3.03  |

Sensors: Video (RGB), Audio, IMU, Monochrome, Gaze.

While easily adaptable for a fixed budget, this policy is inherently problematic for online TAS, as an exhausted budget midway through a sequence may preclude the model from capturing subsequent steps.

**Random** [21]: This policy (Fig. 3 (c)) randomly drops  $\tau$  percentage of each modality. If all modalities are dropped at the same timestep, feature extraction is bypassed, but the TAS model still executes a forward pass, incurring the corresponding computational cost. We train this policy at different dropout rates ( $\tau_{train}$ ) and evaluate the performance varying dropout at inference time ( $\tau_{inf}$ ). We report a plain ( $c = 0$ ) and a cost-aware ( $c = 1$ ) version, which biases selection prioritizing low-power modalities (details at Sec. A.6).

**AdaMML** [34]: This policy (3 (d) left) learns to select which modalities to use per timestep. For this policy, we take their original implementation and adapt its loss to handle our TAS setting. We use low-energy counterparts of RGB video as input to this policy to reduce computational overhead. Specifically, we use gaze crops for Ego-Exo4D [21], and the low-resolution version of RGB (192×192) for the rest of the datasets.

**HCMS** [51]: This policy (3 (d) right) sorts modalities by computational cost, and chooses the cheapest modality to be always on. A gating module decides whether to subsequently activate the remaining inactive modalities or not, ordered by cost. Therefore, the most expensive modality is never active on its own.

## 5 Experiments

We establish the first performance reference for the Multimodal Energy-efficient online Temporal Action Segmentation (METAS) task by discussing baseline performance, energy-accuracy trade-offs, and qualitative examples. Detailed experimental settings are reported in Sec. A.5.

### 5.1 Baseline comparison

**Ego-Exo4D**: Table 1 summarizes performance on Ego-Exo4D [21]. To establish an upper bound, the best unconstrained baseline at 30 FPS (row 1) achieves 45.08% accuracy but consumes a massive 11.53 W. Under the 2.8 W budget, static Frame Rate policies demonstrate that combining complementary modalities surpasses isolated sensors. While fusing the two strongest individual modalities (RGB and audio) unexpectedly degrades performance (row 7), pairing lightweight signals like gaze and monochrome (row 9) yields an efficient balance. The best static baseline (RGB, audio,

Table 2: Performance on CMU [12] across energy budgets and policies. For learned policies, split cells report the modality usage of the TAS model (left) and the policy model (right).

| Budget    | #  | Policy     | Parameters                            | ■         | 👂         | 🌀         | Energy   | Acc   | mAP   | Edit  | F1@10 | F1@25 | F1@50 |
|-----------|----|------------|---------------------------------------|-----------|-----------|-----------|----------|-------|-------|-------|-------|-------|-------|
| No Budget | 1  | Frame Rate | 30.00 FPS                             | 100.0     | 100.0     | ✗         | 09.72 W  | 85.63 | 53.73 | 30.41 | 35.93 | 32.77 | 26.18 |
|           | 2  | Frame Rate | 06.00 FPS                             | 20.0      | ✗         | ✗         | 01.91 W  | 84.34 | 50.97 | 39.07 | 42.90 | 40.72 | 33.47 |
|           | 3  | Frame Rate | 01.00 FPS                             | 3.3       | 3.3       | ✗         | 00.32 W  | 83.17 | 44.23 | 44.75 | 46.95 | 43.34 | 37.12 |
|           | 4  | Frame Rate | 01.00 FPS                             | 3.3       | 3.3       | 3.3       | 00.32 W  | 82.47 | 43.79 | 40.57 | 44.14 | 40.17 | 33.26 |
|           | 5  | Greedy     |                                       | 26.7      | 26.7      | 26.7      | 02.60 W  | 79.49 | 29.38 | 27.08 | 25.95 | 24.48 | 19.92 |
|           | 6  | Random     | $\tau_{lr}0.70, \tau_{inj}0.900, c=0$ | 10.1      | 10.1      | 10.0      | 00.98 W  | 80.97 | 39.30 | 28.34 | 30.35 | 27.82 | 24.95 |
|           | 7  | AdaMML     |                                       | 0.0 / 100 | 0.2 / 100 | 5.3 / 100 | 02.21 W  | 66.71 | 18.69 | 20.62 | 16.91 | 11.38 | 5.50  |
|           | 8  | HCMS       |                                       | 25.0      | 67.4      | 100.0     | 02.61 W  | 83.94 | 47.44 | 30.06 | 34.27 | 32.53 | 28.20 |
| 2.8 W     | 9  | Frame Rate | 00.05 FPS                             | 0.2       | ✗         | ✗         | 16.05 mW | 69.37 | 21.54 | 24.15 | 28.18 | 23.80 | 14.79 |
|           | 10 | Frame Rate | 00.05 FPS                             | 0.2       | 0.2       | ✗         | 16.76 mW | 70.27 | 23.00 | 24.62 | 29.18 | 24.94 | 15.12 |
|           | 11 | Frame Rate | 00.05 FPS                             | 0.2       | 0.2       | 0.2       | 16.77 mW | 70.84 | 22.86 | 24.20 | 29.09 | 26.80 | 17.38 |
|           | 12 | Greedy     |                                       | 0         | 0         | 0         | 00.41 mW | 66.26 | 5.34  | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 13 | Random     | $\tau_{lr}0.97, \tau_{inj}1.000, c=0$ | 0         | 0         | 0         | 00.40 mW | 66.26 | 5.86  | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 14 | AdaMML     |                                       | 0.0 / 100 | 0.2 / 100 | 5.3 / 100 | 22.13 mW | 66.71 | 18.69 | 20.62 | 16.91 | 11.38 | 5.50  |
|           | 15 | HCMS       |                                       | 0         | 0         | 100.0     | 05.02 mW | 42.80 | 12.78 | 13.24 | 10.62 | 6.72  | 3.68  |
|           | 16 | Greedy     |                                       | 0         | 0         | 0         | 00.41 mW | 66.26 | 5.34  | 0.00  | 0.00  | 0.00  | 0.00  |

Sensors: ■ Video (RGB), 👂 Audio, 🌀 IMU, 📺 Monochrome, 👁 Gaze.

gaze; row 11) reaches 45.99% accuracy. A uniform Random policy ( $c = 0$ , row 13) proves superior, surpassing the 11.53 W upper bound using only 1.78 W. Furthermore, a cost-aware Random policy ( $c = 1$ , row 14) shifts usage toward cheaper sensors like IMU (20.5%) over RGB (5.1%), maintaining 46.80% accuracy while slashing energy to 1.14 W. While robust to high training dropouts, Random degrades at extreme values (rows 15-16). In contrast, complex learned policies like AdaMML (rows 17-18) struggle: the overhead of maintaining always-on modalities (gaze, audio, IMU) for routing decisions forces the model to sense other inputs  $< 1\%$  of the time, crippling performance. HCMS (row 19) achieves strong accuracy but lags behind Random in efficiency; its hierarchical design forces all cheaper modalities to remain active merely to trigger RGB. Ultimately, as static policies confirm, naively activating all complementary sensors is often detrimental compared to dynamic, sparse sampling.

Under the extreme 20 mW budget, system dynamics invert. To adhere to the budget, static Frame Rate policies must sample rarely (e.g., RGB active 0.2% of the time in row 21). Yet, lightweight combinations like gaze and monochrome (row 22) still extract enough context to achieve 28.41% accuracy. This extreme constraint completely breaks discrete dynamic models. To avoid exceeding 20 mW, the Greedy policy (row 24) operates entirely blind (0% usage). Cost-aware Random policies ( $c = 1$ ) are highly sensitive at this scale, either heavily violating the budget (row 25) or collapsing to 0% usage (row 26). AdaMML (row 27) fails fundamentally; its intrinsic architectural overhead exceeds 2.2 W even when no inputs are sensed, rendering it structurally unsuited for ultra-low power applications. Meanwhile, HCMS (row 28) adapts to the strict budget by devolving to use only IMU.

**CMU-MMAC:** Table 2 highlights how different dataset dynamics impact optimal routing. Unlike Ego-Exo4D, high frame rates on CMU-MMAC cause severe over-segmentation; thus, the unconstrained 30 FPS baseline (row 1) suffers a low edit score (30.41) despite consuming 9.72 W. Instead, sparse sampling acts as a natural regularizer, allowing static policies to strictly dominate. Under the 2.8 W budget, a 6 FPS RGB baseline (row 2) performs strongly, but dropping to 1 FPS and adding audio (row 3) maximizes the edit score (44.75) using just 0.32 W, slightly edging out full IMU fusion (row 4). Although evaluated at 30 FPS, processing at a lower frame rate implicitly simplifies the temporal task: by holding predictions constant across intermediate frames, the model avoids noisy temporal boundaries. This is especially beneficial in smooth scenarios where repeated predictions remain valid for longer periods. Consequently, dynamic routing models lag significantly, with Greedy (row 5) and Random (row 6) achieving substantially lower accuracy.

Under the extreme 20 mW constraint, dynamic policies collapse entirely to 0% modality usage (rows 12, 13), effectively operating blind by merely predicting the most common action sequence prior. In contrast, extreme sparse static sampling (0.05 FPS) successfully extracts sufficient context—whether using RGB alone, RGB and audio, or full IMU fusion (rows 9–11)—peaking at 70.84% accuracy. Finally, learned policies exhibit severe structural flaws in this regime. AdaMML’s intrinsic overhead triggers massive energy violations, consuming between 2.2 W and 6.8 W regardless of actual sensor usage, rendering it fundamentally unsuited for ultra-low power applications (row 14). While HCMS (row 15) performs reasonably well under the 2.8 W budget, it is outcompeted by naive

Table 3: Performance on CaptainCook4D [35] across energy budgets and policies. Split cells report the modality usage of the TAS model (left) and the policy model (right).

| Budget    | #  | Policy     | Parameters                              | ■         | 🔊         | 🌀         | Energy   | Acc   | mAP   | Edit  | F1@10 | F1@25 | F1@50 |
|-----------|----|------------|-----------------------------------------|-----------|-----------|-----------|----------|-------|-------|-------|-------|-------|-------|
| No Budget | 1  | Frame Rate | 30.00 FPS                               | 100.0     | ✗         | ✗         | 09.30 W  | 44.88 | 19.39 | 18.05 | 18.64 | 14.82 | 7.51  |
|           | 2  | Frame Rate | 01.00 FPS                               | 3.3       | ✗         | ✗         | 00.31 W  | 43.23 | 18.45 | 36.48 | 32.62 | 28.88 | 14.08 |
|           | 3  | Frame Rate | 00.50 FPS                               | 1.7       | 1.7       | ✗         | 00.16 W  | 42.22 | 18.05 | 36.17 | 31.91 | 23.79 | 10.32 |
|           | 4  | Frame Rate | 00.10 FPS                               | 0.3       | ✗         | 0.3       | 00.03 W  | 39.23 | 17.31 | 35.53 | 31.10 | 23.68 | 14.21 |
|           | 5  | Frame Rate | 00.50 FPS                               | 1.6       | 1.6       | 1.6       | 00.16 W  | 40.62 | 17.47 | 37.01 | 29.81 | 22.62 | 11.54 |
|           | 6  | Greedy     |                                         | 26.1      | 26.1      | ✗         | 02.54 W  | 33.80 | 13.14 | 21.04 | 22.18 | 18.24 | 9.45  |
|           | 7  | Random     | $\tau_{tr} 0.97, \tau_{inj} 0.970, c=0$ | 2.9       | 2.9       | 3.0       | 00.29 W  | 29.74 | 10.84 | 25.58 | 24.96 | 20.87 | 12.44 |
|           | 8  | AdaMML     |                                         | 1.4 / 100 | 1.4 / 100 | 1.4 / 100 | 02.30 W  | 22.60 | 7.49  | 5.96  | 9.68  | 1.32  | 0.00  |
|           | 9  | HCMS       |                                         | 24.1      | 66.0      | 98.0      | 02.52 W  | 42.13 | 16.21 | 17.72 | 15.16 | 11.38 | 6.61  |
| 2.8 W     | 10 | Frame Rate | 00.05 FPS                               | 0.2       | ✗         | ✗         | 15.63 mW | 38.51 | 16.05 | 26.52 | 24.72 | 18.35 | 7.68  |
|           | 11 | Frame Rate | 00.05 FPS                               | 0.2       | 0.2       | ✗         | 16.33 mW | 40.94 | 18.21 | 32.80 | 28.39 | 18.42 | 8.21  |
|           | 12 | Frame Rate | 00.05 FPS                               | 0.2       | 0.2       | 0.2       | 16.02 mW | 38.25 | 16.84 | 31.24 | 25.02 | 16.56 | 8.10  |
|           | 13 | Greedy     |                                         | 0         | 0         | ✗         | 00.35 mW | 23.67 | 8.29  | 5.96  | 9.68  | 1.32  | 0.00  |
|           | 14 | Random     | $\tau_{tr} 0.99, \tau_{inj} 1.000, c=0$ | 0         | 0         | 0         | 00.35 mW | 20.33 | 9.12  | 6.31  | 6.12  | 0.00  | 0.00  |
|           | 15 | AdaMML     |                                         | 0.0 / 100 | 0.0 / 100 | 0.0 / 100 | 2168 mW  | 23.86 | 7.84  | 5.96  | 9.68  | 1.32  | 0.00  |
|           | 16 | HCMS       |                                         | 0         | 0         | 98.0      | 04.92 mW | 18.75 | 10.50 | 23.73 | 9.87  | 2.92  | 2.12  |
|           |    |            |                                         |           |           |           |          |       |       |       |       |       |       |

Sensors: ■ Video (RGB), 🔊 Audio, 🌀 IMU, 📷 Monochrome, 👁 Gaze.

static baselines; at 20 mW, it is forced to rely exclusively on the IMU and fails to compete with sparsely sampled multimodal cues.

**CaptainCook4D:** Table 3 evaluates CaptainCook4D [35]. Similar to CMU-MMAC, this scenario is highly susceptible to over-segmentation at high framerates; consequently, the unconstrained 30 FPS baseline (row 1) achieves a low edit score of 18.05 despite consuming 9.30 W. As a result, also in this case, static policies strictly dominate. Under the 2.8 W budget, a minimal 1 FPS RGB-only policy (row 2) achieves best overall performance with an edit score of 36.48 and accuracy of 42.23 using just 0.31 W, outperforming various multimodal fusions at lower framerates (rows 3–5). Dynamic approaches lose their advantage, with Greedy (row 6) and Random (row 7) lagging significantly at 33.80% and 29.74% accuracy, respectively. Complex models like AdaMML fail entirely: the severe tuning required to comply with the budget (2.30 W, row 8) collapses performance to 22.60% accuracy. Meanwhile, HCMS obtains reasonable accuracy but a low edit score, mirroring the over-segmentation issues of the unconstrained baseline.

Under the extreme 20 mW constraint, dynamic approaches drop to 0% modality usage to conserve energy, operating blind and plummeting to an edit score of roughly 6.00 (rows 13, 14). In contrast, drastically reduced 0.05 FPS static baselines using RGB alone or fused with audio and IMU (rows 10–12) successfully extract sufficient context to maintain robust edit scores up to 32.80. AdaMML’s architectural overhead again proves fundamentally unsuited for ultra-low power applications, consuming over 2.1 W even with 0% sensor usage (row 15). Finally, HCMS adapts to the severe power constraints by relying exclusively on the cheapest available modality.

## 5.2 Energy-Accuracy trade-offs

Figure 4 illustrates stark domain differences in routing efficiency. On Ego-Exo4D (Fig. 4a), Random routing ( $\tau_{tr} = 0.70, c = 1$ ) strictly dominates, defining the Pareto frontier across the energy spectrum. Static policies exhibit a clear transition: ultra-lightweight IMU performs reasonably at extreme constraints ( $< 1$  mW), while multimodal fusions (e.g., RGB+Audio) overtake them as budgets approach 2.8 W. Conversely, CMU-MMAC (Fig. 4b) inverts this dynamic. Here, a static Framerate (RGB) policy establishes the optimal frontier, rapidly saturating to peak accuracy at minimal energy expenditure. A random policy fails to match this efficiency, collapsing entirely at ultra-low scales. Finally, the trade-off curves expose the severe architectural penalties of learned models. While AdaMML clusters at the extreme right of the energy axis, HCMS is competitive at high energy costs, but collapses when a strict budget is enforced. Sec. A.9 reports trade-off curves for CaptainCook4D.

## 5.3 Qualitative Examples and Collapse of Learned Policies

Figure 5 illustrates the qualitative limitations of AdaMML [34]. To offset its massive computational overhead, the policy overcompensates by heavily biasing activation toward the cheapest sensors. As shown on the left, the system frequently relies exclusively on the IMU even when visual context

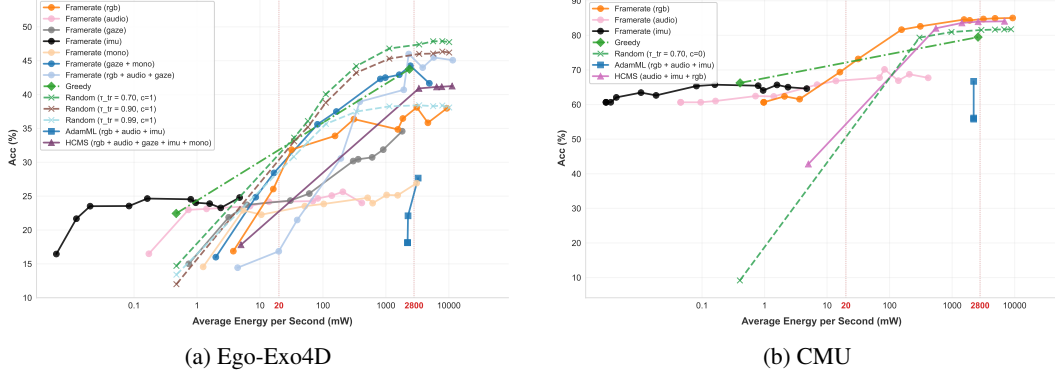


Figure 4: Energy-Accuracy trade-offs on Ego-Exo4D (a) and CMU (b).

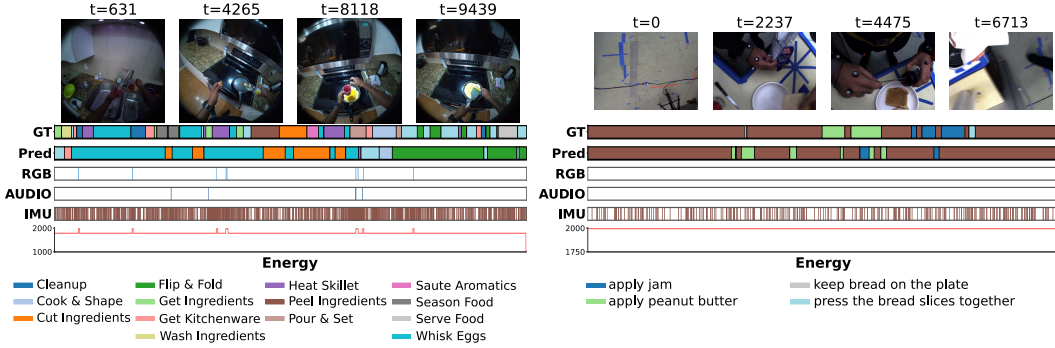


Figure 5: Qualitative examples of success and failure cases for AdaMML [34] in Ego-Exo4D [21] and CMU-MMAC [12] respectively

is strictly required, causing severe segmentation errors. This stems from a structural flaw in the AdaMML loss formulation: it penalizes energy usage during successful predictions but fails to penalize the routing policy when conservative sensor choices cause misclassifications. The right panel shows a complete policy collapse, where the model permanently deactivates RGB and audio, relying solely on the IMU.

## 6 Conclusions

We introduced Ego-METAS, the first benchmark for online, energy-efficient multimodal temporal action segmentation. By unifying three diverse datasets, we shift the focus of egocentric perception from compute-heavy offline analysis to the energy constraints of real-world wearables. Our evaluations reveal a critical gap: while dynamic modality routing is essential for efficiency, existing policies designed for trimmed clips struggle to adapt to untrimmed, continuous sequences. Ego-METAS provides the foundation to drive the development of adaptive, always-on perception systems.

**Limitations:** While comprehensive, our energy constraints rely on hardware-profiled estimates rather than physical edge deployment, potentially omitting some complex real-time processing overheads. Additionally, our framework evaluates routing on pre-extracted features; future architectures should jointly optimize the routing policy and feature extractors to maximize true on-device efficiency.

**Broader Impact:** Energy-efficient perception enables always-on wearable assistive technologies without severe battery drain while simultaneously reducing the carbon footprint of continuous AI. However, always-on cameras inherently raise bystander privacy concerns, dictating that real-world deployment must be strictly coupled with robust on-device privacy-preserving mechanisms.

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## A Supplementary material

### A.1 Motivation for the new benchmark

The proposed benchmark builds upon the initial Energy-efficient multimodal keystone recognition benchmark introduced in Ego-Exo4D [21]. The lack of a publicly released implementation challenges reproducibility and comparative evaluation, and the provided annotations present several important limitations for online Temporal Action Segmentation (TAS). In particular, the initial benchmark utilizes a trimmed subset of annotations extracted from the Keystep benchmark, excluding a substantial number of action segments, particularly those belonging to background class. While this design choice may be reasonable for keystone prediction in pre-segmented clips, in METAS we aim to evaluate continuous perception in long clips, requiring models to handle long background transitions between keysteps. For long procedural videos in the wild, it is crucial to learn to distinguish between true background and foreground, which we found a profound limitation of the existing annotations. Additionally, we observed that the original dataset has some inconsistencies in the annotations, including duplicated classes (see Fig. 7). Taking all these into account, motivated us to perform the re-annotation of the dataset, which is described in Sec. A.2.

We also found that the main evaluation metric chosen for the task in the previous benchmark, the *mean calibrated Average Precision (mcAP)*, proposed initially by De Geest et al. [11], while commonly used in online action detection, is suboptimal for our online setting. Let us analyze Eq. 2:

$$\text{cAP} = \frac{1}{P} \sum_k \frac{w \cdot TP(k)}{w \cdot TP(k) + FP(k)} I(k), \quad \text{with } w = \frac{\# \text{ negative frames}}{\# \text{ positive frames}} \quad (2)$$

where  $TP(k)$  and  $FP(k)$  are the cumulative true and false positives at rank  $k$ ,  $I(k)$  is an indicator function equal to 1 if the  $k$ -th prediction is a true positive and 0 otherwise, and  $P$  is the total number of positive frames in the dataset. Notice that it can be rewritten as:

$$\text{cAP} = \frac{1}{P} \sum_k \frac{TP(k)}{TP(k) + \frac{FP(k)}{w}} I(k). \quad (3)$$

As the proportion of background frames (negative frames) increases,  $w$  grows accordingly, causing the term  $\frac{FP}{w}$  to approach zero. Consequently, the reported performance in datasets with lots of background, such as Ego-Exo4D, can be artificially inflated, potentially approaching 100%, despite limited true recognition capability. When training on the full dataset, models tend to collapse to predict the background class, as this circumstance produces deceptively high metric scores while failing to capture relevant actions. This issue becomes more pronounced when training across all scenarios. As shown in Fig. 6, the background class dominates the data distribution, clearly separating itself from the remaining classes. As a result, the metric is biased toward background predictions, masking the model’s actual ability to recognize meaningful actions.

To address this issue, our benchmark uses other metrics as primary metrics: Frame Accuracy (Acc), segmental Edit Distance (Edit), and segmental F1 scores. Besides, we make publicly available all our code, baselines and new multimodal annotations for three datasets, including the re-annotated Ego-Exo4D [21].

### A.2 Re-annotation pipeline

#### A.2.1 Ego-Exo4D

Despite the limitations discussed in Sec. A.1, Ego-Exo4D remains the largest egocentric dataset with procedural annotations and multiple modalities. It is also the first to introduce the challenging task of Energy-efficient online multimodal keystone recognition. Building upon this benchmark, we conducted a rigorous revision of the dataset and developed a re-annotation pipeline to adapt it for our requirements.

We first performed a detailed analysis of the structure of the dataset, including number of classes, distribution, frequency of appearance, and length. We covered both the original annotations and the subset of classes for the most similar benchmark, the Keystep recognition benchmark. Following

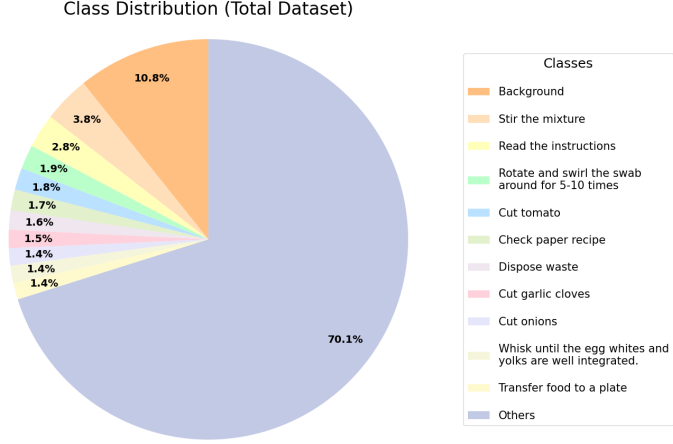


Figure 6: Original Dataset Labels

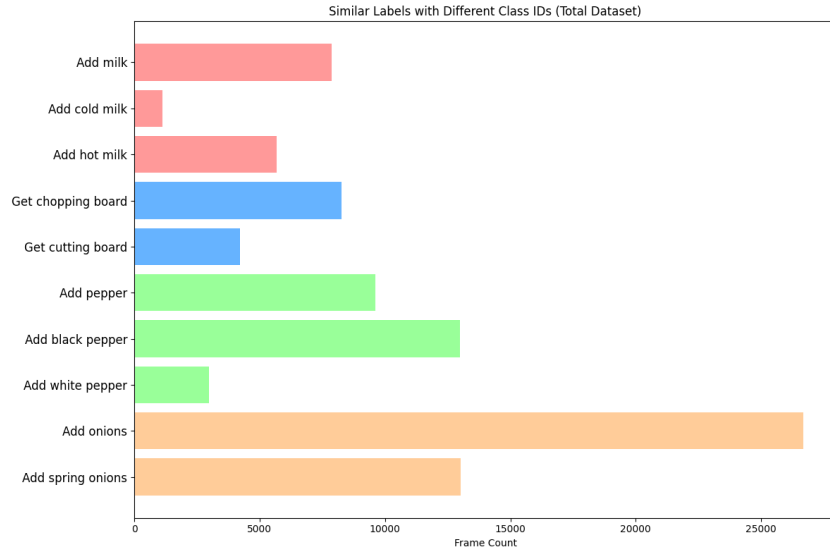


Figure 7: Examples of duplicated classes

prior works [41], we determined the most appropriate strategy for this benchmark was to train a model per scenario and average the results to obtain the global dataset metrics. This methodology prevents the background class to dominate the dataset, discouraging models to trivial collapse.

Even opting for this training strategy, the class granularity was too fine for the limited number of videos per scenario. Besides, many classes were too semantically similar to each other, and background was still the dominant class most of the time, effectively making this task extremely difficult to tackle. In the original annotations, only scenario had enough balance between number of classes and data samples (“First Aid -CPR”). Furthermore, due too lack of sufficient data, we excluded scenarios “Cooking Pasta” and “Cooking Sushi Rolls” since the former is only composed by 3 videos in the training set and the latter is composed of only one video and is impossible to split between training, validation and test.

After identifying the problems, we proceed with class clustering, in order to group similar classes together into a single class. We extracted statistics per class and fed them into an LLM [48] to perform the clustering using this statistical report, the frequency of appearance, and the coherence. This was repeated per scenario, and subsequently human supervision was needed for final corrections and verifications. Final reannotations can be found from Tab. 5 to Tab. 18.

Furthermore, our analysis revealed that, for some videos, the actors were narrating in different languages the steps of the procedure, and other videos have loud music in the background preventing the audio from being informative. Although initial results did not show significant impact in the results, we have documented the names of the compromised videos where this happens (see Tab. 4). To ensure the narrations do not help models to actually predict the correct steps, we made sure that none of this spotted videos are on the test set by swapping the compromised ones with another uncompromised from another split. In addition, we enforce that the training set has all available classes in that split.

Table 4: Actor narration per scenario

| Scenario                         | Found | Total | Percentage (%) |
|----------------------------------|-------|-------|----------------|
| First Aid - CPR                  | 26    | 51    | 50.98          |
| Clean and Lubricate the Chain    | 6     | 26    | 23.08          |
| Install a Wheel                  | 9     | 62    | 14.52          |
| Remove a Wheel                   | 12    | 64    | 18.75          |
| Fix a Flat Tire                  | 24    | 79    | 30.38          |
| Covid-19 Rapid Antigen Test      | 29    | 182   | 15.93          |
| Cooking Noodles                  | 7     | 34    | 20.59          |
| Making Coffee latte              | 2     | 14    | 14.29          |
| Making Cucumber & Tomato Salad   | 14    | 55    | 25.45          |
| Cooking Scrambled Eggs           | 7     | 25    | 28.00          |
| Cooking an Omelet                | 9     | 54    | 16.67          |
| Making Milk Tea                  | 16    | 45    | 35.56          |
| Making Sesame-Ginger Asian Salad | 7     | 29    | 24.14          |
| Cooking Tomato & Eggs            | 5     | 44    | 11.36          |
| Making Chai Tea                  | 2     | 9     | 22.22          |

Since the test set annotations from Ego-Exo4D were never publicly released, we decided to use the validation set from their benchmark as our test set and split in a 80-20% the training set to obtain a validation set (see Tab. 20). For more information on how to process each modality, please refer to Sec. A.7.

### A.2.2 CaptainCook4D

Since we propose a new multimodal egocentric action segmentation benchmark, for CaptainCook4D [35] we selected only those videos that contain video, audio and IMU information. Therefore, a considerable amount of videos were discarded, and neither of the official proposed splits were large enough for training and evaluating (see Tab. 42). For that reason, we do not train on a per-scenario basis, and instead the full dataset had to be trained jointly. Thus, we create a new split for this benchmark, where 20% of the data is used for testing and the rest is divided 80-20% for train and validation sets. We ensure that the training split contains all different classes, which was not possible for test and validation.

Similarly to Ego-Exo4D, background class was predominant, representing 18% of the dataset, again causing the training models to collapse. Consequently, a new re-annotation process, following previously defined steps was applied. The new annotations can be found at Tab. 19.

Table 5: Label mapping for “Clean and Lubricate the Chain”

| New ID | Old ID | Step Name             | Old Step Name                         |
|--------|--------|-----------------------|---------------------------------------|
| 0      | 616    | Background            | Background                            |
| 1      | 436    | Prepare Tools & Setup | Check the brake                       |
|        | 442    |                       | Get the brush                         |
|        | 443    |                       | Get the degreaser                     |
|        | 444    |                       | Get the cleaning tool                 |
|        | 445    |                       | Fill the cleaning tool with degreaser |
|        | 452    |                       | Get a cloth                           |
|        | 455    |                       | Clean the brush                       |

Continued on next page...

| New ID | Old ID | Step Name              | Old Step Name                                              |
|--------|--------|------------------------|------------------------------------------------------------|
| 2      | 446    | Apply Degreaser        | Add degreaser to the brush                                 |
|        | 456    |                        | Add degreaser to the chain                                 |
| 3      | 447    | Move Chain (Backpedal) | Slowly move the chain by backpedaling with hand            |
| 4      | 448    | Scrub Chain            | Hold the brush to the chain as you backpedal with hand     |
|        | 450    |                        | brush the chain while it remains stationary                |
| 5      | 449    | Dry Chain              | Use a cloth to dry off any liquid on the chain             |
| 6      | 451    | Lubricate Chain        | Get the chain lube                                         |
|        | 453    |                        | Place a cloth underneath the chain to catch lube droppings |

Table 6: Label mapping for “Install a Wheel”

| New ID | Old ID | Step Name            | Old Step Name                            |
|--------|--------|----------------------|------------------------------------------|
| 0      | 413    | Background           | Put the bike on a bike repair stand      |
|        | 414    |                      | Put the bike upside down                 |
|        | 420    |                      | Return wrenches                          |
|        | 439    |                      | Loosen the axle nuts                     |
|        | 616    |                      | Background                               |
|        | 956    |                      | Deflate the tire                         |
| 1      | 408    | Get Tools            | Get appropriate wrenches                 |
|        | 409    |                      | Get bike tire lever                      |
|        | 419    |                      | Get bike air pump                        |
|        | 613    |                      | Get thru-axle                            |
| 2      | 421    | Mount Wheel to Frame | Get bike wheel                           |
|        | 422    |                      | Put the wheel fully seated into the fork |
|        | 423    |                      | Get the wheel                            |
|        | 433    |                      | Clip the wheel to the hub and axle       |
|        | 440    |                      | Lift the wheel into the fork             |
| 3      | 425    | Tighten Axle Nuts    | Tighten both axle nuts using a wrench    |
| 4      | 424    | Secure Quick Release | Push the release lever inwards           |
|        | 426    |                      | Push level inward and turn axle CW       |
| 5      | 427    | Secure Brakes        | Tighten brake cable to rear axle         |
|        | 428    |                      | Tighten the brake pads                   |
|        | 431    |                      | Connect the brake noodle                 |
|        | 437    |                      | Clip the brake to the wheel              |
| 6      | 412    | Adjust Derailleur    | Pivot the derailleur back                |
| 7      | 429    | Check Function       | Roll the wheel                           |
|        | 430    |                      | Rotate the wheel                         |
|        | 432    |                      | Check the both hand brakes               |
|        | 434    |                      | Rotate the pedal                         |
|        | 435    |                      | Squeeze the brake                        |
|        | 436    |                      | Check the brake                          |
|        | 438    |                      | Hold hand brake                          |
|        | 441    |                      | Check the wheel                          |

Table 7: Label mapping for “Remove a Wheel”

| New ID | Old ID | Step Name         | Old Step Name                        |
|--------|--------|-------------------|--------------------------------------|
| 0      | 410    | Background        | Shift the gear to the smallest cog   |
|        | 413    |                   | Put the bike on a bike repair stand  |
|        | 414    |                   | Put the bike upside down             |
|        | 419    |                   | Get bike air pump                    |
|        | 616    |                   | Background                           |
| 1      | 408    | Get Tools         | Get appropriate wrenches             |
| 2      | 412    | Adjust Derailleur | Pivot the derailleur back            |
| 3      | 411    | Open Release      | Release caliper rim brake pads       |
|        | 415    |                   | Pull the release lever outwards      |
|        | 416    |                   | Loosen both axle nuts using a wrench |
|        | 439    |                   | Loosen the axle nuts                 |
|        | 602    |                   | Pull lever outward and turn axle CCW |

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| New ID | Old ID | Step Name    | Old Step Name                     |
|--------|--------|--------------|-----------------------------------|
| 4      | 418    | Remove Wheel | Pull out the wheel from the frame |
| 5      | 417    | Deflate Tube | Deflate the tube                  |

Table 8: Label mapping for “Fix a Flat Tire”

| New ID | Old ID | Step Name              | Old Step Name                      |
|--------|--------|------------------------|------------------------------------|
| 0      | 409    | Background             | Get bike tire lever                |
|        | 616    |                        | Background                         |
| 1      | 379    | Remove Valve Hardware  | Remove dust cap from the wheel     |
|        | 380    |                        | Remove presta valve from the wheel |
|        | 381    |                        | Remove stem nut from the wheel     |
|        | 382    |                        | Loosen lock nut at the valve stem  |
|        | 396    |                        | Remove the valve cap               |
|        | 398    |                        | Remove the dust cap                |
| 2      | 383    | Deflate Tube           | Squeeze out air inside the tube    |
|        | 417    |                        | Deflate the tube                   |
| 3      | 384    | Detach Tire Bead       | Engage tire lever on the rim       |
|        | 385    |                        | Pull back and lift bead out of rim |
|        | 386    |                        | Use tire lever around the rim      |
| 4      | 387    | Remove Old Tube        | Push the valve through its hole    |
|        | 388    |                        | Pull the inner tube out            |
| 5      | 389    | Inspect Tire           | Check for damage/splits in thread  |
|        | 407    |                        | Check for damage/splits in tube    |
| 6      | 391    | Prepare New Tube       | Get a new inner tube               |
|        | 399    |                        | Unpack the new tube                |
| 7      | 400    | Install Inner Tube     | Engage the valve stem into the rim |
|        | 401    |                        | Fit the tube to the wheel          |
| 8      | 402    | Mount Tire Bead        | Fit the tire to the wheel          |
| 9      | 392    | Install Valve Hardware | Fix stem nut to the wheel          |
|        | 394    |                        | Tighten the Dust Cap               |
|        | 395    |                        | Attach dust cap to the valve       |
|        | 397    |                        | Apply the dust cap                 |
|        | 393    |                        | Inflate the inner tube a little    |
| 10     | 403    | Inflate Tire           | Partially inflate to check seating |
|        | 406    |                        | Fully inflate the tire             |
| 11     | 405    | Check Bead Seating     | Inspect the bead seat line         |

Table 9: Label mapping for “Covid-19 Rapid Antigen Test”

| New ID | Old ID | Step Name         | Old Step Name                    |
|--------|--------|-------------------|----------------------------------|
| 0      | 133    | Background        | Set a timer                      |
|        | 334    |                   | Check the expiration date        |
|        | 350    |                   | Blow nose                        |
|        | 357    |                   | Get a timer                      |
|        | 363    |                   | Wait for testing incubation time |
|        | 365    |                   | Fold package box                 |
|        | 371    |                   | Stop timer                       |
|        | 373    |                   | Wait for 15 minutes              |
|        | 377    |                   | Collect saliva sample            |
|        | 378    |                   | Spit into the testing tube       |
|        | 615    |                   | Mistake                          |
|        | 616    |                   | Background                       |
|        | 873    |                   | Slowly extract tube from mouth   |
| 1      | 355    | Unbox package     | Unbox package                    |
| 2      | 337    | Arrange Materials | Arrange test material            |
|        | 370    |                   | Position tube on the box         |
| 3      | 335    | Prepare Tube      | Locate test tube                 |
|        | 336    |                   | Locate and unwrap test tube cap  |
|        | 338    |                   | Unwrap the testing tube          |

Continued on next page...

| New ID | Old ID                          | Step Name          | Old Step Name                                                                                                                     |
|--------|---------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------|
|        | 342                             |                    | Get the testing tube                                                                                                              |
|        | 345                             |                    | Carefully open tube seal paper                                                                                                    |
|        | 364                             |                    | Open the test tube lid                                                                                                            |
| 4      | 343<br>349                      | Fill Tube          | Fill tube with testing solution<br>Add sealed solution                                                                            |
| 5      | 344<br>348                      | Prepare Swab       | Locate and unwrap swab<br>Wrap/Unwrap Antigen Test Pack                                                                           |
| 6      | 351<br>353<br>354               | Swab Nose          | Slowly insert swab into nose<br>Slowly extract swab from nostril<br>Repeat with other nostril                                     |
| 7      | 352                             | Swab Nose (Rotate) | Rotate swab 5-10 times                                                                                                            |
| 8      | 347<br>358<br>367<br>369<br>372 | Mix Swab           | Cover the test tube<br>Dip swab into the testing tube<br>Remove swab from the tube<br>Shake the test tube<br>Check solutions tube |
| 9      | 346<br>356<br>366<br>368        | Prepare Cassette   | Locate and unwrap test plate<br>Unwrap testing plate<br>Locate and unwrap test strip<br>Dip test strip into tube                  |
| 10     | 360<br>361<br>362               | Apply Sample       | Apply drops to testing plate<br>Set swab into testing plate<br>Cover the testing plate                                            |
| 11     | 359<br>374                      | Check Result       | Visually inspect testing tube<br>Check the testing plate                                                                          |
| 12     | 339<br>375                      | Dispose Waste      | Return items to the box<br>Dispose waste                                                                                          |
| 13     | 340<br>341<br>376               | Read Instructions  | Read the instructions<br>Fold the instruction paper<br>Unfold the instruction paper                                               |

Table 10: Label mapping for “Cooking Noodles”

| New ID | Old ID | Step Name       | Old Step Name               |
|--------|--------|-----------------|-----------------------------|
|        | 123    |                 | Visually inspect the recipe |
|        | 132    |                 | Check paper recipe          |
|        | 133    |                 | Set a timer                 |
|        | 616    |                 | Background                  |
|        | 207    |                 | Get chopping board          |
|        | 208    |                 | Get chopsticks              |
|        | 210    |                 | Get cutting board           |
|        | 211    |                 | Get fork                    |
|        | 213    |                 | Get kitchen tong            |
|        | 214    |                 | Get kitchen towel           |
|        | 215    |                 | Get knife                   |
|        | 217    |                 | Get measuring tool          |
|        | 218    |                 | Get napkin                  |
|        | 221    |                 | Get plate                   |
|        | 222    |                 | Get pot holder              |
|        | 223    |                 | Get sieve                   |
|        | 224    |                 | Get skillet/pan/wok         |
|        | 225    |                 | Get spatula                 |
|        | 226    |                 | Get spoon                   |
|        | 229    |                 | Get a cup or mug            |
|        | 357    |                 | Get a timer                 |
|        | 493    |                 | Get a pot or saucepan       |
|        | 495    |                 | Get an electric kettle      |
|        | 548    |                 | Get a bowl                  |
| 1      | 140    | Get Kitchenware | Get cabbage                 |
|        | 143    |                 | Get celeries                |
|        | 144    |                 | Get cheese                  |

Continued on next page...

| New ID | Old ID | Step Name           | Old Step Name                     |
|--------|--------|---------------------|-----------------------------------|
|        | 153    |                     | Get curry powder                  |
|        | 160    |                     | Get garlic cloves                 |
|        | 161    |                     | Get garlic paste                  |
|        | 166    |                     | Get honey                         |
|        | 167    |                     | Get hot sauce                     |
|        | 168    |                     | Get ketchup                       |
|        | 176    |                     | Get noodles                       |
|        | 177    |                     | Get nutmeg                        |
|        | 178    |                     | Get oil                           |
|        | 179    |                     | Get olive oil                     |
|        | 181    |                     | Get onions                        |
|        | 182    |                     | Get oyster sauce                  |
|        | 184    |                     | Get peanut butter                 |
|        | 185    |                     | Get pepper                        |
|        | 189    |                     | Get salt                          |
|        | 190    |                     | Get scallion                      |
|        | 191    |                     | Get sesame oil                    |
|        | 192    |                     | Get sesame seeds                  |
|        | 193    |                     | Get soy sauce                     |
|        | 195    |                     | Get spring onions                 |
|        | 197    |                     | Get tamari                        |
|        | 200    |                     | Get toasted sesame oil            |
|        | 202    |                     | Get tomato sauce                  |
|        | 205    |                     | Get water                         |
|        | 69     |                     | Wash garlic cloves                |
|        | 72     |                     | Wash noodles                      |
|        | 75     |                     | Wash scallion                     |
|        | 76     |                     | Wash spring onions                |
|        | 83     |                     | Peel garlic cloves                |
|        | 84     |                     | Peel fresh ginger                 |
|        | 86     |                     | Peel onions                       |
|        | 87     |                     | Peel spring onions                |
| 3      | 93     | Prepare Ingredients | Cut cabbage                       |
|        | 95     |                     | Cut celeries                      |
|        | 98     |                     | Cut chili powder                  |
|        | 102    |                     | Cut garlic cloves                 |
|        | 107    |                     | Cut mustard leaves                |
|        | 108    |                     | Cut noodles                       |
|        | 110    |                     | Cut onions                        |
|        | 111    |                     | Cut scallion                      |
|        | 113    |                     | Cut spring onions                 |
|        | 529    |                     | Cut ginger                        |
| 4      | 494    | Heat Water          | Wait until water boils            |
|        | 532    |                     | Check heat of the pot             |
| 5      | 533    | Boil Noodles        | Add the noodles to boiling water  |
|        | 534    |                     | Stir noodles in the pot           |
|        | 535    |                     | Taste for doneness                |
| 6      | 536    | Drain Noodles       | Drain the noodles into a strainer |
|        | 38     |                     | Add oil                           |
|        | 124    |                     | Turn on the stove                 |
|        | 125    |                     | Adjust the stove heat             |
|        | 126    |                     | Turn off the stove                |
|        | 127    |                     | Lift the lid                      |
| 7      | 128    | Prepare Skillet     | Close the lid                     |
|        | 129    |                     | Place skillet on the stove        |
|        | 130    |                     | Remove skillet from the stove     |
|        | 477    |                     | Check heat of the skillet         |
|        | 486    |                     | Tilt and rotate the skillet       |
|        | 518    |                     | Add olive oil to a skillet        |
| 8      | 22     | Stir-Fry Aromatics  | Add garlic cloves                 |
|        | 23     |                     | Add garlic paste                  |
|        | 24     |                     | Add garlic powder                 |

Continued on next page...

| New ID | Old ID | Step Name        | Old Step Name                  |
|--------|--------|------------------|--------------------------------|
|        | 39     |                  | Add onions                     |
|        | 44     |                  | Add fresh ginger               |
|        | 46     |                  | Add scallion                   |
|        | 50     |                  | Add spring onions              |
|        | 517    |                  | Add mustard leaves to skillet  |
|        | 36     |                  | Add noodles                    |
| 9      | 512    | Stir-Fry Noodles | Stir the mixture               |
|        | 1748   |                  | Add onion powder to skillet    |
|        | 537    |                  | Add mix of ingredients to bowl |
| 10     | 538    | Mix Sauce (Bowl) | Add mustard leaves to a bowl   |
|        | 539    |                  | Add olive oil to a bowl        |
|        | 540    |                  | Add oyster sauce to a bowl     |
|        | 5      |                  | Add black pepper               |
|        | 6      |                  | Add cabbage                    |
|        | 8      |                  | Add celeries                   |
|        | 9      |                  | Add cheese                     |
|        | 11     |                  | Add chili powder               |
|        | 17     |                  | Add curry powder               |
|        | 18     |                  | Add dried herbs                |
|        | 28     |                  | Add honey                      |
|        | 29     |                  | Add hot sauce                  |
|        | 30     |                  | Add ketchup                    |
|        | 37     |                  | Add nutmeg                     |
| 11     | 41     | Season Food      | Add peanut butter              |
|        | 42     |                  | Add pepper                     |
|        | 45     |                  | Add salt                       |
|        | 47     |                  | Add sesame oil                 |
|        | 48     |                  | Add sesame seeds               |
|        | 49     |                  | Add soy sauce                  |
|        | 52     |                  | Add tamari                     |
|        | 55     |                  | Add tomato sauce               |
|        | 56     |                  | Add turmeric powder            |
|        | 57     |                  | Add vinegar                    |
|        | 58     |                  | Add water                      |
|        | 121    |                  | Taste the recipe               |
|        | 523    |                  | Add oyster sauce to skillet    |
| 12     | 228    | Serve Food       | Transfer food to a plate       |
|        | 0      |                  | Wash hands                     |
|        | 1      |                  | Wipe hands                     |
|        | 235    |                  | Wash bowl                      |
|        | 238    |                  | Wash cutting board             |
|        | 242    |                  | Wash knife                     |
| 13     | 244    | Wash Items       | Wash measuring tool            |
|        | 247    |                  | Wash plate                     |
|        | 248    |                  | Wash pot or saucepan           |
|        | 249    |                  | Wash sieve                     |
|        | 250    |                  | Wash skillet/frying pan/wok    |
|        | 251    |                  | Wash spatula                   |
|        | 252    |                  | Wash spoon                     |
|        | 253    |                  | Put away bowl                  |
|        | 254    |                  | Put away chopping board        |
|        | 263    |                  | Put away knife                 |
|        | 264    |                  | Put away measuring tool        |
|        | 265    |                  | Put away napkin                |
|        | 267    |                  | Put away plate                 |
| 14     | 269    | Put Away Items   | Put away pot or saucepan       |
|        | 271    |                  | Put away sieve                 |
|        | 272    |                  | Put away skillet/pan/wok       |
|        | 273    |                  | Put away spatula               |
|        | 274    |                  | Put away spoon                 |
|        | 275    |                  | Put away timer                 |
|        | 284    |                  | Put away cheese                |

Continued on next page...

| New ID | Old ID | Step Name | Old Step Name               |
|--------|--------|-----------|-----------------------------|
|        | 296    |           | Put away garlic cloves      |
|        | 297    |           | Put away garlic paste       |
|        | 300    |           | Put away honey              |
|        | 301    |           | Put away hot sauce          |
|        | 302    |           | Put away ketchup            |
|        | 306    |           | Put away noodles            |
|        | 308    |           | Put away oil                |
|        | 309    |           | Put away olive oil          |
|        | 310    |           | Put away onions             |
|        | 311    |           | Put away oyster sauce       |
|        | 313    |           | Put away pepper             |
|        | 315    |           | Put away salt               |
|        | 316    |           | Put away scallion           |
|        | 317    |           | Put away sesame oil         |
|        | 318    |           | Put away sesame seeds       |
|        | 319    |           | Put away soy sauce          |
|        | 321    |           | Put away spring onions      |
|        | 323    |           | Put away tamari             |
|        | 326    |           | Put away toasted sesame oil |
|        | 328    |           | Put away tomato sauce       |
|        | 332    |           | Wipe kitchen countertop     |
|        | 333    |           | Throw out trash or waste    |

Table 11: Label mapping for “Making Coffee latte”

| New ID | Old ID | Step Name                 | Old Step Name               |
|--------|--------|---------------------------|-----------------------------|
| 0      | 529    | Background                | Cut ginger                  |
|        | 616    |                           | Background                  |
| 1      | 207    | Get Kitchenware           | Get chopping board          |
|        | 209    |                           | Get coffee filter           |
|        | 211    |                           | Get fork                    |
|        | 214    |                           | Get kitchen towel           |
|        | 215    |                           | Get knife                   |
|        | 217    |                           | Get measuring tool          |
|        | 218    |                           | Get napkin                  |
|        | 226    |                           | Get spoon                   |
|        | 229    |                           | Get a cup or mug            |
|        | 493    |                           | Get a pot or saucepan       |
|        | 495    |                           | Get an electric kettle      |
| 2      | 548    | Get Ingredients           | Get a bowl                  |
|        | 582    |                           | Get french press            |
|        | 589    |                           | Get a milk frothing pitcher |
|        | 147    |                           | Get chocolate               |
|        | 150    |                           | Get coffee beans            |
|        | 151    |                           | Get coffee grounds          |
| 3      | 174    | Operate Heating Appliance | Get milk                    |
|        | 196    |                           | Get sugar                   |
|        | 205    |                           | Get water                   |
|        | 531    |                           | Get cinnamon stick          |
|        | 124    |                           | Turn on the stove           |
| 3      | 125    | Operate Heating Appliance | Adjust the stove heat       |
|        | 126    |                           | Turn off the stove          |
|        | 127    |                           | Lift the lid                |
|        | 128    |                           | Close the lid               |
|        | 129    |                           | Place skillet on stove      |
|        | 130    |                           | Remove skillet from stove   |
|        | 526    |                           | Turn on electric kettle     |
|        | 558    |                           | Fill electric kettle        |
|        | 559    |                           | Heat the saucepan           |
|        | 560    |                           | Reduce water level          |
|        | 561    |                           | Place water on the stove    |

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| New ID | Old ID | Step Name                | Old Step Name                 |
|--------|--------|--------------------------|-------------------------------|
| 4      | 494    | Wait for Boiling         | Wait until water boils        |
|        | 597    |                          | Wait until milk boils         |
| 5      | 15     | Prepare Filter & Grounds | Add coffee grounds            |
|        | 557    |                          | Operate the grinder           |
|        | 563    |                          | Add instant coffee            |
|        | 568    |                          | Open instant coffee jar       |
|        | 569    |                          | Close instant coffee jar      |
|        | 570    |                          | Add freshly ground coffee     |
|        | 572    |                          | Place the coffee filter       |
|        | 573    |                          | Set pour-over device          |
|        | 574    |                          | Place filter bag over bowl    |
|        | 575    |                          | Rinse the filter paper        |
|        | 576    |                          | Measure coffee grounds        |
| 6      | 577    | Pour Water for Brewing   | Add grounds to filter         |
|        | 584    |                          | Add grounds to french press   |
|        | 58     |                          | Add water                     |
|        | 562    |                          | Pour hot water to mug         |
|        | 579    |                          | Pour hot water through filter |
| 7      | 583    | Process Milk             | Preheat french press          |
|        | 591    |                          | Submerge steam wand tip       |
|        | 592    |                          | Turn on steam wand            |
|        | 593    |                          | Steam to desired temperature  |
|        | 594    |                          | Remove steam wand             |
|        | 595    |                          | Fill a pot with milk          |
| 8      | 596    | Pour Milk                | Stir the milk in the pot      |
|        | 564    |                          | Add cold milk                 |
|        | 565    |                          | Add hot milk                  |
|        | 590    |                          | Pour milk into pitcher        |
|        | 598    |                          | Pour cold milk into mug       |
| 9      | 599    | Extract & Mix Coffee     | Pour frothed milk into mug    |
|        | 566    |                          | Mix instant coffee in water   |
|        | 567    |                          | Mix instant coffee in milk    |
|        | 571    |                          | Wait for extraction           |
|        | 580    |                          | Strain coffee into a cup      |
|        | 581    |                          | Remove pour-over device       |
|        | 585    |                          | Stir coffee/water mixture     |
|        | 586    |                          | Place lid on french press     |
| 10     | 587    | Pour Coffee to Serve     | Press the plunger             |
|        | 588    |                          | Pour coffee into a mug        |
| 11     | 12     | Add Sweetener & Spices   | Add chocolate                 |
|        | 25     |                          | Add fresh ginger              |
|        | 51     |                          | Add sugar                     |
|        | 99     |                          | Cut chocolate                 |
|        | 121    |                          | Taste the recipe              |
|        | 233    |                          | Add cinnamon stick            |
|        | 600    |                          | Stir the coffee               |
| 12     | 601    | Wash Items               | Remove cinnamon               |
|        | 237    |                          | Wash cup or mug               |
|        | 242    |                          | Wash knife                    |
|        | 244    |                          | Wash measuring tool           |
|        | 245    |                          | Wash napkin                   |
|        | 248    |                          | Wash pot or saucepan          |
| 13     | 252    | Put Away Items           | Wash spoon                    |
|        | 253    |                          | Put away bowl                 |
|        | 256    |                          | Put away coffee filter        |
|        | 257    |                          | Put away cup or mug           |
|        | 265    |                          | Put away napkin               |
|        | 269    |                          | Put away pot or saucepan      |
|        | 270    |                          | Put away pour-over device     |
|        | 274    |                          | Put away spoon                |
|        | 287    |                          | Put away chocolate            |
|        | 289    |                          | Put away cinnamon sticks      |

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| New ID | Old ID | Step Name  | Old Step Name           |
|--------|--------|------------|-------------------------|
|        | 290    |            | Put away coffee grounds |
|        | 305    |            | Put away milk           |
|        | 322    |            | Put away sugar          |
|        | 331    |            | Put away water          |
|        | 1      |            | Wipe hands              |
| 14     | 332    | Wipe Clean | Wipe kitchen countertop |
|        | 333    |            | Throw out trash         |

Table 12: Label mapping for “Making Cucumber & Tomato Salad”

| New ID | Old ID                                                                                                                                                                      | Step Name             | Old Step Name                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0      | 124<br>616                                                                                                                                                                  | Background            | Turn on the stove<br>Background                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1      | 129<br>207<br>210<br>215<br>219<br>221<br>223<br>225<br>226<br>495<br>548                                                                                                   | Get Kitchenware       | Place the skillet/pan/pot on the stove<br>Get chopping board<br>Get cutting board<br>Get knife<br>Get peeler<br>Get plate<br>Get sieve<br>Get spatula<br>Get spoon<br>Get an electric kettle<br>Get a bowl                                                                                                                                                                                                                              |
| 2      | 134<br>138<br>145<br>152<br>154<br>165<br>170<br>172<br>173<br>174<br>178<br>179<br>180<br>181<br>185<br>188<br>189<br>191<br>193<br>196<br>200<br>201<br>203<br>204<br>206 | Get Ingredients       | Get agave<br>Get black pepper<br>Get cherry tomatoes<br>Get cucumber<br>Get dried herbs<br>Get green chilies<br>Get lemon juice<br>Get lime<br>Get mayonnaise<br>Get milk<br>Get oil<br>Get olive oil<br>Get onion powder<br>Get onions<br>Get pepper<br>Get rice vinegar<br>Get salt<br>Get sesame oil<br>Get soy sauce<br>Get sugar<br>Get toasted sesame oil<br>Get tomato<br>Get turmeric powder<br>Get vinegar<br>Get white pepper |
| 3      | 64<br>66<br>73<br>78<br>150                                                                                                                                                 | Wash Vegetables       | Wash cherry tomatoes<br>Wash cucumber<br>Wash onions<br>Wash tomato<br>Wash lemon                                                                                                                                                                                                                                                                                                                                                       |
| 4      | 81<br>85<br>86                                                                                                                                                              | Peel Ingredients      | Peel cucumber<br>Peel ice<br>Peel onions                                                                                                                                                                                                                                                                                                                                                                                                |
| 5      | 97<br>115                                                                                                                                                                   | Cut Tomato            | Cut cherry tomatoes<br>Cut tomato                                                                                                                                                                                                                                                                                                                                                                                                       |
| 6      | 101                                                                                                                                                                         | Cut Cucumber          | Cut cucumber                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 7      | 91                                                                                                                                                                          | Cut Other Ingredients | Cut black pepper                                                                                                                                                                                                                                                                                                                                                                                                                        |

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| New ID | Old ID | Step Name             | Old Step Name                         |
|--------|--------|-----------------------|---------------------------------------|
|        | 103    |                       | Cut green chilies                     |
|        | 105    |                       | Cut lime                              |
|        | 110    |                       | Cut onions                            |
|        | 118    |                       | Remove seeds from cucumber            |
|        | 120    |                       | Remove seeds from tomato              |
|        | 10     |                       | Add cherry tomatoes                   |
|        | 16     |                       | Add cucumber                          |
| 8      | 39     | Add Solid Ingredients | Add onions                            |
|        | 54     |                       | Add tomato                            |
|        | 537    |                       | Add a mix of ingredients to a bowl    |
|        | 5      |                       | Add black pepper                      |
|        | 11     |                       | Add chili powder                      |
|        | 18     |                       | Add dried herbs                       |
|        | 31     |                       | Add lemon juice                       |
|        | 33     |                       | Add mayonnaise                        |
|        | 34     |                       | Add milk                              |
|        | 38     |                       | Add oil                               |
|        | 42     |                       | Add pepper                            |
|        | 45     |                       | Add salt                              |
|        | 47     |                       | Add sesame oil                        |
| 9      | 48     | Add Dressing & Spices | Add sesame seeds                      |
|        | 49     |                       | Add soy sauce                         |
|        | 51     |                       | Add sugar                             |
|        | 56     |                       | Add turmeric powder                   |
|        | 57     |                       | Add vinegar                           |
|        | 59     |                       | Add white pepper                      |
|        | 467    |                       | Add agave to a mixing bowl            |
|        | 471    |                       | Add lime juice to a mixing bowl       |
|        | 472    |                       | Add onion powder to a mixing bowl     |
|        | 510    |                       | Pour the dressing                     |
|        | 539    |                       | Add olive oil to a bowl               |
|        | 474    |                       | Stir the salad mixture                |
| 10     | 475    | Mix Salad             | Stir the dressing mixture             |
|        | 476    |                       | Toss the salad in a mixing bowl       |
| 11     | 228    | Serve Salad           | Transfer food to a plate              |
|        | 235    |                       | Wash bowl                             |
|        | 236    |                       | Wash chopsticks                       |
|        | 238    |                       | Wash cutting board                    |
| 12     | 242    | Wash Dishes           | Wash knife                            |
|        | 247    |                       | Wash plate                            |
|        | 250    |                       | Wash skillet or frying pan or wok     |
|        | 252    |                       | Wash spoon                            |
|        | 253    |                       | Put away bowl                         |
|        | 254    |                       | Put away chopping board               |
|        | 255    |                       | Put away chopsticks                   |
|        | 258    |                       | Put away cutting board                |
|        | 263    |                       | Put away knife                        |
|        | 266    |                       | Put away peeler                       |
|        | 267    |                       | Put away plate                        |
|        | 269    |                       | Put away pot or saucepan              |
|        | 272    |                       | Put away skillet or frying pan or wok |
|        | 273    |                       | Put away spatula                      |
|        | 279    |                       | Put away black pepper                 |
| 13     | 291    | Put Away Items        | Put away cucumber                     |
|        | 293    |                       | Put away dried herbs                  |
|        | 303    |                       | Put away lemon juice                  |
|        | 304    |                       | Put away mayonnaise                   |
|        | 308    |                       | Put away oil                          |
|        | 310    |                       | Put away onions                       |
|        | 313    |                       | Put away pepper                       |
|        | 315    |                       | Put away salt                         |
|        | 319    |                       | Put away soy sauce                    |

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| New ID | Old ID | Step Name    | Old Step Name            |
|--------|--------|--------------|--------------------------|
|        | 322    |              | Put away sugar           |
|        | 327    |              | Put away tomato          |
|        | 329    |              | Put away turmeric powder |
|        | 330    |              | Put away vinegar         |
|        | 0      |              | Wash hands               |
| 14     | 1      | Wipe & Clean | Wipe hands               |
|        | 332    |              | Wipe kitchen countertop  |
|        | 333    |              | Throw out trash or waste |

Table 13: Label mapping for “Cooking Scrambled Eggs”

| New ID | Old ID | Step Name                         | Old Step Name                                                            |
|--------|--------|-----------------------------------|--------------------------------------------------------------------------|
| 0      | 130    | Background                        | Remove the skillet or pan or pot from the stove                          |
|        | 131    |                                   | Check phone                                                              |
|        | 616    |                                   | Background                                                               |
| 1      | 138    | Get Items                         | Get black pepper                                                         |
|        | 139    |                                   | Get butter                                                               |
|        | 144    |                                   | Get cheese                                                               |
|        | 145    |                                   | Get cherry tomatoes                                                      |
|        | 146    |                                   | Get chili powder                                                         |
|        | 153    |                                   | Get curry powder                                                         |
|        | 154    |                                   | Get dried herbs                                                          |
|        | 156    |                                   | Get eggs                                                                 |
|        | 174    |                                   | Get milk                                                                 |
|        | 178    |                                   | Get oil                                                                  |
|        | 181    |                                   | Get onions                                                               |
|        | 185    |                                   | Get pepper                                                               |
|        | 189    |                                   | Get salt                                                                 |
|        | 195    |                                   | Get spring onions                                                        |
|        | 201    |                                   | Get tomato                                                               |
|        | 207    |                                   | Get chopping board                                                       |
|        | 208    |                                   | Get chopsticks                                                           |
|        | 211    |                                   | Get fork                                                                 |
|        | 215    |                                   | Get knife                                                                |
|        | 216    |                                   | Get ladle                                                                |
|        | 221    |                                   | Get plate                                                                |
|        | 224    |                                   | Get skillet or frying pan or wok                                         |
|        | 225    |                                   | Get spatula                                                              |
|        | 226    |                                   | Get spoon                                                                |
|        | 227    |                                   | Get whisk                                                                |
|        | 548    |                                   | Get a bowl                                                               |
| 2      | 38     | Heat Skillet                      | Add oil                                                                  |
|        | 124    |                                   | Turn on the stove                                                        |
|        | 125    |                                   | Adjust the stove heat                                                    |
|        | 126    |                                   | Turn off the stove                                                       |
|        | 129    |                                   | Place the skillet or pan or pot on the stove                             |
|        | 477    |                                   | Check heat of the skillet                                                |
|        | 478    |                                   | Add butter to skillet                                                    |
|        | 486    |                                   | Tilt and rotate the skillet to allow oil or butter flow into empty space |
| 3      | 76     | Prepare Ingredients<br>(Cut/Peel) | Wash spring onions                                                       |
|        | 78     |                                   | Wash tomato                                                              |
|        | 86     |                                   | Peel onions                                                              |
|        | 87     |                                   | Peel spring onions                                                       |
|        | 92     |                                   | Cut butter                                                               |
|        | 96     |                                   | Cut cheese                                                               |
|        | 97     |                                   | Cut cherry tomatoes                                                      |
|        | 110    |                                   | Cut onions                                                               |
|        | 113    |                                   | Cut spring onions                                                        |
|        | 115    |                                   | Cut tomato                                                               |
| 4      | 34     | Whisk Eggs                        | Add milk                                                                 |
|        | 479    |                                   | Crack eggs into a mixing bowl                                            |

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| New ID | Old ID | Step Name      | Old Step Name                                                |
|--------|--------|----------------|--------------------------------------------------------------|
|        | 480    |                | Whisk until the egg whites and yolks are well integrated.    |
| 5      | 10     | Fry Vegetables | Add cherry tomatoes                                          |
|        | 39     |                | Add onions                                                   |
|        | 50     |                | Add spring onions                                            |
|        | 54     |                | Add tomato                                                   |
|        | 481    |                | Stir the mix of vegetables in the skillet                    |
| 6      | 9      | Scramble Eggs  | Add cheese                                                   |
|        | 12     |                | Add chocolate                                                |
|        | 33     |                | Add eggs                                                     |
|        | 482    |                | Gently push cooked portions toward the center of the skillet |
|        | 485    |                | Cook until the egg is cooked                                 |
|        | 487    |                | Pour egg mixture into a skillet                              |
| 7      | 5      | Season Food    | Add black pepper                                             |
|        | 11     |                | Add chili powder                                             |
|        | 17     |                | Add curry powder                                             |
|        | 18     |                | Add dried herbs                                              |
|        | 42     |                | Add pepper                                                   |
|        | 45     |                | Add salt                                                     |
|        | 48     |                | Add sesame seeds                                             |
|        | 614    |                | Add ingredients to egg mixture                               |
| 1      | 228    | Serve Food     | Transfer food to a plate                                     |
| 9      | 0      | Cleanup        | Wash hands                                                   |
|        | 1      |                | Wipe hands                                                   |
|        | 127    |                | Lift the lid of a skillet or pan or pot                      |
|        | 128    |                | Close the lid of a skillet or pan or pot                     |
|        | 235    |                | Wash bowl                                                    |
|        | 240    |                | Wash fork                                                    |
|        | 242    |                | Wash knife                                                   |
|        | 247    |                | Wash plate                                                   |
|        | 250    |                | Wash skillet or frying pan or wok                            |
|        | 251    |                | Wash spatula                                                 |
|        | 252    |                | Wash spoon                                                   |
|        | 253    |                | Put away bowl                                                |
|        | 254    |                | Put away chopping board                                      |
|        | 263    |                | Put away knife                                               |
|        | 272    |                | Put away skillet or frying pan or wok                        |
|        | 273    |                | Put away spatula                                             |
|        | 279    |                | Put away black pepper                                        |
|        | 280    |                | Put away butter                                              |
|        | 286    |                | Put away chili powder                                        |
|        | 292    |                | Put away curry powder                                        |
|        | 293    |                | Put away dried herbs                                         |
|        | 294    |                | Put away eggs                                                |
|        | 305    |                | Put away milk                                                |
|        | 308    |                | Put away oil                                                 |
|        | 315    |                | Put away salt                                                |
|        | 332    |                | Wipe kitchen countertop                                      |
|        | 333    |                | Throw out trash or waste                                     |

Table 14: Label mapping for “Cooking an Omelet”

| New ID | Old ID | Step Name       | Old Step Name                            |
|--------|--------|-----------------|------------------------------------------|
| 0      | 130    | Background      | Remove skillet/pan/pot from stove        |
|        | 132    |                 | Check paper recipe                       |
|        | 489    |                 | Pour cooked egg mixture into a plate     |
|        | 492    |                 | Cook until the omelet is cooked          |
|        | 616    |                 | Background                               |
|        | 2322   |                 | Soak up the excess oil from the skillet. |
| 1      | 207    | Get Kitchenware | Get chopping board                       |
|        | 208    |                 | Get chopsticks                           |
|        | 210    |                 | Get cutting board                        |

Continued on next page...

| New ID | Old ID | Step Name        | Old Step Name                    |
|--------|--------|------------------|----------------------------------|
|        | 211    |                  | Get fork                         |
|        | 213    |                  | Get kitchen tong                 |
|        | 215    |                  | Get knife                        |
|        | 216    |                  | Get ladle                        |
|        | 217    |                  | Get measuring tool               |
|        | 218    |                  | Get napkin                       |
|        | 221    |                  | Get plate                        |
|        | 224    |                  | Get skillet or frying pan or wok |
|        | 225    |                  | Get spatula                      |
|        | 226    |                  | Get spoon                        |
|        | 229    |                  | Get a cup or mug                 |
|        | 548    |                  | Get a bowl                       |
|        | 632    |                  | Get whisk                        |
|        | 135    |                  | Get almonds                      |
|        | 136    |                  | Get beef                         |
|        | 138    |                  | Get black pepper                 |
|        | 139    |                  | Get butter                       |
|        | 144    |                  | Get cheese                       |
|        | 146    |                  | Get chili powder                 |
|        | 148    |                  | Get cilantro                     |
|        | 153    |                  | Get curry powder                 |
|        | 154    |                  | Get dried herbs                  |
|        | 155    |                  | Get dried seaweed                |
|        | 156    |                  | Get eggs                         |
|        | 160    |                  | Get garlic cloves                |
|        | 161    |                  | Get garlic paste                 |
|        | 163    |                  | Get ginger garlic paste          |
|        | 164    |                  | Get ginger paste                 |
|        | 165    |                  | Get green chilies                |
| 2      | 168    | Get Ingredients  | Get ketchup                      |
|        | 174    |                  | Get milk                         |
|        | 178    |                  | Get oil                          |
|        | 180    |                  | Get onion powder                 |
|        | 181    |                  | Get onions                       |
|        | 183    |                  | Get parsley                      |
|        | 185    |                  | Get pepper                       |
|        | 186    |                  | Get radish                       |
|        | 188    |                  | Get rice vinegar                 |
|        | 189    |                  | Get salt                         |
|        | 190    |                  | Get scallion                     |
|        | 193    |                  | Get soy sauce                    |
|        | 196    |                  | Get sugar                        |
|        | 201    |                  | Get tomato                       |
|        | 202    |                  | Get tomato sauce                 |
|        | 203    |                  | Get turmeric powder              |
|        | 204    |                  | Get vinegar                      |
|        | 206    |                  | Get white pepper                 |
|        | 60     |                  | Wash bell peppers                |
|        | 67     |                  | Wash eggs                        |
| 3      | 70     | Wash Ingredients | Wash green chilies               |
|        | 73     |                  | Wash onions                      |
|        | 74     |                  | Wash radish                      |
|        | 78     |                  | Wash tomato                      |
| 4      | 83     | Peel Ingredients | Peel garlic cloves               |
|        | 86     |                  | Peel onions                      |
|        | 89     |                  | Cut beef                         |
|        | 90     |                  | Cut bell peppers                 |
|        | 92     |                  | Cut butter                       |
|        | 102    |                  | Cut garlic cloves                |
| 5      | 103    | Cut Ingredients  | Cut green chilies                |
|        | 110    |                  | Cut onions                       |
|        | 111    |                  | Cut scallion                     |
|        | 115    |                  | Cut tomato                       |

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| New ID | Old ID | Step Name       | Old Step Name                         |
|--------|--------|-----------------|---------------------------------------|
|        | 119    |                 | Remove seeds from green chilies       |
|        | 529    |                 | Cut ginger                            |
|        | 38     |                 | Add oil                               |
|        | 124    |                 | Turn on the stove                     |
|        | 125    |                 | Adjust the stove heat                 |
|        | 126    |                 | Turn off the stove                    |
| 6      | 127    | Heat Skillet    | Lift the lid of a skillet/pan/pot     |
|        | 128    |                 | Close the lid of a skillet/pan/pot    |
|        | 129    |                 | Place skillet/pan/pot on the stove    |
|        | 477    |                 | Check heat of the skillet             |
|        | 478    |                 | Add butter to skillet                 |
|        | 486    |                 | Tilt and rotate the skillet           |
|        | 34     |                 | Add milk                              |
| 7      | 469    | Whisk Eggs      | Add ginger garlic paste to a bowl     |
|        | 472    |                 | Add onion powder to a mixing bowl     |
|        | 479    |                 | Crack eggs into a mixing bowl         |
|        | 480    |                 | Whisk egg whites and yolks            |
|        | 3      |                 | Add beef                              |
|        | 9      |                 | Add cheese                            |
|        | 13     |                 | Add cilantro                          |
|        | 22     |                 | Add garlic cloves                     |
|        | 23     |                 | Add garlic paste                      |
|        | 26     |                 | Add ginger paste                      |
|        | 27     |                 | Add green chilies                     |
|        | 30     |                 | Add ketchup                           |
| 8      | 39     | Saute Aromatics | Add onions                            |
|        | 40     |                 | Add parsley                           |
|        | 43     |                 | Add radish                            |
|        | 46     |                 | Add scallion                          |
|        | 49     |                 | Add soy sauce                         |
|        | 54     |                 | Add tomato                            |
|        | 55     |                 | Add tomato sauce                      |
|        | 57     |                 | Add vinegar                           |
|        | 58     |                 | Add water                             |
|        | 488    |                 | Stir fry egg mixture                  |
| 9      | 483    | Pour & Set      | Tilt/rotate skillet for uncooked egg  |
|        | 487    |                 | Pour egg mixture into a skillet       |
| 10     | 482    | Cook & Shape    | Gently push cooked portions to center |
| 11     | 490    | Flip & Fold     | Flip the egg to cook evenly           |
|        | 491    |                 | Fold the omelet in half               |
|        | 2      |                 | Add almonds                           |
|        | 5      |                 | Add black pepper                      |
|        | 11     |                 | Add chili powder                      |
|        | 14     |                 | Add cinnamon powder                   |
|        | 17     |                 | Add curry powder                      |
| 12     | 18     | Season Food     | Add dried herbs                       |
|        | 37     |                 | Add nutmeg                            |
|        | 42     |                 | Add pepper                            |
|        | 45     |                 | Add salt                              |
|        | 51     |                 | Add sugar                             |
|        | 56     |                 | Add turmeric powder                   |
|        | 59     |                 | Add white pepper                      |
| 13     | 228    | Serve Food      | Transfer food to a plate              |
|        | 0      |                 | Wash hands                            |
|        | 1      |                 | Wipe hands                            |
|        | 235    |                 | Wash bowl                             |
| 14     | 236    | Cleanup         | Wash chopsticks                       |
|        | 237    |                 | Wash cup or mug                       |
|        | 238    |                 | Wash cutting board                    |
|        | 242    |                 | Wash knife                            |
|        | 244    |                 | Wash measuring tool                   |
|        | 247    |                 | Wash plate                            |

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| New ID | Old ID | Step Name | Old Step Name               |
|--------|--------|-----------|-----------------------------|
|        | 248    |           | Wash pot or saucepan        |
|        | 250    |           | Wash skillet/frying pan/wok |
|        | 251    |           | Wash spatula                |
|        | 252    |           | Wash spoon                  |
|        | 253    |           | Put away bowl               |
|        | 254    |           | Put away chopping board     |
|        | 258    |           | Put away cutting board      |
|        | 259    |           | Put away fork               |
|        | 263    |           | Put away knife              |
|        | 264    |           | Put away measuring tool     |
|        | 268    |           | Put away pot holder         |
|        | 273    |           | Put away spatula            |
|        | 276    |           | Put away almonds            |
|        | 277    |           | Put away beef               |
|        | 279    |           | Put away black pepper       |
|        | 286    |           | Put away chili powder       |
|        | 292    |           | Put away curry powder       |
|        | 294    |           | Put away eggs               |
|        | 297    |           | Put away garlic paste       |
|        | 299    |           | Put away green chilies      |
|        | 305    |           | Put away milk               |
|        | 308    |           | Put away oil                |
|        | 310    |           | Put away onions             |
|        | 313    |           | Put away pepper             |
|        | 315    |           | Put away salt               |
|        | 320    |           | Put away spinach            |
|        | 327    |           | Put away tomato             |
|        | 329    |           | Put away turmeric powder    |
|        | 332    |           | Wipe kitchen countertop     |
|        | 333    |           | Throw out trash or waste    |
| 15     | 615    | Mistake   | Mistake                     |

Table 15: Label mapping for “Making Milk Tea”

| New ID | Old ID | Step Name  | Old Step Name               |
|--------|--------|------------|-----------------------------|
| 0      | 114    | Background | Cut tea bag                 |
|        | 121    |            | Taste the recipe            |
|        | 122    |            | Measure the temperature     |
|        | 123    |            | Visually inspect the recipe |
|        | 616    |            | Background                  |
| 1      | 138    | Get Items  | Get black pepper            |
|        | 151    |            | Get coffee grounds          |
|        | 154    |            | Get dried herbs             |
|        | 166    |            | Get honey                   |
|        | 174    |            | Get milk                    |
|        | 177    |            | Get nutmeg                  |
|        | 181    |            | Get onions                  |
|        | 190    |            | Get scallion                |
|        | 196    |            | Get sugar                   |
|        | 198    |            | Get tea bag                 |
|        | 199    |            | Get tea leaves              |
|        | 205    |            | Get water                   |
|        | 208    |            | Get chopsticks              |
|        | 211    |            | Get fork                    |
|        | 213    |            | Get kitchen tong            |
|        | 215    |            | Get knife                   |
|        | 218    |            | Get napkin                  |
|        | 221    |            | Get plate                   |
|        | 222    |            | Get pot holder              |
|        | 223    |            | Get sieve                   |
|        | 226    |            | Get spoon                   |

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| New ID | Old ID | Step Name              | Old Step Name                      |
|--------|--------|------------------------|------------------------------------|
|        | 229    |                        | Get a cup or mug                   |
|        | 231    |                        | Get a fine-mesh sieve              |
|        | 493    |                        | Get a pot or saucepan              |
|        | 495    |                        | Get an electric kettle             |
|        | 527    |                        | Get ginger                         |
|        | 531    |                        | Get cinnamon stick                 |
|        | 548    |                        | Get a bowl                         |
|        | 58     |                        | Add water                          |
|        | 124    |                        | Turn on the stove                  |
|        | 125    |                        | Adjust the stove heat              |
|        | 126    |                        | Turn off the stove                 |
| 2      | 127    | Heat Water             | Lift the lid                       |
|        | 128    |                        | Close the lid                      |
|        | 129    |                        | Place the skillet/pan/pot on stove |
|        | 494    |                        | Wait until water boils             |
|        | 558    |                        | Fill an electric kettle with water |
|        | 15     |                        | Add coffee grounds                 |
|        | 18     |                        | Add dried herbs                    |
| 3      | 53     | Brew Tea               | Add tea leaves                     |
|        | 496    |                        | Add tea bags to hot water          |
|        | 562    |                        | Pour hot water to a cup or mug     |
|        | 497    |                        | Steep the tea until ready          |
| 4      | 502    | Steep & Simmer         | Simmer the tea over low heat       |
|        | 498    |                        | Remove the tea bag                 |
| 5      | 499    | Remove Tea             | Drain the tea                      |
|        | 34     |                        | Add milk                           |
| 6      | 500    | Add Milk               | Pour milk into the tea             |
|        | 12     |                        | Add chocolate                      |
|        | 28     |                        | Add honey                          |
|        | 37     |                        | Add nutmeg                         |
| 7      | 51     | Add Sweetener & Spices | Add sugar                          |
|        | 109    |                        | Cut nutmeg                         |
|        | 233    |                        | Add cinnamon stick                 |
|        | 234    |                        | Stir the tea                       |
| 8      | 501    | Stir Drink             | Stir milk with a spoon             |
|        | 230    |                        | Warm up a cup                      |
| 9      | 232    | Pour to Serve          | Pour the tea into the cup          |
|        | 0      |                        | Wash hands                         |
|        | 1      |                        | Wipe hands                         |
|        | 77     |                        | Wash tea bag                       |
|        | 236    |                        | Wash chopsticks                    |
|        | 237    |                        | Wash cup or mug                    |
|        | 238    |                        | Wash cutting board                 |
|        | 239    |                        | Wash electric kettle               |
|        | 241    |                        | Wash kitchen tong                  |
|        | 242    |                        | Wash knife                         |
|        | 248    |                        | Wash pot or saucepan               |
|        | 249    |                        | Wash sieve                         |
|        | 250    |                        | Wash skillet/frying pan/wok        |
|        | 252    |                        | Wash spoon                         |
|        | 255    |                        | Put away chopsticks                |
|        | 257    |                        | Put away cup or mug                |
|        | 260    |                        | Put away kettle                    |
| 10     | 261    | Cleanup                | Put away kitchen tong              |
|        | 263    |                        | Put away knife                     |
|        | 269    |                        | Put away pot or saucepan           |
|        | 274    |                        | Put away spoon                     |
|        | 290    |                        | Put away coffee grounds            |
|        | 293    |                        | Put away dried herbs               |
|        | 305    |                        | Put away milk                      |
|        | 306    |                        | Put away noodles                   |
|        | 307    |                        | Put away nutmeg                    |

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| New ID | Old ID | Step Name | Old Step Name            |
|--------|--------|-----------|--------------------------|
|        | 310    |           | Put away onions          |
|        | 316    |           | Put away scallion        |
|        | 322    |           | Put away sugar           |
|        | 324    |           | Put away tea bag         |
|        | 325    |           | Put away tea leaves      |
|        | 331    |           | Put away water           |
|        | 332    |           | Wipe kitchen countertop  |
|        | 333    |           | Throw out trash or waste |
| 11     | 615    | Mistake   | Mistake                  |

Table 16: Label mapping for “Making Sesame-Ginger Asian Salad”

| New ID | Old ID | Step Name  | Old Step Name                             |
|--------|--------|------------|-------------------------------------------|
| 0      | 122    | Background | Measure the temperature                   |
|        | 131    |            | Check phone                               |
|        | 132    |            | Check paper recipe                        |
|        | 496    |            | Place the skillet/pan/pot on the stove    |
|        | 497    |            | Remove the skillet/pan/pot from the stove |
|        | 615    |            | Mistake                                   |
|        | 616    |            | Background                                |
| 1      | 135    | Get Items  | Get almonds                               |
|        | 137    |            | Get bell peppers                          |
|        | 142    |            | Get carrots                               |
|        | 143    |            | Get celeries                              |
|        | 145    |            | Get cherry tomatoes                       |
|        | 146    |            | Get chili powder                          |
|        | 148    |            | Get cilantro                              |
|        | 149    |            | Get cinnamon powder                       |
|        | 152    |            | Get cucumber                              |
|        | 157    |            | Get feta cheese                           |
|        | 160    |            | Get garlic cloves                         |
|        | 161    |            | Get garlic paste                          |
|        | 162    |            | Get garlic powder                         |
|        | 165    |            | Get green chilies                         |
|        | 166    |            | Get honey                                 |
|        | 169    |            | Get lemon                                 |
|        | 170    |            | Get lemon juice                           |
|        | 171    |            | Get lettuce                               |
|        | 173    |            | Get mayonnaise                            |
|        | 175    |            | Get mustard                               |
|        | 178    |            | Get oil                                   |
|        | 179    |            | Get olive oil                             |
|        | 181    |            | Get onions                                |
|        | 182    |            | Get oyster sauce                          |
|        | 183    |            | Get parsley                               |
|        | 184    |            | Get peanut butter                         |
|        | 185    |            | Get pepper                                |
|        | 188    |            | Get rice vinegar                          |
|        | 189    |            | Get salt                                  |
|        | 190    |            | Get scallion                              |
|        | 191    |            | Get sesame oil                            |
|        | 192    |            | Get sesame seeds                          |
|        | 193    |            | Get soy sauce                             |
|        | 194    |            | Get spinach                               |
|        | 195    |            | Get spring onions                         |
|        | 197    |            | Get tamari                                |
|        | 200    |            | Get toasted sesame oil                    |
|        | 201    |            | Get tomato                                |
|        | 206    |            | Get white pepper                          |
|        | 207    |            | Get chopping board                        |
|        | 208    |            | Get chopsticks                            |

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| New ID                    | Old ID | Step Name             | Old Step Name                  |
|---------------------------|--------|-----------------------|--------------------------------|
|                           | 210    |                       | Get cutting board              |
|                           | 211    |                       | Get fork                       |
|                           | 213    |                       | Get kitchen tong               |
|                           | 214    |                       | Get kitchen towel              |
|                           | 215    |                       | Get knife                      |
|                           | 216    |                       | Get ladle                      |
|                           | 217    |                       | Get measuring tool             |
|                           | 218    |                       | Get napkin                     |
|                           | 219    |                       | Get peeler                     |
|                           | 220    |                       | Get pitcher                    |
|                           | 221    |                       | Get plate                      |
|                           | 226    |                       | Get spoon                      |
|                           | 229    |                       | Get a cup or mug               |
|                           | 527    |                       | Get ginger                     |
|                           | 548    |                       | Get a bowl                     |
| 2                         | 60     | Wash Ingredients      | Wash bell peppers              |
|                           | 61     |                       | Wash cabbage                   |
|                           | 62     |                       | Wash carrots                   |
|                           | 63     |                       | Wash celeries                  |
|                           | 64     |                       | Wash cherry tomatoes           |
|                           | 65     |                       | Wash cilantro                  |
|                           | 68     |                       | Wash fresh parsley             |
|                           | 69     |                       | Wash garlic cloves             |
|                           | 71     |                       | Wash lettuce                   |
|                           | 76     |                       | Wash spring onions             |
|                           | 528    |                       | Wash ginger                    |
| 3                         | 79     | Peel Ingredients      | Peel carrots                   |
|                           | 80     |                       | Peel coriander leaves          |
|                           | 82     |                       | Peel fresh cilantro            |
|                           | 83     |                       | Peel garlic cloves             |
|                           | 84     |                       | Peel fresh ginger              |
|                           | 86     |                       | Peel onions                    |
| 4                         | 503    | Grate Ingredients     | Grate carrots                  |
|                           | 504    |                       | Grate ginger                   |
|                           | 603    |                       | Grate garlic                   |
| 5                         | 90     | Cut Bell Peppers      | Cut bell peppers               |
|                           | 116    |                       | Remove seeds from bell peppers |
| 6                         | 88     | Cut Other Veg         | Cut almonds                    |
|                           | 93     |                       | Cut cabbage                    |
|                           | 94     |                       | Cut carrots                    |
|                           | 95     |                       | Cut celeries                   |
|                           | 98     |                       | Cut chili powder               |
|                           | 100    |                       | Cut cilantro                   |
|                           | 101    |                       | Cut cucumber                   |
|                           | 102    |                       | Cut garlic cloves              |
|                           | 104    |                       | Cut lettuce                    |
|                           | 110    |                       | Cut onions                     |
|                           | 111    |                       | Cut scallion                   |
|                           | 112    |                       | Cut spinach                    |
|                           | 113    |                       | Cut spring onions              |
|                           | 117    |                       | Remove seeds from celeries     |
|                           | 529    |                       | Cut ginger                     |
| 7                         | 4      | Add Solid Ingredients | Add bell peppers               |
|                           | 7      |                       | Add carrots                    |
|                           | 8      |                       | Add celeries                   |
|                           | 10     |                       | Add cherry tomatoes            |
|                           | 13     |                       | Add cilantro                   |
|                           | 16     |                       | Add cucumber                   |
|                           | 20     |                       | Add feta cheese                |
|                           | 22     |                       | Add garlic cloves              |
|                           | 25     |                       | Add fresh ginger               |
|                           | 27     |                       | Add green chilies              |
| Continued on next page... |        |                       |                                |

| New ID | Old ID | Step Name            | Old Step Name                            |
|--------|--------|----------------------|------------------------------------------|
|        | 32     |                      | Add lettuce                              |
|        | 39     |                      | Add onions                               |
|        | 46     |                      | Add scallion                             |
|        | 473    |                      | Add spinach to a mixing bowl             |
|        | 508    |                      | Add prepared mix of vegetables to a bowl |
|        | 2      |                      | Add almonds                              |
|        | 5      |                      | Add black pepper                         |
|        | 11     |                      | Add chili powder                         |
|        | 14     |                      | Add cinnamon powder                      |
|        | 23     |                      | Add garlic paste                         |
|        | 24     |                      | Add garlic powder                        |
|        | 26     |                      | Add ginger paste                         |
|        | 28     |                      | Add honey                                |
|        | 29     |                      | Add hot sauce                            |
|        | 31     |                      | Add lemon juice                          |
|        | 33     |                      | Add mayonnaise                           |
|        | 35     |                      | Add mustard                              |
|        | 38     |                      | Add oil                                  |
| 8      | 41     | Make Dressing        | Add peanut butter                        |
|        | 42     |                      | Add pepper                               |
|        | 45     |                      | Add salt                                 |
|        | 47     |                      | Add sesame oil                           |
|        | 48     |                      | Add sesame seeds                         |
|        | 49     |                      | Add soy sauce                            |
|        | 52     |                      | Add tamari                               |
|        | 56     |                      | Add turmeric powder                      |
|        | 57     |                      | Add vinegar                              |
|        | 59     |                      | Add white pepper                         |
|        | 468    |                      | Add cardamom to a mixing bowl            |
|        | 469    |                      | Add ginger garlic paste to a mixing bowl |
|        | 470    |                      | Add lemon to a mixing bowl               |
|        | 539    |                      | Add olive oil to a bowl                  |
|        | 540    |                      | Add oyster sauce to a bowl               |
|        | 506    |                      | Whisk the mix of ingredients             |
| 9      | 507    | Whisk/Shake Dressing | Shake the mix of ingredients             |
|        | 509    |                      | Whisk the dressing before pouring        |
| 10     | 510    | Pour Dressing        | Pour the dressing                        |
|        | 511    |                      | Toss the mixture                         |
| 11     | 512    | Toss Salad           | Stir the mixture                         |
|        | 121    |                      | Taste the recipe                         |
| 12     | 228    | Serve Salad          | Transfer food to a plate                 |
|        | 0      |                      | Wash hands                               |
|        | 1      |                      | Wipe hands                               |
|        | 235    |                      | Wash bowl                                |
|        | 238    |                      | Wash cutting board                       |
|        | 240    |                      | Wash fork                                |
|        | 242    |                      | Wash knife                               |
|        | 244    |                      | Wash measuring tool                      |
|        | 246    |                      | Wash peeler                              |
|        | 247    |                      | Wash plate                               |
|        | 252    |                      | Wash spoon                               |
|        | 253    |                      | Put away bowl                            |
|        | 254    |                      | Put away chopping board                  |
|        | 255    |                      | Put away chopsticks                      |
|        | 263    |                      | Put away knife                           |
|        | 264    |                      | Put away measuring tool                  |
|        | 265    |                      | Put away napkin                          |
|        | 267    |                      | Put away plate                           |
|        | 274    |                      | Put away spoon                           |
|        | 278    |                      | Put away bell peppers                    |
|        | 280    |                      | Put away butter                          |
| 13     | 282    | Cleanup              | Put away carrots                         |

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| New ID | Old ID | Step Name | Old Step Name            |
|--------|--------|-----------|--------------------------|
|        | 283    |           | Put away celeries        |
|        | 285    |           | Put away cherry tomatoes |
|        | 288    |           | Put away cinnamon powder |
|        | 296    |           | Put away garlic cloves   |
|        | 298    |           | Put away fresh ginger    |
|        | 299    |           | Put away green chilies   |
|        | 300    |           | Put away honey           |
|        | 303    |           | Put away lemon juice     |
|        | 308    |           | Put away oil             |
|        | 310    |           | Put away onions          |
|        | 313    |           | Put away pepper          |
|        | 314    |           | Put away rice vinegar    |
|        | 315    |           | Put away salt            |
|        | 317    |           | Put away sesame oil      |
|        | 318    |           | Put away sesame seeds    |
|        | 319    |           | Put away soy sauce       |
|        | 325    |           | Put away tea leaves      |
|        | 327    |           | Put away tomato          |
|        | 332    |           | Wipe kitchen countertop  |
|        | 333    |           | Throw out trash or waste |

Table 17: Label mapping for Cooking Tomato & Eggs

| New ID | Old ID | Step Name       | Old Step Name                    |
|--------|--------|-----------------|----------------------------------|
| 0      | 132    | Background      | Check paper recipe               |
|        | 489    |                 | Visually inspect the recipe      |
|        | 615    |                 | Mistake                          |
|        | 616    |                 | Background                       |
| 1      | 207    | Get Kitchenware | Get chopping board               |
|        | 208    |                 | Get chopsticks                   |
|        | 210    |                 | Get cutting board                |
|        | 211    |                 | Get fork                         |
|        | 214    |                 | Get kitchen towel                |
|        | 215    |                 | Get knife                        |
|        | 217    |                 | Get measuring tool               |
|        | 218    |                 | Get napkin                       |
|        | 221    |                 | Get plate                        |
|        | 224    |                 | Get skillet or frying pan or wok |
|        | 225    |                 | Get spatula                      |
|        | 226    |                 | Get spoon                        |
|        | 227    |                 | Get whisk                        |
|        | 229    |                 | Get a cup or mug                 |
| 2      | 493    | Get Ingredients | Get a pot or saucepan            |
|        | 548    |                 | Get a bowl                       |
|        | 138    |                 | Get black pepper                 |
|        | 139    |                 | Get butter                       |
|        | 143    |                 | Get celeries                     |
|        | 145    |                 | Get cherry tomatoes              |
|        | 146    |                 | Get chili powder                 |
|        | 156    |                 | Get eggs                         |
|        | 158    |                 | Get flour                        |
|        | 160    |                 | Get garlic cloves                |
|        | 165    |                 | Get green chilies                |
|        | 178    |                 | Get oil                          |
|        | 179    |                 | Get olive oil                    |
|        | 183    |                 | Get parsley                      |
|        | 185    |                 | Get pepper                       |
|        | 189    |                 | Get salt                         |
|        | 190    |                 | Get scallion                     |
|        | 195    |                 | Get spring onions                |
|        | 196    |                 | Get sugar                        |

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| New ID | Old ID | Step Name           | Old Step Name                               |
|--------|--------|---------------------|---------------------------------------------|
|        | 201    |                     | Get tomato                                  |
|        | 203    |                     | Get turmeric powder                         |
|        | 205    |                     | Get water                                   |
|        | 206    |                     | Get white pepper                            |
|        | 527    |                     | Get ginger                                  |
| 3      | 63     | Wash Ingredients    | Wash celeries                               |
|        | 75     |                     | Wash scallion                               |
|        | 76     |                     | Wash spring onions                          |
|        | 78     |                     | Wash tomato                                 |
| 4      | 83     | Peel Ingredients    | Peel garlic cloves                          |
|        | 120    |                     | Remove seeds from tomato                    |
|        | 513    |                     | Cut a cross on the tomato                   |
| 5      | 92     | Cut Aromatics       | Cut butter                                  |
|        | 95     |                     | Cut celeries                                |
|        | 102    |                     | Cut garlic cloves                           |
|        | 103    |                     | Cut green chilies                           |
|        | 111    |                     | Cut scallion                                |
|        | 113    |                     | Cut spring onions                           |
|        | 529    |                     | Cut ginger                                  |
| 6      | 97     | Cut Tomato          | Cut cherry tomatoes                         |
|        | 115    |                     | Cut tomato                                  |
| 7      | 479    | Crack Eggs          | Crack eggs into a mixing bowl               |
| 8      | 480    | Whisk Eggs          | Whisk until egg whites/yolks are integrated |
| 9      | 38     | Heat Skillet        | Add oil                                     |
|        | 124    |                     | Turn on the stove                           |
|        | 125    |                     | Adjust the stove heat                       |
|        | 126    |                     | Turn off the stove                          |
|        | 127    |                     | Lift the lid of a skillet/pan/pot           |
|        | 128    |                     | Close the lid of a skillet/pan/pot          |
|        | 129    |                     | Place the skillet/pan/pot on the stove      |
|        | 477    |                     | Check heat of the skillet                   |
|        | 478    |                     | Add butter to skillet                       |
|        | 486    |                     | Tilt and rotate the skillet                 |
| 10     | 482    | Scramble Eggs       | Gently push cooked portions toward center   |
|        | 483    |                     | Tilt/rotate skillet for uncooked egg        |
|        | 485    |                     | Cook until the egg is cooked                |
|        | 487    |                     | Pour egg mixture into a skillet             |
|        | 514    |                     | Break scrambled egg into small pieces       |
| 11     | 515    | Transfer Eggs (Out) | Transfer scrambled egg                      |
| 12     | 8      | Add & Mush Tomatoes | Add celeries                                |
|        | 22     |                     | Add garlic cloves                           |
|        | 25     |                     | Add fresh ginger                            |
|        | 27     |                     | Add green chilies                           |
|        | 39     |                     | Add onions                                  |
|        | 40     |                     | Add parsley                                 |
|        | 46     |                     | Add scallion                                |
|        | 50     |                     | Add spring onions                           |
|        | 54     |                     | Add tomato                                  |
|        | 58     |                     | Add water                                   |
|        | 516    |                     | Add cloves to a skillet                     |
|        | 518    |                     | Add olive oil to a skillet                  |
|        | 519    |                     | Leave to cook until tomato becomes mushy    |
|        | 521    |                     | Add cooked tomatoes into the skillet        |
| 13     | 512    | Stir-Fry Mixture    | Stir the mixture                            |
| 14     | 19     | Combine Eggs        | Add eggs                                    |
|        | 520    |                     | Add the scrambled egg into the skillet      |
| 15     | 5      | Season Food         | Add black pepper                            |
|        | 11     |                     | Add chili powder                            |
|        | 42     |                     | Add pepper                                  |
|        | 45     |                     | Add salt                                    |
|        | 51     |                     | Add sugar                                   |
|        | 56     |                     | Add turmeric powder                         |

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| New ID | Old ID | Step Name  | Old Step Name                         |
|--------|--------|------------|---------------------------------------|
|        | 59     |            | Add white pepper                      |
|        | 121    |            | Taste the recipe                      |
|        | 522    |            | Add flour to the skillet              |
| 16     | 228    | Serve Food | Transfer food to a plate              |
|        | 0      |            | Wash hands                            |
|        | 1      |            | Wipe hands                            |
|        | 235    |            | Wash bowl                             |
|        | 236    |            | Wash chopsticks                       |
|        | 237    |            | Wash cup or mug                       |
|        | 238    |            | Wash cutting board                    |
|        | 240    |            | Wash fork                             |
|        | 242    |            | Wash knife                            |
|        | 243    |            | Wash ladle                            |
|        | 247    |            | Wash plate                            |
|        | 250    |            | Wash skillet or frying pan or wok     |
|        | 251    |            | Wash spatula                          |
|        | 252    |            | Wash spoon                            |
|        | 253    |            | Put away bowl                         |
|        | 254    |            | Put away chopping board               |
|        | 257    |            | Put away cup or mug                   |
|        | 259    |            | Put away fork                         |
| 17     | 262    | Cleanup    | Put away kitchen towel                |
|        | 263    |            | Put away knife                        |
|        | 264    |            | Put away measuring tool               |
|        | 265    |            | Put away napkin                       |
|        | 272    |            | Put away skillet or frying pan or wok |
|        | 273    |            | Put away spatula                      |
|        | 274    |            | Put away spoon                        |
|        | 294    |            | Put away eggs                         |
|        | 296    |            | Put away garlic cloves                |
|        | 308    |            | Put away oil                          |
|        | 312    |            | Put away parsley                      |
|        | 315    |            | Put away salt                         |
|        | 317    |            | Put away sesame oil                   |
|        | 322    |            | Put away sugar                        |
|        | 327    |            | Put away tomato                       |
|        | 331    |            | Put away water                        |
|        | 332    |            | Wipe kitchen countertop               |
|        | 333    |            | Throw out trash or waste              |

Table 18: Label mapping for “Making Chai Tea”

| New ID | Old ID | Step Name  | Old Step Name         |
|--------|--------|------------|-----------------------|
|        | 132    |            | Check paper recipe    |
|        | 133    |            | Set a timer           |
| 0      | 357    | Background | Get a timer           |
|        | 525    |            | Check the heat        |
|        | 616    |            | Background            |
|        | 141    |            | Get cardamom          |
|        | 154    |            | Get dried herbs       |
|        | 159    |            | Get fresh thyme       |
|        | 174    |            | Get milk              |
|        | 196    |            | Get sugar             |
|        | 198    |            | Get tea bag           |
|        | 199    |            | Get tea leaves        |
| 1      | 212    | Get Items  | Get kitchen shears    |
|        | 217    |            | Get measuring tool    |
|        | 223    |            | Get sieve             |
|        | 226    |            | Get spoon             |
|        | 229    |            | Get a cup or mug      |
|        | 231    |            | Get a fine-mesh sieve |

Continued on next page...

| New ID | Old ID | Step Name        | Old Step Name                             |
|--------|--------|------------------|-------------------------------------------|
|        | 493    |                  | Get a pot or saucepan                     |
|        | 527    |                  | Get ginger                                |
|        | 531    |                  | Get cinnamon stick                        |
| 2      | 124    | Stove Operations | Turn on the stove                         |
|        | 125    |                  | Adjust the stove heat                     |
|        | 126    |                  | Turn off the stove                        |
|        | 129    |                  | Place the skillet/pan/pot on the stove    |
|        | 130    |                  | Remove the skillet/pan/pot from the stove |
| 3      | 58     | Add Water & Boil | Add water                                 |
|        | 494    |                  | Wait until water boils                    |
|        | 524    |                  | Fill a cup or mug with water              |
| 4      | 18     | Add Spices       | Add dried herbs                           |
|        | 21     |                  | Add fresh thyme                           |
|        | 25     |                  | Add fresh ginger                          |
|        | 48     |                  | Add sesame seeds                          |
|        | 233    |                  | Add cinnamon stick                        |
|        | 530    |                  | Add cardamom to tea                       |
| 5      | 53     | Add Tea Leaves   | Add tea leaves                            |
|        | 496    |                  | Add tea bags to hot water                 |
| 6      | 502    | Simmering        | Simmer the tea over low heat              |
| 7      | 34     | Add Milk         | Add milk                                  |
|        | 106    |                  | Cut milk                                  |
|        | 500    |                  | Pour milk into the tea                    |
| 8      | 51     | Add Sugar        | Add sugar                                 |
| 9      | 234    | Stir Mixture     | Stir the tea                              |
|        | 501    |                  | Stir milk with a spoon                    |
| 10     | 232    | Strain & Serve   | Pour the tea into the cup                 |
| 11     | 237    | Wash Items       | Wash cup or mug                           |
|        | 244    |                  | Wash measuring tool                       |
|        | 248    |                  | Wash pot or saucepan                      |
|        | 249    |                  | Wash sieve                                |
|        | 252    |                  | Wash spoon                                |
| 12     | 1      | Put Away Items   | Wipe hands                                |
|        | 264    |                  | Put away measuring tool                   |
|        | 269    |                  | Put away pot or saucepan                  |
|        | 271    |                  | Put away sieve                            |
|        | 274    |                  | Put away spoon                            |
|        | 275    |                  | Put away timer                            |
|        | 281    |                  | Put away cardamom                         |
|        | 289    |                  | Put away cinnamon sticks                  |
|        | 293    |                  | Put away dried herbs                      |
|        | 305    |                  | Put away milk                             |
|        | 318    |                  | Put away sesame seeds                     |
|        | 322    |                  | Put away sugar                            |
|        | 325    |                  | Put away tea leaves                       |
|        | 332    |                  | Wipe kitchen countertop                   |
|        | 333    |                  | Throw out trash or waste                  |

Table 19: Label mapping for CaptainCook4D [35]

| New ID                    | Old ID | Step Name       | Old Step Name               |
|---------------------------|--------|-----------------|-----------------------------|
| 0                         | 0      | Background      | Background                  |
| 1                         | 16     | Cut Ingredients | Continue slicing with floss |
|                           | 20     |                 | Slice one tomato            |
|                           | 39     |                 | Peel 4 large garlic cloves  |
|                           | 41     |                 | Chop tomato roughly         |
|                           | 42     |                 | Mince the garlic            |
|                           | 58     |                 | Crack one egg in the bowl   |
|                           | 61     |                 | Chop 1/4 tomato             |
|                           | 62     |                 | Chop 1 tsp cilantro         |
|                           | 63     |                 | Chop 1/4 medium onion       |
| Continued on next page... |        |                 |                             |

| New ID                    | Old ID | Step Name       | Old Step Name                      |
|---------------------------|--------|-----------------|------------------------------------|
|                           | 64     |                 | Peel 2 garlic cloves               |
|                           | 65     |                 | Chop 1 green chilli                |
|                           | 66     |                 | Mince peeled garlic cloves         |
|                           | 87     |                 | Peel the cucumber                  |
|                           | 89     |                 | Chop or grate the cucumber         |
|                           | 92     |                 | Peel 1 garlic clove                |
|                           | 93     |                 | Spiralize 1 medium zucchini        |
|                           | 105    |                 | Cut English muffin into two pieces |
|                           | 117    |                 | Chop 1 shallot                     |
|                           | 120    |                 | Peel 2 garlic cloves               |
|                           | 122    |                 | Mince garlic cloves                |
|                           | 123    |                 | Slice the mushrooms                |
|                           | 139    |                 | Chop 1 strawberry                  |
|                           | 149    |                 | Cut tomato into two pieces         |
|                           | 150    |                 | Chop 2 tbsp cilantro               |
|                           | 152    |                 | Crack one egg in a bowl            |
|                           | 169    |                 | Peel 2 cloves of garlic            |
|                           | 171    |                 | Mince garlic                       |
|                           | 174    |                 | Slice mushrooms                    |
|                           | 178    |                 | Slice 1/3 of the bell pepper       |
|                           | 188    |                 | Cut tofu into large cubes          |
|                           | 225    |                 | Use scissors to cut corner         |
|                           | 227    |                 | Chop 1/4 red bell pepper           |
|                           | 245    |                 | Chop 1 scallion                    |
|                           | 247    |                 | Cut avocado into thin slices       |
|                           | 255    |                 | Slice two 1/2 inch rounds          |
|                           | 259    |                 | Cut 1/4 cup cherry tomatoes        |
|                           | 266    |                 | Peel one medium onion              |
|                           | 268    |                 | Cut onion into two pieces          |
|                           | 269    |                 | Peel 1 garlic clove                |
|                           | 270    |                 | Slice 1/8 medium onion             |
|                           | 271    |                 | Cut 1/8 garlic clove               |
|                           | 273    |                 | Mince 1/8 garlic clove             |
|                           | 295    |                 | Peel 1 garlic clove                |
|                           | 296    |                 | Peel 1 medium onion                |
|                           | 297    |                 | Chop 1 garlic clove                |
|                           | 298    |                 | Slice 1/4 medium onion             |
|                           | 311    |                 | Prepare the filter insert          |
|                           | 313    |                 | Grind the coffee beans             |
| 2                         | 29     | Add Ingredients | Add corn into bowl                 |
|                           | 31     |                 | Add 1 tsp pepper powder            |
|                           | 32     |                 | Add 1 tsp softened butter          |
|                           | 36     |                 | Add 1 tsp salt to the bowl         |
|                           | 37     |                 | Add lime juice to the bowl         |
|                           | 45     |                 | Add 1/2 tsp cumin seeds            |
|                           | 46     |                 | Add 1/4 tsp mustard to the pan     |
|                           | 50     |                 | Add 2 tbsp red chili powder        |
|                           | 52     |                 | Add tomato puree to the pan        |
|                           | 53     |                 | Add 1/2 tsp salt to the pan        |
|                           | 59     |                 | Add 1 tbsp milk to the bowl        |
|                           | 60     |                 | Add 1/3 tsp salt to the bowl       |
|                           | 69     |                 | Add 1/3 tsp salt to the pan        |
|                           | 70     |                 | Add chopped onions to the pan      |
|                           | 72     |                 | Add chilli to the pan              |
|                           | 73     |                 | Add garlic to the pan              |
|                           | 75     |                 | Add 1/8 tsp of turmeric            |
|                           | 76     |                 | Add tomatoes to the pan            |
|                           | 78     |                 | Pour whisked eggs into pan         |
|                           | 82     |                 | Add 1 tsp cumin powder             |
|                           | 83     |                 | Add 1 tbsp chopped cilantro        |
|                           | 85     |                 | Add 1/2 tsp chaat masala           |
|                           | 86     |                 | Add 1/4 tsp salt to the bowl       |
|                           | 88     |                 | Add 1/4 tsp red chilli powder      |
| Continued on next page... |        |                 |                                    |

| New ID | Old ID | Step Name | Old Step Name                   |
|--------|--------|-----------|---------------------------------|
|        | 90     |           | Add cucumber to whisked curd    |
|        | 96     |           | Add 1 minced garlic clove       |
|        | 98     |           | Add the zucchini noodles        |
|        | 99     |           | Add 1/6 cup parmesan cheese     |
|        | 107    |           | Pour 1 egg into ramekin cup     |
|        | 127    |           | Add chopped shallot to pan      |
|        | 129    |           | Add 1/4 tbsp balsamic vinegar   |
|        | 130    |           | Add 2 cloves minced garlic      |
|        | 134    |           | Add 1/2 tsp baking powder       |
|        | 135    |           | Add 1 egg to blender            |
|        | 136    |           | Add 1 banana to blender         |
|        | 137    |           | Add 1 heaped tbsp flour         |
|        | 141    |           | Pour three puddles into pan     |
|        | 147    |           | Splash maple syrup on plate     |
|        | 151    |           | Add chopped cilantro to bowl    |
|        | 153    |           | Add 1/2 tsp black pepper        |
|        | 159    |           | Pour egg mixture into pan       |
|        | 164    |           | Add 1/8 tsp black pepper        |
|        | 165    |           | Add 1/8 cup soy sauce           |
|        | 167    |           | Add 1/6 cup water to bowl       |
|        | 168    |           | Add 1 tsp cornstarch to bowl    |
|        | 170    |           | Add 1/2 tbsp minced ginger      |
|        | 173    |           | Add 2 cloves minced garlic      |
|        | 175    |           | Add 1 tbsp honey to bowl        |
|        | 180    |           | Add sliced mushrooms to skillet |
|        | 181    |           | Add broccoli to skillet         |
|        | 183    |           | Add bell pepper to skillet      |
|        | 186    |           | Pour the sauce into skillet     |
|        | 190    |           | Add 1 tbsp olive oil to pan     |
|        | 191    |           | Add tofu cubes to the pan       |
|        | 192    |           | Add 1/4 tsp salt to the pan     |
|        | 207    |           | Measure and add 2 tbsp flour    |
|        | 208    |           | Add pinch of salt to bowl       |
|        | 210    |           | Add 1/4 tsp baking powder       |
|        | 211    |           | Add 1.5 tbsp sugar to bowl      |
|        | 214    |           | Add 2 tsp vegetable oil         |
|        | 215    |           | Add 2 tbsp water to bowl        |
|        | 216    |           | Add 1/4 tsp vanilla extract     |
|        | 218    |           | Pour batter into mug            |
|        | 228    |           | Add 1/3 cup cheddar cheese      |
|        | 230    |           | Add 1 tbsp water to bowl        |
|        | 234    |           | Add 1/4 tsp pepper to bowl      |
|        | 235    |           | Add 1/4 tsp salt to bowl        |
|        | 236    |           | Add 1/2 tbsp butter to bowl     |
|        | 238    |           | Add 1 tsp Sriracha sauce        |
|        | 242    |           | Add 1/4 cup mayonnaise to bowl  |
|        | 243    |           | Add 1 can drained tuna          |
|        | 246    |           | Add chopped scallion to bowl    |
|        | 256    |           | Add 1/8 cup mozzarella          |
|        | 257    |           | Add 1/16 cup basil to bowl      |
|        | 258    |           | Add 1/4 tsp salt to bowl        |
|        | 262    |           | Add 1/4 tsp pepper to bowl      |
|        | 276    |           | Pour sauce over meatballs       |
|        | 281    |           | Add 1/16 tsp baking soda        |
|        | 282    |           | Add 4 tbsp flour to mug         |
|        | 283    |           | Add 1/8 tsp salt to mug         |
|        | 284    |           | Add 1/8 tsp baking powder       |
|        | 286    |           | Add 3 tbsp milk to mug          |
|        | 287    |           | Add 1 tbsp olive oil to mug     |
|        | 300    |           | Add noodles to the bowl         |
|        | 304    |           | Add basil to the bowl           |
|        | 306    |           | Add chopped cilantro to bowl    |
|        | 320    |           | Pour water into filter          |

Continued on next page...

| New ID                    | Old ID | Step Name       | Old Step Name                   |
|---------------------------|--------|-----------------|---------------------------------|
|                           | 322    |                 | Pour rest of water over grounds |
|                           | 1      |                 | Place tortilla on board         |
|                           | 2      |                 | Scoop nut butter from jar       |
|                           | 5      |                 | Scoop jelly from the jar        |
|                           | 11     |                 | Discard ends of tortilla        |
|                           | 13     |                 | Place floss between toothpicks  |
|                           | 17     |                 | Place pinwheels on a plate      |
|                           | 40     |                 | Take 1 tomato                   |
|                           | 56     |                 | Take pan off the heat           |
|                           | 57     |                 | Transfer to serving bowl        |
|                           | 103    |                 | Remove from heat                |
|                           | 115    |                 | Place egg over the lettuce      |
|                           | 133    |                 | Transfer to serving dish        |
|                           | 145    |                 | Transfer to a plate             |
|                           | 146    |                 | Serve with strawberries         |
|                           | 148    |                 | Take a tomato                   |
|                           | 157    |                 | Scoop tomatoes from pan         |
|                           | 158    |                 | Put tomatoes on serving plate   |
|                           | 162    |                 | Transfer omelette to plate      |
|                           | 163    |                 | Take 5 broccoli florets         |
|                           | 166    |                 | Take 2 cremini mushrooms        |
| 3                         | 172    | Transfer Items  | Take 1 bell pepper              |
|                           | 195    |                 | Remove pan from heat            |
|                           | 199    |                 | Briefly remove from heat        |
|                           | 206    |                 | Transfer to a serving dish      |
|                           | 209    |                 | Place liner inside mug          |
|                           | 222    |                 | Carefully remove liner          |
|                           | 229    |                 | Place pepper in bowl            |
|                           | 240    |                 | Open a can of tuna              |
|                           | 244    |                 | Take 1 ripe avocado             |
|                           | 248    |                 | Place avocado slices on leaf    |
|                           | 265    |                 | Spoon mixture onto bread        |
|                           | 274    |                 | Place 5 meatballs on plate      |
|                           | 280    |                 | Take a microwavable mug         |
|                           | 289    |                 | Take 1 tbsp marinara sauce      |
|                           | 294    |                 | Remove noodles from package     |
|                           | 299    |                 | Put vegetables in bowl          |
|                           | 310    |                 | Place dripper on mug            |
|                           | 312    |                 | Place paper filter in dripper   |
|                           | 316    |                 | Transfer grounds to filter      |
|                           | 317    |                 | Transfer water to a kettle      |
|                           | 324    |                 | Discard filter and grounds      |
|                           | 33     |                 | Stir the bowl                   |
|                           | 38     |                 | Mix contents of bowl well       |
|                           | 43     |                 | Puree tomatoes in blender       |
|                           | 51     |                 | Mix contents of pan well        |
|                           | 54     |                 | Mix tomato puree with pan       |
|                           | 67     |                 | Whisk egg mixture in bowl       |
|                           | 79     |                 | Mix with spatula for 3 mins     |
|                           | 91     |                 | Combine ingredients in bowl     |
|                           | 109    |                 | Stir the ramekin cup            |
|                           | 138    |                 | Blitz blender for 20 seconds    |
|                           | 154    |                 | Beat the contents of bowl       |
| 4                         | 160    | Mix Ingredients | Stir gently with wooden spoon   |
|                           | 176    |                 | Whisk contents of bowl          |
|                           | 185    |                 | Whisk sauce again               |
|                           | 217    |                 | Whisk batter until no lumps     |
|                           | 233    |                 | Mix cheese and bell pepper      |
|                           | 237    |                 | Mix ingredients of bowl well    |
|                           | 249    |                 | Mix contents of bowl            |
|                           | 264    |                 | Combine contents of bowl        |
|                           | 267    |                 | Mix sauce in small bowl         |
|                           | 278    |                 | Stir contents in microwave      |
| Continued on next page... |        |                 |                                 |

| New ID | Old ID | Step Name           | Old Step Name                |
|--------|--------|---------------------|------------------------------|
|        | 285    |                     | Stir contents in mug well    |
|        | 307    |                     | Mix in flavour packet        |
|        | 308    |                     | Stir noodles until dissolved |
|        | 30     |                     | Microwave corn for 2 mins    |
|        | 34     |                     | Microwave corn for 3 mins    |
|        | 44     |                     | Heat 3 tbsp oil in pan       |
|        | 48     |                     | Lower the heat               |
|        | 49     |                     | Saute garlic for 2-3 mins    |
|        | 55     |                     | Simmer over low heat         |
|        | 68     |                     | Heat 2 tbsp oil in pan       |
|        | 71     |                     | Saute onions until soft      |
|        | 77     |                     | Cook covered for 1 minute    |
|        | 94     |                     | Heat large pan on medium     |
|        | 95     |                     | Melt 1 tbsp softened butter  |
|        | 97     |                     | Cook garlic until fragrant   |
|        | 102    |                     | Cook for 2 mins for zoodles  |
|        | 108    |                     | Microwave ramekin uncovered  |
|        | 110    |                     | Microwave until egg is set   |
|        | 124    |                     | Heat 1 tbsp oil in skillet   |
| 5      | 140    | Cook Food           | Melt butter in frying pan    |
|        | 142    |                     | Cook until tops bubble       |
|        | 144    |                     | Cook for 20-30 seconds more  |
|        | 155    |                     | Heat 1 tbsp oil in pan       |
|        | 156    |                     | Cook tomatoes cut-side down  |
|        | 179    |                     | Heat 2 tbsp oil in skillet   |
|        | 193    |                     | Turn on heat to medium       |
|        | 194    |                     | Cook until tofu is browned   |
|        | 197    |                     | Return heat to medium        |
|        | 198    |                     | Cook until tofu turns brown  |
|        | 202    |                     | Return to low heat           |
|        | 203    |                     | Cook pan for 2 minutes       |
|        | 205    |                     | Cook pan until darkened      |
|        | 219    |                     | Microwave batter for 60s     |
|        | 231    |                     | Melt cheese in microwave     |
|        | 263    |                     | Toast bread until charred    |
|        | 279    |                     | Microwave for 1 more minute  |
|        | 303    |                     | Microwave ramen for 4 mins   |
| 6      | 27     | Measure Ingredients | Measure 2 cups of corn       |
|        | 309    |                     | Weigh the coffee beans       |
|        | 314    |                     | Measure 12 oz cold water     |
|        | 4      |                     | Clean knife with paper towel |
|        | 7      |                     | Clean knife with paper towel |
|        | 18     |                     | Rinse a tomato               |
|        | 19     |                     | Gently dry with towel        |
| 7      | 84     | Wash Items          | Rinse 1 medium cucumber      |
|        | 118    |                     | Rinse 3 mushrooms            |
|        | 119    |                     | Pat mushrooms dry            |
|        | 189    |                     | Pat tofu dry with towel      |
|        | 241    |                     | Drain excess water from can  |
|        | 323    |                     | Let coffee drain into mug    |
|        | 22     |                     | Garnish with seasoning       |
|        | 23     |                     | Season tomato slices         |
|        | 24     |                     | Season with black pepper     |
|        | 25     |                     | Sprinkle mozzarella on top   |
|        | 80     |                     | Garnish with cilantro        |
|        | 100    |                     | Pepper to taste              |
|        | 101    |                     | Season with salt             |
|        | 104    |                     | Top with more parmesan       |
|        | 111    |                     | Top cup with salsa           |
| 8      | 113    | Season Food         | Sprinkle cheese on cup       |
|        | 131    |                     | Pepper on pan to taste       |
|        | 132    |                     | Season pan with salt         |
|        | 200    |                     | Drizzle soy sauce on pan     |

Continued on next page...

| New ID | Old ID | Step Name          | Old Step Name                |
|--------|--------|--------------------|------------------------------|
|        | 201    |                    | Drizzle with sesame oil      |
|        | 250    |                    | Season pepper on bowl        |
|        | 251    |                    | Season bowl with salt        |
|        | 252    |                    | Top lettuce with tuna        |
|        | 291    |                    | Sprinkle cheese on sauce     |
|        | 292    |                    | Sprinkle herbs inside mug    |
| 9      | 6      | Spread Ingredients | Spread jelly over nut butter |
|        | 106    |                    | Coat ramekin with spray      |
|        | 112    |                    | Line muffin with lettuce     |
|        | 239    |                    | Lay out 2 lettuce leaves     |
|        | 261    |                    | Brush bread with olive oil   |
|        | 290    |                    | Spread sauce around surface  |
|        | 315    |                    | Spread open filter to cone   |
| 10     | 14     | Manipulate Items   | Cross the floss ends         |
|        | 15     |                    | Pull floss ends to slice     |
|        | 35     |                    | Extract lime juice           |
|        | 121    |                    | Pull out mushroom stems      |
|        | 143    |                    | Flip pancakes with spatula   |
|        | 196    |                    | Flip tofu with tongs         |
|        | 204    |                    | Flip tofu on the pan         |
|        | 220    |                    | Invert mug to release cake   |
|        | 226    |                    | Squeeze frosting dollops     |
|        | 253    |                    | Roll up the lettuce wraps    |
| 11     | 9      | Wait               | Secure roll with toothpicks  |
|        | 28     |                    | Thaw corn under cold water   |
|        | 116    |                    | Replace top of muffin        |
|        | 161    |                    | Stop stirring to let it set  |
|        | 177    |                    | Set aside sauce mixture      |
|        | 212    |                    | Set aside the lined mug      |
|        | 221    |                    | Allow to cool until safe     |
|        | 254    |                    | Secure wrap with toothpick   |
|        | 301    |                    | Cover noodles with water     |
|        | 302    |                    | Cover with lid or towel      |
|        | 305    |                    | Let noodles sit for 1 minute |
|        | 321    |                    | Wait 30 seconds for bloom    |

In Fig. 8 and Fig. 9 detail the total duration (in hours), number of classes, and average video length (in seconds) across all the scenarios for all three datasets after the re-annotation process described in Sec. A.2. In this section, we outline the final details and statistics for the released multi-dataset, comprising the following three collections:



Table 20: Ego-Exo4D [21] per scenario statistics

| Scenario                              | N. Videos (Narrations) / duration (min) |                |               | Classes | Total Time (h) | Avg Len (s)       | Min Len (s) | Max Len (s) |
|---------------------------------------|-----------------------------------------|----------------|---------------|---------|----------------|-------------------|-------------|-------------|
|                                       | Train                                   | Val            | Test          |         |                |                   |             |             |
| Covid-19 Rapid Antigen Test           | 110 (16)/443.20                         | 27 (13)/113.96 | 45 (0)/191.56 | 14      | 12.48          | 246.8 $\pm$ 83.2  | 78.6        | 594.3       |
| Clean and Lubricate the Chain         | 17 (5)/57.88                            | 4 (1)/9.40     | 5 (0)/13.77   | 7       | 1.35           | 187.0 $\pm$ 73.9  | 57.8        | 366.8       |
| Cooking Scrambled Eggs                | 16 (5)/112.65                           | 3 (2)/31.24    | 6 (0)/51.36   | 10      | 3.25           | 468.6 $\pm$ 244.0 | 62.6        | 1000.6      |
| Cooking Noodles                       | 22 (3)/324.94                           | 5 (4)/91.43    | 7 (0)/12.91   | 15      | 8.82           | 934.0 $\pm$ 407.3 | 333.1       | 1704.2      |
| Cooking an Omelet                     | 35 (6)/219.40                           | 8 (3)/35.64    | 11 (0)/55.68  | 16      | 5.18           | 345.2 $\pm$ 231.2 | 109.8       | 1245.0      |
| Cooking Tomato & Eggs                 | 28 (5)/330.92                           | 7 (0)/72.66    | 9 (0)/94.97   | 18      | 8.31           | 679.8 $\pm$ 411.2 | 43.1        | 2079.7      |
| Fix a Flat Tire - Replace a Bike Tube | 55 (16)/269.38                          | 13 (8)/64.21   | 11 (0)/47.91  | 12      | 6.36           | 289.7 $\pm$ 137.3 | 28.3        | 669.2       |
| Install a Wheel                       | 41 (8)/91.33                            | 10 (1)/15.06   | 11 (0)/18.30  | 8       | 2.08           | 120.7 $\pm$ 103.1 | 24.3        | 621.6       |
| Making Chai Tea                       | 6 (2)/57.42                             | 1 (0)/8.84     | 2 (0)/18.14   | 13      | 1.41           | 562.7 $\pm$ 135.4 | 441.8       | 922.2       |
| Making Coffee latte                   | 10 (1)/69.46                            | 2 (1)/15.53    | 2 (0)/9.86    | 15      | 1.58           | 406.5 $\pm$ 227.0 | 29.1        | 987.9       |
| Making Cucumber & Tomato Salad        | 34 (8)/109.89                           | 8 (6)/35.35    | 13 (0)/43.61  | 15      | 3.15           | 206.0 $\pm$ 138.4 | 37.2        | 509.6       |
| Making Milk Tea                       | 27 (10)/131.28                          | 6 (6)/27.25    | 12 (0)/41.95  | 12      | 3.34           | 267.3 $\pm$ 173.1 | 74.8        | 732.5       |
| Making Sesame-Ginger Asian Salad      | 18 (6)/240.90                           | 4 (1)/50.31    | 7 (0)/102.49  | 14      | 6.56           | 814.6 $\pm$ 340.4 | 207.1       | 1383.3      |
| Remove a Wheel                        | 43 (9)/59.14                            | 10 (3)/11.21   | 11 (0)/14.59  | 6       | 1.42           | 79.6 $\pm$ 62.0   | 19.6        | 322.6       |
| First Aid - CPR                       | 31 (19)/44.00                           | 7 (7)/9.39     | 13 (0)/14.57  | 11      | 1.13           | 80.0 $\pm$ 23.4   | 45.5        | 184.9       |

**Ego-Exo4D [21]** This dataset was derived from the energy-efficient set, which we further processed and trimmed as described in Sec. A.2.1. Videos without audio or IMU were removed, resulting in a dataset with 777 total videos. The shortest video has 589 frames (19.6 seconds) and the largest has 62,392 frames (34 minutes, and 39 seconds), yielding a total duration of 66 hours, 24 minutes, and 43.8 seconds. See Tab. 20 for additional per-scenario statistics, including global and independent statistics per split. Regarding number of classes, the dataset has an average of 12.4 classes per scenario. Tabs. 21 to 35 show additional per-class statistics for all scenarios independently.

**CMU-MMAC [12]:** This dataset was obtained from the official project website. We built upon the framework provided by [30], further partitioning their training set into a 80-20% split to create a validation set. See Tab. 36 for additional per-scenario statistics, including splits. For final data preparation, we isolated the egocentric videos for this study, excluding the third-person perspectives as they contain redundant information and deviate from a true wearable device setting. Since the audio and IMU streams were not originally synchronized with the video and annotations, we aligned them following the methodology of [33]. Videos lacking audio or IMU data were also removed, resulting in a total dataset duration of 14 hours, 22 minutes, and 48 seconds. Regarding video duration, shortest video has 99.1 seconds, whereas longer one has 850.2 seconds, yielding generally shorter video durations compared to Ego-Exo4D. Moreover, average number of classes per scenario is 7.8. See Tabs. 37 to 41 for further per-class statistics for all scenarios independently.

**CaptainCook4D [35]:** This dataset was taken from the official webpage of [35]. As explained in Sec. A.2.2, videos without Audio or IMU were excluded, resulting in a total of 78 videos and 23 hours and 49 minutes (Tab. 43). The used modalities include video, audio and IMU. Video and audio data were acquired from a GoPro camera system, whereas IMU data were collected with a HoloLens. Due to the synchronization between both devices is not available in the original dataset we had to calibrate this synchronization. After an initial preprocessing step based on visual frame alignment we had to manually refine this synchronization. As shown in Tab. 43), shortest video duration has 247.1 seconds, whereas larger video is 2142.6 seconds long (35 minutes and 43 seconds), similar to Ego-Exo4D. Regarding number of classes, CaptainCook4D has 6.1 unique classes per activity.

Table 21: Per-class stats for “Covid-19 Rapid Antigen Test”

| Class ID | Class Name                 | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|----------------------------|-----------|---------|---------|---------|---------|
| 0        | Background                 | 2680      | 0.03    | 163.20  | 2.22    | 7.68    |
| 1        | Unbox package              | 133       | 1.47    | 53.73   | 15.63   | 9.58    |
| 2        | Unbox & Arrange Materials  | 303       | 0.70    | 95.93   | 11.93   | 14.57   |
| 3        | Prepare Tube               | 310       | 0.70    | 60.87   | 8.23    | 6.76    |
| 4        | Fill Tube with Solution    | 36        | 10.30   | 47.10   | 26.69   | 8.97    |
| 5        | Prepare Swab               | 192       | 1.60    | 44.30   | 12.91   | 7.55    |
| 6        | Swab Nose (Insert/Extract) | 411       | 0.07    | 155.93  | 5.42    | 12.52   |
| 7        | Swab Nose (Rotate)         | 305       | 0.47    | 127.40  | 15.57   | 16.61   |
| 8        | Mix Swab in Tube           | 431       | 0.40    | 60.97   | 6.35    | 6.55    |
| 9        | Prepare Test Cassette      | 184       | 1.33    | 63.10   | 12.47   | 7.17    |
| 10       | Apply Sample to Cassette   | 186       | 1.00    | 67.17   | 14.84   | 8.62    |
| 11       | Check Result               | 109       | 1.03    | 65.57   | 7.27    | 9.43    |
| 12       | Dispose Waste              | 448       | 0.83    | 74.17   | 10.39   | 10.19   |
| 13       | Read Instructions          | 557       | 0.03    | 145.53  | 12.67   | 14.38   |

Table 22: Per-class stats for “Clean and Lubricate the Chain”

| Class ID | Class Name             | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|------------------------|-----------|---------|---------|---------|---------|
| 0        | Background             | 211       | 0.03    | 43.70   | 3.04    | 6.50    |
| 1        | Prepare Tools & Setup  | 56        | 0.93    | 41.77   | 6.95    | 8.94    |
| 2        | Apply Degreaser        | 45        | 0.10    | 45.00   | 7.55    | 10.54   |
| 3        | Move Chain (Backpedal) | 56        | 0.53    | 109.33  | 9.87    | 17.96   |
| 4        | Scrub Chain            | 71        | 1.60    | 110.90  | 14.26   | 17.06   |
| 5        | Dry Chain              | 22        | 4.60    | 139.97  | 51.17   | 42.37   |
| 6        | Lubricate Chain        | 35        | 1.13    | 110.07  | 22.89   | 22.82   |

Table 23: Per-class stats for “Cooking Scrambled Eggs”

| Class ID | Class Name                     | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|--------------------------------|-----------|---------|---------|---------|---------|
| 0        | Background                     | 694       | 0.03    | 20.97   | 1.55    | 3.05    |
| 1        | Get Items                      | 188       | 0.70    | 24.20   | 5.10    | 4.12    |
| 2        | Heat Skillet                   | 160       | 0.47    | 88.57   | 10.08   | 12.22   |
| 3        | Prepare Ingredients (Cut/Peel) | 35        | 2.83    | 181.47  | 36.48   | 39.41   |
| 4        | Whisk Eggs                     | 79        | 1.50    | 88.83   | 14.06   | 11.94   |
| 5        | Fry Vegetables                 | 50        | 2.57    | 61.27   | 17.01   | 13.43   |
| 6        | Scramble Eggs                  | 91        | 1.83    | 301.73  | 35.97   | 45.65   |
| 7        | Season Food                    | 41        | 2.17    | 48.23   | 7.74    | 7.61    |
| 8        | Serve Food                     | 24        | 4.67    | 42.67   | 19.40   | 9.25    |
| 9        | Cleanup                        | 132       | 0.50    | 39.73   | 5.83    | 6.29    |

Table 24: Per-class stats for “Cooking Noodles”

| Class ID | Class Name          | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|---------------------|-----------|---------|---------|---------|---------|
| 0        | Background          | 1554      | 0.03    | 239.13  | 3.47    | 10.16   |
| 1        | Get Kitchenware     | 192       | 0.37    | 68.70   | 5.90    | 7.22    |
| 2        | Get Ingredients     | 258       | 0.30    | 108.83  | 6.19    | 8.72    |
| 3        | Prepare Ingredients | 147       | 1.63    | 250.93  | 39.39   | 38.48   |
| 4        | Heat Water          | 25        | 2.07    | 97.47   | 19.38   | 22.92   |
| 5        | Boil Noodles        | 208       | 1.00    | 138.83  | 15.43   | 17.22   |
| 6        | Drain Noodles       | 38        | 1.43    | 53.07   | 19.56   | 11.75   |
| 7        | Prepare Skillet     | 351       | 0.07    | 67.17   | 6.13    | 6.76    |
| 8        | Stir-Fry Aromatics  | 71        | 1.60    | 31.50   | 10.69   | 6.00    |
| 9        | Stir-Fry Noodles    | 265       | 0.03    | 155.13  | 17.71   | 19.51   |
| 10       | Mix Sauce (Bowl)    | 35        | 1.03    | 64.33   | 16.85   | 14.08   |
| 11       | Season Food         | 203       | 0.67    | 72.70   | 13.98   | 10.55   |
| 12       | Serve Food          | 38        | 4.23    | 76.30   | 21.37   | 14.42   |
| 13       | Wash Items          | 90        | 0.77    | 39.50   | 8.49    | 7.38    |
| 14       | Put Away Items      | 178       | 0.20    | 32.13   | 4.48    | 4.22    |

Table 25: Per-class stats for “Cooking an Omelet”

| Class ID | Class Name       | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|------------------|-----------|---------|---------|---------|---------|
| 0        | Background       | 1376      | 0.03    | 49.53   | 1.38    | 4.34    |
| 1        | Get Kitchenware  | 294       | 0.37    | 22.07   | 4.54    | 3.17    |
| 2        | Get Ingredients  | 312       | 0.20    | 34.50   | 4.05    | 4.48    |
| 3        | Wash Ingredients | 15        | 1.80    | 39.40   | 9.86    | 9.66    |
| 4        | Peel Ingredients | 33        | 1.43    | 44.93   | 13.33   | 10.94   |
| 5        | Cut Ingredients  | 116       | 1.73    | 115.30  | 17.21   | 17.56   |
| 6        | Heat Skillet     | 341       | 0.43    | 61.20   | 5.89    | 6.31    |
| 7        | Whisk Eggs       | 173       | 0.77    | 55.83   | 11.54   | 9.42    |
| 8        | Saute Aromatics  | 129       | 0.83    | 99.37   | 12.96   | 15.88   |
| 9        | Pour & Set       | 83        | 1.53    | 33.43   | 10.36   | 6.54    |
| 10       | Cook & Shape     | 50        | 2.20    | 105.87  | 17.82   | 17.57   |
| 11       | Flip & Fold      | 96        | 0.70    | 82.07   | 15.99   | 15.67   |
| 12       | Season Food      | 125       | 0.87    | 38.93   | 6.82    | 4.27    |
| 13       | Serve Food       | 61        | 1.30    | 40.23   | 10.63   | 8.30    |
| 14       | Cleanup          | 213       | 0.40    | 112.93  | 5.03    | 9.51    |
| 15       | Mistake          | 2         | 7.83    | 12.33   | 10.08   | 2.25    |

Table 26: Per-class stats for “Cooking Tomato &amp; Eggs”

| Class ID | Class Name          | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|---------------------|-----------|---------|---------|---------|---------|
| 0        | Background          | 1797      | 0.03    | 126.73  | 2.49    | 6.72    |
| 1        | Get Kitchenware     | 266       | 0.10    | 23.70   | 4.44    | 3.48    |
| 2        | Get Ingredients     | 277       | 0.60    | 64.20   | 4.83    | 5.53    |
| 3        | Wash Ingredients    | 24        | 1.97    | 33.30   | 10.24   | 7.46    |
| 4        | Peel Ingredients    | 25        | 1.57    | 187.90  | 32.87   | 36.94   |
| 5        | Cut Aromatics       | 86        | 2.00    | 152.07  | 24.22   | 22.56   |
| 6        | Cut Tomato          | 68        | 2.00    | 162.00  | 34.44   | 37.32   |
| 7        | Crack Eggs          | 47        | 4.43    | 39.07   | 10.98   | 6.49    |
| 8        | Whisk Eggs          | 81        | 1.80    | 129.20  | 18.34   | 18.92   |
| 9        | Heat Skillet        | 459       | 0.50    | 57.53   | 6.02    | 7.22    |
| 10       | Scramble Eggs       | 110       | 2.63    | 91.47   | 19.50   | 17.86   |
| 11       | Transfer Eggs (Out) | 16        | 5.60    | 46.93   | 18.22   | 11.88   |
| 12       | Add & Mush Tomatoes | 198       | 0.97    | 83.00   | 10.24   | 10.80   |
| 13       | Stir-Fry Mixture    | 224       | 0.53    | 184.23  | 19.39   | 20.15   |
| 14       | Combine Eggs        | 28        | 2.63    | 22.13   | 9.54    | 4.98    |
| 15       | Season Food         | 97        | 1.53    | 36.10   | 9.18    | 6.48    |
| 16       | Serve Food          | 50        | 4.70    | 123.50  | 20.48   | 18.24   |
| 17       | Cleanup             | 320       | 0.23    | 43.03   | 5.21    | 5.35    |

Table 27: Per-class stats for “Fix a Flat Tire - Replace a Bike Tube”

| Class ID | Class Name             | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|------------------------|-----------|---------|---------|---------|---------|
| 0        | Background             | 894       | 0.03    | 163.57  | 2.39    | 7.15    |
| 1        | Remove Valve Hardware  | 86        | 0.80    | 26.53   | 8.09    | 4.33    |
| 2        | Deflate Tube           | 53        | 1.83    | 117.47  | 25.80   | 18.81   |
| 3        | Detach Tire Bead       | 106       | 0.53    | 101.03  | 15.80   | 16.84   |
| 4        | Remove Old Tube        | 96        | 1.10    | 80.03   | 15.59   | 13.72   |
| 5        | Inspect Tire           | 64        | 1.00    | 124.63  | 27.73   | 27.03   |
| 6        | Prepare New Tube       | 88        | 0.87    | 81.90   | 10.76   | 13.13   |
| 7        | Install Inner Tube     | 136       | 1.97    | 134.63  | 28.87   | 28.89   |
| 8        | Mount Tire Bead        | 98        | 5.60    | 184.13  | 30.89   | 26.28   |
| 9        | Install Valve Hardware | 52        | 1.33    | 32.50   | 8.70    | 5.53    |
| 10       | Inflate Tire           | 166       | 3.93    | 139.13  | 25.38   | 24.10   |
| 11       | Check Bead Seating     | 59        | 2.23    | 75.77   | 19.92   | 17.89   |

Table 28: Per-class stats for “Install a Wheel”

| Class ID | Class Name                  | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|-----------------------------|-----------|---------|---------|---------|---------|
| 0        | Background                  | 374       | 0.03    | 132.97  | 2.69    | 8.53    |
| 1        | Get Tools                   | 37        | 0.97    | 89.60   | 8.17    | 18.45   |
| 2        | Mount Wheel to Frame        | 93        | 0.90    | 110.87  | 14.87   | 18.62   |
| 3        | Tighten Axle Nuts           | 106       | 1.13    | 106.73  | 25.40   | 22.67   |
| 4        | Secure Quick Release        | 32        | 1.63    | 38.57   | 16.49   | 9.98    |
| 5        | Secure Brakes               | 54        | 2.23    | 314.40  | 20.46   | 44.68   |
| 6        | Adjust Derailleur           | 14        | 2.97    | 18.63   | 7.50    | 4.60    |
| 7        | Check Function (Spin/Brake) | 83        | 0.53    | 11.80   | 4.33    | 2.52    |

Table 29: Per-class stats for “Making Chai Tea”

| Class ID | Class Name       | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|------------------|-----------|---------|---------|---------|---------|
| 0        | Background       | 317       | 0.03    | 86.73   | 3.64    | 9.60    |
| 1        | Get Items        | 81        | 0.87    | 32.57   | 5.45    | 4.84    |
| 2        | Stove Operations | 70        | 0.77    | 28.50   | 7.04    | 6.46    |
| 3        | Add Water & Boil | 25        | 2.57    | 44.20   | 10.59   | 8.17    |
| 4        | Add Spices       | 22        | 1.83    | 40.97   | 11.68   | 9.98    |
| 5        | Add Tea Leaves   | 12        | 2.33    | 24.07   | 11.92   | 6.14    |
| 6        | Simmering        | 8         | 1.57    | 77.63   | 29.73   | 24.83   |
| 7        | Add Milk         | 14        | 3.90    | 39.00   | 21.07   | 11.66   |
| 8        | Add Sugar        | 8         | 8.70    | 54.00   | 24.33   | 13.53   |
| 9        | Stir Mixture     | 46        | 3.43    | 259.47  | 21.74   | 37.68   |
| 10       | Strain & Serve   | 9         | 8.57    | 39.90   | 25.93   | 9.95    |
| 11       | Wash Items       | 11        | 3.50    | 19.53   | 10.18   | 4.22    |
| 12       | Put Away Items   | 51        | 1.20    | 16.33   | 4.65    | 3.03    |

Table 30: Per-class stats for “Making Coffee latte”

| Class ID | Class Name                     | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|--------------------------------|-----------|---------|---------|---------|---------|
| 0        | Background                     | 302       | 0.03    | 131.40  | 4.04    | 12.75   |
| 1        | Get Kitchenware                | 62        | 0.63    | 28.57   | 6.53    | 5.53    |
| 2        | Get Ingredients                | 30        | 1.17    | 25.37   | 5.96    | 5.13    |
| 3        | Operate Heating Appliance      | 53        | 0.67    | 31.20   | 8.68    | 6.62    |
| 4        | Wait for Boiling               | 26        | 3.50    | 91.37   | 26.07   | 18.25   |
| 5        | Prepare Filter & Grounds       | 20        | 2.73    | 96.73   | 19.93   | 21.45   |
| 6        | Pour Water for Brewing         | 25        | 0.03    | 48.77   | 15.13   | 12.24   |
| 7        | Process Milk (Heat/Stir/Froth) | 35        | 3.80    | 64.03   | 16.76   | 12.51   |
| 8        | Pour Milk                      | 20        | 1.83    | 96.23   | 19.66   | 21.21   |
| 9        | Extract & Mix Coffee           | 16        | 4.73    | 91.43   | 20.98   | 21.86   |
| 10       | Pour Coffee to Serve           | 6         | 8.57    | 42.37   | 20.05   | 12.38   |
| 11       | Add Sweetener & Spices         | 26        | 3.17    | 20.07   | 9.92    | 5.23    |
| 12       | Wash Items                     | 8         | 2.20    | 29.03   | 11.45   | 8.89    |
| 13       | Put Away Items                 | 25        | 0.60    | 19.73   | 4.63    | 3.66    |
| 14       | Wipe Clean                     | 9         | 1.33    | 29.67   | 7.85    | 9.11    |

Table 31: Per-class stats for “Making Cucumber &amp; Tomato Salad”

| Class ID | Class Name            | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|-----------------------|-----------|---------|---------|---------|---------|
| 0        | Background            | 801       | 0.03    | 33.27   | 1.24    | 3.81    |
| 1        | Get Kitchenware       | 121       | 0.47    | 27.97   | 4.43    | 4.16    |
| 2        | Get Ingredients       | 189       | 0.30    | 42.80   | 5.28    | 7.11    |
| 3        | Wash Vegetables       | 39        | 0.60    | 34.80   | 11.71   | 7.59    |
| 4        | Peel Ingredients      | 23        | 2.53    | 61.23   | 18.82   | 11.70   |
| 5        | Cut Tomato            | 97        | 1.40    | 89.77   | 16.62   | 18.05   |
| 6        | Cut Cucumber          | 86        | 0.03    | 87.43   | 24.34   | 22.57   |
| 7        | Cut Other Ingredients | 25        | 1.97    | 61.00   | 14.55   | 15.28   |
| 8        | Add Solid Ingredients | 131       | 0.47    | 36.67   | 6.93    | 7.56    |
| 9        | Add Dressing & Spices | 117       | 1.53    | 53.33   | 8.93    | 6.69    |
| 10       | Mix Salad             | 59        | 0.87    | 40.00   | 14.45   | 9.75    |
| 11       | Serve Salad           | 10        | 3.13    | 24.47   | 11.05   | 7.49    |
| 12       | Wash Dishes           | 22        | 3.20    | 65.63   | 11.68   | 12.90   |
| 13       | Put Away Items        | 72        | 0.33    | 30.87   | 5.07    | 6.74    |
| 14       | Wipe & Clean          | 46        | 0.40    | 45.57   | 6.76    | 8.35    |

Table 32: Per-class stats for “Making Milk Tea”

| Class ID | Class Name             | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|------------------------|-----------|---------|---------|---------|---------|
| 0        | Background             | 736       | 0.03    | 171.13  | 3.90    | 12.12   |
| 1        | Get Items              | 249       | 0.57    | 104.83  | 7.19    | 9.09    |
| 2        | Heat Water             | 191       | 0.20    | 147.43  | 9.76    | 16.30   |
| 3        | Brew Tea               | 66        | 3.60    | 42.97   | 12.48   | 7.83    |
| 4        | Steep & Simmer         | 53        | 1.20    | 151.03  | 20.93   | 25.73   |
| 5        | Remove Tea             | 10        | 1.17    | 15.67   | 7.05    | 3.66    |
| 6        | Add Milk               | 48        | 2.73    | 22.00   | 10.34   | 4.60    |
| 7        | Add Sweetener & Spices | 50        | 1.47    | 23.23   | 9.91    | 5.37    |
| 8        | Stir Drink             | 102       | 1.67    | 59.33   | 9.17    | 8.32    |
| 9        | Pour to Serve          | 29        | 2.27    | 41.17   | 16.66   | 9.30    |
| 10       | Cleanup                | 156       | 0.57    | 56.80   | 7.00    | 7.72    |
| 11       | Mistake                | 1         | 2.20    | 2.20    | 2.20    | 0.00    |

Table 33: Per-class stats for “Making Sesame-Ginger Asian Salad”

| Class ID | Class Name            | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|-----------------------|-----------|---------|---------|---------|---------|
| 0        | Background            | 964       | 0.03    | 88.10   | 2.83    | 6.7     |
| 1        | Get Items             | 357       | 0.67    | 99.83   | 6.12    | 8.64    |
| 2        | Wash Ingredients      | 55        | 0.53    | 29.97   | 10.19   | 7.81    |
| 3        | Peel Ingredients      | 47        | 3.40    | 108.93  | 28.82   | 22.15   |
| 4        | Grate Ingredients     | 37        | 7.53    | 220.37  | 70.40   | 47.96   |
| 5        | Cut Bell Peppers      | 51        | 2.23    | 183.50  | 30.39   | 32.99   |
| 6        | Cut Other Veg         | 149       | 2.20    | 181.20  | 33.77   | 30.07   |
| 7        | Add Solid Ingredients | 154       | 0.90    | 77.43   | 12.73   | 10.90   |
| 8        | Make Dressing         | 171       | 0.07    | 112.53  | 14.97   | 14.05   |
| 9        | Whisk/Shake Dressing  | 23        | 2.17    | 37.17   | 16.15   | 8.65    |
| 10       | Pour Dressing         | 22        | 3.07    | 50.63   | 13.22   | 9.55    |
| 11       | Toss Salad            | 50        | 2.13    | 61.00   | 18.88   | 10.80   |
| 12       | Serve Salad           | 16        | 1.53    | 44.77   | 15.28   | 12.05   |
| 13       | Cleanup               | 192       | 0.30    | 49.70   | 6.45    | 6.66    |

Table 34: Per-class stats for “Remove a Wheel”

| Class ID | Class Name         | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|--------------------|-----------|---------|---------|---------|---------|
| 0        | Background         | 241       | 0.03    | 98.07   | 6.85    | 13.00   |
| 1        | Get Tools          | 32        | 0.23    | 16.87   | 3.54    | 3.16    |
| 2        | Adjust Derailleur  | 14        | 3.37    | 38.57   | 9.47    | 8.72    |
| 3        | Open Quick Release | 129       | 1.60    | 188.10  | 22.67   | 28.01   |
| 4        | Remove Wheel       | 24        | 1.70    | 21.03   | 6.83    | 4.73    |
| 5        | Deflate Tube       | 3         | 14.60   | 70.73   | 36.84   | 24.35   |

Table 35: Per-class stats for “First Aid - CPR”

| Class ID | Class Name                              | Instances | Min (s) | Max (s) | Avg (s) | Std (s) |
|----------|-----------------------------------------|-----------|---------|---------|---------|---------|
| 0        | Background                              | 388       | 0.03    | 26.07   | 2.12    | 4.64    |
| 1        | Have a conversation asking different qu | 24        | 0.53    | 16.13   | 3.34    | 4.08    |
| 2        | Check if the patient is responding at a | 18        | 1.17    | 17.23   | 4.16    | 3.58    |
| 3        | Tap patient to confirm consciousness    | 75        | 0.83    | 12.70   | 3.82    | 2.35    |
| 4        | Confirm patient consciousness from the  | 98        | 0.73    | 13.07   | 3.82    | 2.13    |
| 5        | Kneel on the side of the patient’s neck | 20        | 1.27    | 16.07   | 4.77    | 3.30    |
| 6        | Place the lower part of the palm (heel) | 45        | 0.17    | 43.47   | 1.88    | 6.33    |
| 7        | Place the other hand on top of the fir  | 65        | 0.17    | 8.97    | 1.38    | 1.39    |
| 8        | Press hard at a rate of 100 to 120 comp | 109       | 0.23    | 57.40   | 18.39   | 8.98    |
| 9        | Do artificial respirations two times af | 4         | 2.20    | 6.53    | 4.84    | 1.62    |
| 10       | Call for help                           | 23        | 1.77    | 24.93   | 6.31    | 5.74    |

Table 36: CMU Kitchens Dataset Per-Scenario Statistics  
N. Videos / duration (min)

| Scenario | Train     | Val     | Test     | Classes | Total Time (h) | Avg Len (s)   | Min Len (s) | Max Len (s) |
|----------|-----------|---------|----------|---------|----------------|---------------|-------------|-------------|
| Salad    | 19/105.21 | 6/36.00 | 6/29.45  | 10      | 2.84           | 330.3 ± 79.8  | 129.6       | 527.6       |
| Sandwich | 20/60.08  | 4/10.06 | 3/9.94   | 5       | 1.33           | 177.9 ± 32.6  | 99.1        | 224.4       |
| Eggs     | 19/97.65  | 3/16.53 | 7/38.21  | 9       | 2.54           | 315.3 ± 74.5  | 179.0       | 471.6       |
| Brownie  | 15/95.61  | 5/37.11 | 11/77.41 | 9       | 3.50           | 406.7 ± 96.7  | 213.4       | 683.7       |
| Pizza    | 20/167.29 | 5/34.97 | 5/47.17  | 6       | 4.16           | 498.9 ± 136.0 | 242.3       | 850.2       |

Table 37: Per-Class Statistics for “Salad”

| ID | Class Name      | Inst. | Min (s) | Max (s) | Avg (s) | Std (s) |
|----|-----------------|-------|---------|---------|---------|---------|
| 0  | Background      | 252   | 0.07    | 111.20  | 27.79   | 27.14   |
| 1  | peel carrot     | 32    | 2.83    | 34.43   | 15.51   | 7.12    |
| 2  | cut carrot      | 33    | 2.10    | 113.87  | 18.65   | 21.01   |
| 3  | cut lettuce     | 20    | 2.70    | 42.00   | 13.44   | 9.58    |
| 4  | peel cucumber   | 32    | 4.70    | 40.37   | 19.52   | 9.70    |
| 5  | cut cucumber    | 35    | 3.07    | 27.33   | 11.29   | 6.41    |
| 6  | cut white onion | 33    | 2.87    | 43.03   | 17.49   | 10.29   |
| 7  | add pepper      | 15    | 2.23    | 10.93   | 6.30    | 2.41    |
| 8  | add mayonnaise  | 13    | 3.17    | 13.17   | 6.35    | 2.99    |
| 9  | mix             | 8     | 2.93    | 22.10   | 10.20   | 6.48    |

Table 38: Per-Class Statistics for “Sandwich”

| ID | Class Name          | Inst. | Min (s) | Max (s) | Avg (s) | Std (s) |
|----|---------------------|-------|---------|---------|---------|---------|
| 0  | Background          | 194   | 0.10    | 89.30   | 18.58   | 19.30   |
| 1  | keep bread on plate | 26    | 0.37    | 5.60    | 1.45    | 1.09    |
| 2  | apply peanut butter | 62    | 1.53    | 28.00   | 8.72    | 6.16    |
| 3  | apply jam           | 52    | 0.70    | 43.80   | 10.13   | 8.94    |
| 4  | press bread slices  | 27    | 1.30    | 8.17    | 3.50    | 1.71    |

Table 39: Per-Class Statistics for “Eggs”

| ID | Class Name     | Inst. | Min (s) | Max (s) | Avg (s) | Std (s) |
|----|----------------|-------|---------|---------|---------|---------|
| 0  | Background     | 331   | 0.23    | 200.90  | 20.37   | 26.22   |
| 1  | break egg      | 47    | 2.67    | 12.73   | 5.25    | 1.72    |
| 2  | mix            | 70    | 1.13    | 51.57   | 12.64   | 10.06   |
| 3  | add salt       | 32    | 1.07    | 11.87   | 3.92    | 2.15    |
| 4  | add pepper     | 28    | 1.50    | 12.73   | 4.03    | 2.40    |
| 5  | pour oil       | 31    | 1.50    | 10.33   | 3.98    | 2.05    |
| 6  | pour mixture   | 29    | 3.57    | 28.33   | 10.13   | 4.82    |
| 7  | flip           | 38    | 1.47    | 33.47   | 11.10   | 6.97    |
| 8  | plate omelette | 29    | 1.47    | 13.30   | 6.61    | 3.34    |

Table 40: Per-Class Statistics for “Brownie”

| ID | Class Name      | Inst. | Min (s) | Max (s) | Avg (s) | Std (s) |
|----|-----------------|-------|---------|---------|---------|---------|
| 0  | Background      | 318   | 0.07    | 176.43  | 22.24   | 24.60   |
| 1  | break egg       | 45    | 3.43    | 27.83   | 9.51    | 5.18    |
| 2  | mix eggs        | 14    | 6.03    | 32.80   | 19.56   | 8.48    |
| 3  | add water       | 32    | 1.90    | 7.20    | 3.41    | 1.17    |
| 4  | add oil         | 33    | 2.07    | 10.10   | 4.60    | 1.91    |
| 5  | add brownie mix | 45    | 2.47    | 33.63   | 13.15   | 7.68    |
| 6  | mix contents    | 67    | 0.80    | 133.03  | 36.96   | 28.80   |
| 7  | spray oil       | 28    | 3.57    | 12.13   | 6.77    | 2.05    |
| 8  | pour mixture    | 39    | 5.97    | 66.07   | 33.70   | 15.14   |

Table 41: Per-Class Statistics for “Pizza”

| ID | Class Name    | Inst. | Min (s) | Max (s) | Avg (s) | Std (s) |
|----|---------------|-------|---------|---------|---------|---------|
| 0  | Background    | 235   | 0.10    | 167.60  | 29.42   | 32.53   |
| 1  | spread batter | 40    | 2.77    | 128.93  | 34.05   | 23.21   |
| 2  | apply sauce   | 36    | 1.73    | 106.23  | 38.44   | 21.64   |
| 3  | add cheese    | 50    | 11.57   | 214.13  | 53.76   | 40.18   |
| 4  | add pepperoni | 49    | 0.70    | 228.13  | 51.61   | 48.61   |
| 5  | to oven       | 30    | 1.00    | 8.50    | 2.97    | 1.62    |

Table 42: Captain Cook 4D: Videos per Scenario across Split Strategies

| Scenario                        | Recordings |          |           | Person    |           |           | Environment |           |           |
|---------------------------------|------------|----------|-----------|-----------|-----------|-----------|-------------|-----------|-----------|
|                                 | Train      | Val      | Test      | Train     | Val       | Test      | Train       | Val       | Test      |
| Blender Banana Pancakes         | 2          | 0        | 1         | 1         | 1         | 1         | 3           | 0         | 0         |
| Broccoli Stir Fry               | 1          | 0        | 1         | 1         | 1         | 0         | 2           | 0         | 0         |
| Butter Corn Cup                 | 2          | 0        | 1         | 1         | 1         | 1         | 3           | 0         | 0         |
| Caprese Bruschetta              | 3          | 0        | 2         | 4         | 0         | 1         | 2           | 0         | 3         |
| Cheese Pimiento                 | 3          | 1        | 0         | 2         | 0         | 2         | 2           | 2         | 0         |
| Coffee                          | 1          | 0        | 0         | 0         | 1         | 0         | 0           | 1         | 0         |
| Cucumber Raita                  | 5          | 1        | 3         | 3         | 3         | 3         | 5           | 4         | 0         |
| Dressed Up Meatballs            | 1          | 1        | 2         | 3         | 1         | 0         | 3           | 0         | 1         |
| Herb Omelet with Fried Tomatoes | 0          | 1        | 2         | 2         | 0         | 1         | 3           | 0         | 0         |
| Microwave Egg Sandwich          | 2          | 1        | 0         | 1         | 0         | 2         | 2           | 0         | 1         |
| Microwave Mug Pizza             | 3          | 0        | 1         | 1         | 2         | 1         | 3           | 1         | 0         |
| Mug Cake                        | 3          | 0        | 2         | 4         | 0         | 1         | 1           | 0         | 4         |
| Pan Fried Tofu                  | 3          | 0        | 2         | 2         | 1         | 2         | 2           | 3         | 0         |
| Pinwheels                       | 1          | 0        | 0         | 0         | 1         | 0         | 0           | 1         | 0         |
| Ramen                           | 3          | 0        | 2         | 3         | 1         | 1         | 4           | 1         | 0         |
| Sauteed Mushrooms               | 2          | 0        | 1         | 2         | 1         | 0         | 3           | 0         | 0         |
| Scrambled Eggs                  | 1          | 1        | 2         | 2         | 1         | 1         | 1           | 0         | 3         |
| Spicy Tuna Avocado Wraps        | 1          | 1        | 0         | 2         | 0         | 0         | 1           | 0         | 1         |
| Tomato Chutney                  | 1          | 0        | 1         | 2         | 0         | 0         | 1           | 1         | 0         |
| Tomato Mozzarella Salad         | 4          | 2        | 1         | 4         | 0         | 3         | 5           | 2         | 0         |
| Zoodles                         | 2          | 0        | 1         | 2         | 0         | 1         | 1           | 0         | 2         |
| <b>TOTAL</b>                    | <b>44</b>  | <b>9</b> | <b>25</b> | <b>42</b> | <b>15</b> | <b>21</b> | <b>47</b>   | <b>16</b> | <b>15</b> |

Table 43: Captain Cook 4D Dataset Per-Activity Statistics

| Activity                        | N. Videos / duration (min) |         |         | Unique classes | Total (h) | Avg Len (s) | Min Len (s) | Max Len (s) |
|---------------------------------|----------------------------|---------|---------|----------------|-----------|-------------|-------------|-------------|
|                                 | Train                      | Val     | Test    |                |           |             |             |             |
| Pan Fried Tofu                  | 3/107.13                   | 0/0.00  | 2/71.42 | 7              | 2.98      | 2142.6      | 2142.6      | 2142.6      |
| Ramen                           | 3/107.13                   | 0/0.00  | 2/71.42 | 6              | 2.98      | 2142.6      | 2142.6      | 2142.6      |
| Cucumber Raita                  | 5/80.17                    | 1/16.03 | 3/48.10 | 4              | 2.41      | 962.1       | 962.1       | 962.1       |
| Dressed Up Meatballs            | 1/35.71                    | 1/35.71 | 2/71.42 | 6              | 2.38      | 2142.6      | 2142.6      | 2142.6      |
| Microwave Mug Pizza             | 3/107.13                   | 0/0.00  | 1/35.71 | 6              | 2.38      | 2142.6      | 2142.6      | 2142.6      |
| Scrambled Eggs                  | 1/18.01                    | 1/18.01 | 2/36.01 | 5              | 1.20      | 1080.4      | 1080.4      | 1080.4      |
| Caprese Bruschetta              | 3/42.89                    | 0/0.00  | 2/28.59 | 6              | 1.19      | 857.8       | 857.8       | 857.8       |
| Mug Cake                        | 3/28.72                    | 0/0.00  | 2/19.14 | 7              | 0.80      | 574.3       | 574.3       | 574.3       |
| Herb Omelet with Fried Tomatoes | 0/0.00                     | 1/15.64 | 2/31.28 | 6              | 0.78      | 938.3       | 938.3       | 938.3       |
| Zoodles                         | 2/28.59                    | 0/0.00  | 1/14.30 | 5              | 0.71      | 857.8       | 857.8       | 857.8       |
| Tomato Chutney                  | 1/20.72                    | 0/0.00  | 1/20.72 | 5              | 0.69      | 1243.3      | 1243.3      | 1243.3      |
| Butter Corn Cup                 | 2/27.25                    | 0/0.00  | 1/13.63 | 6              | 0.68      | 817.6       | 817.6       | 817.6       |
| Microwave Egg Sandwich          | 2/26.57                    | 1/13.28 | 0/0.00  | 8              | 0.66      | 797.0       | 797.0       | 797.0       |
| Blender Banana Pancakes         | 2/26.57                    | 0/0.00  | 1/13.28 | 6              | 0.66      | 797.0       | 797.0       | 797.0       |
| Cheese Pimiento                 | 3/29.18                    | 1/9.73  | 0/0.00  | 5              | 0.65      | 583.5       | 583.5       | 583.5       |
| Coffee                          | 1/35.71                    | 0/0.00  | 0/0.00  | 8              | 0.60      | 2142.6      | 2142.6      | 2142.6      |
| Broccoli Stir Fry               | 1/16.03                    | 0/0.00  | 1/16.03 | 6              | 0.53      | 962.1       | 962.1       | 962.1       |
| Sauteed Mushrooms               | 2/19.62                    | 0/0.00  | 1/9.81  | 7              | 0.49      | 588.7       | 588.7       | 588.7       |
| Tomato Mozzarella Salad         | 4/16.48                    | 2/8.24  | 1/4.12  | 5              | 0.48      | 247.1       | 247.1       | 247.1       |
| Spicy Tuna Avocado Wraps        | 1/13.03                    | 1/13.03 | 0/0.00  | 9              | 0.43      | 781.5       | 781.5       | 781.5       |
| Pinwheels                       | 1/7.71                     | 0/0.00  | 0/0.00  | 6              | 0.13      | 462.5       | 462.5       | 462.5       |

#### A.4 Estimation of Energy Budget per second

For the computation of energy for a video of a given length, we follow Eq. (2).

The constants  $\alpha = 4.6 * 10^{-9}$  mJ/MAC and  $\beta = 80 * 10^{-9}$  mJ/byte convert these operations to energy (mJ) [21]. We adopt  $P_{\text{RGB}} = 15$  mW and  $P_{\text{audio}} = 0.5$  mW from [21], and set  $P_{\text{mono}} = 1$  mW [7],  $P_{\text{IMU}} = 0.2$  mW [15], and  $P_{\text{gaze}} = 0.63$  mW [3].

For the memory energy, we use the PyTorch Profiler tool to compute the memory cost. This profiler returns the CUDA bytes used for a forward pass of it through the model. Since this number of bytes is not linear and cannot be estimated, we use this tool per video to obtain the number of bytes, and from it the  $E_{\text{Vmem}}$ .

The number of Multiply-Accumulate Operations (MACs) is estimated at the beginning of training using the THOP profiler [27]. For feature extractors  $f_{\phi}^{\text{MAC}}$  we use the MACs obtained from a forward pass. For the models used as baselines for this proposed benchmark, we observed that the MAC count scales linearly with the input length, never passing through the origin. Therefore, at the beginning of the training, we pass two inputs of different lengths through the model to estimate the per-frame computational cost and the fixed computational overhead:

$$f_{\Psi}^{\text{MAC}}(L) = m_{\text{frame}}L + n_{\text{fixed}},$$

where  $L$  denotes the input length,  $m_{\text{frame}}$  represents the additional computational cost introduced by each extra frame, and  $n_{\text{fixed}}$  accounts for fixed computations that are independent of the input length. After obtaining the total number of MACs, the computational energy  $E_{\text{Vcomp}}$  can be obtained.

Ultimately, for the computation of the capture energy  $E_{\text{Vcap}}$ , no additional functionality was needed, just the number of active steps  $a_t^{(m)}$  of each sensor, the temporal duration between consecutive timesteps  $\Delta t$  and its power consumption  $P_m$ .

The average energy per second of the video is finally the total energy divided by the number of seconds of the video  $T$ .

#### A.5 Experimental settings

To ensure reproducibility, we provide the specific training configurations used for our baselines in all three datasets. We train per scenario on Ego-Exo4D [21] and CMU-MMAC [12], since joint training causes background class to strongly dominate and can lead to collapse. However, for CaptainCook4D [35], we perform global training across all scenarios simultaneously as there are not enough data to be trained per scenario (details in Sec. A.2). For the energy computation, we follow the formulation and energy values in Sec. 3.2 using the profiler implementation described in Sec. A.4.

All baselines are implemented in PyTorch and trained with the Adam optimizer. We use a weight decay of  $e^{-4}$ , an initial learning rate of  $5e^{-4}$ , and a cosine annealing learning-rate schedule. For non-learned policies, we train for 50 epochs following the training protocol of ProTAS [42]. For learned policies, we adopt the setup of AdaMML [34]: the TAS model is first warmed up for 5 epochs, after which AdaMML alternates policy and task-model training for 20 epochs, followed by 10 epochs of TAS fine-tuning. For HCMS [51], since no specific training pipeline is provided, we replace the alternating scheme with joint training while keeping the remaining AdaMML training specifications unchanged. Experiments were conducted on an NVIDIA RTX 4090 GPU.

For AdaMML [34] and HCMS [51], each trained policy produces a fixed trade-off between performance and energy consumption. Since the target energy budget is not enforced during training within their pipelines, we adapt these policies to different budget constraints at inference time by varying the threshold applied to the Gumbel-Softmax decisions [26]. This allows us to shift the modality-selection behavior of the policy and obtain operating points with different energy-accuracy trade-offs. Similarly, the Random policy was ablated using different thresholds and training and inference time to evaluate their performance under different circumstances.

#### A.6 Random Policy computation

Random policy ( $c = 0$ ) just performs a random dropout of each modality per frame. However, the cost-aware version ( $c = 1$ ), takes into account the cost of each modality, enforcing more expensive ones to be dropped more times than the cheaper ones while maintaining the dropout percentage  $\tau$  of the standard ( $c = 0$ ) version.

The percentage of times the average of the modalities should be on is  $p_{\text{on}} = 1.0 - \tau$

For each modality  $m \in \{1, \dots, M\}$  with associated cost  $c_m$ , we compute log-normalized costs  $\log c_m$ . And define the cost range as  $\Delta \log c = \max_m \log c_m - \min_m \log c_m$

For each modality, we compute relative expensiveness:

$$e_m = \frac{\log c_m - \min \log c}{\Delta \log c} \quad (4)$$

This is transformed into relative selection weights using a weighted interpolation:

$$w_m = \alpha_{\min} + (1 - \alpha_{\min})(1 - e_m) \quad (5)$$

where  $\alpha_{\min} = 0.25$  ensures all modalities remain sampleable even for expensive modalities.

The weights are then normalized and, finally, the selection probability for each modality is  $\tau_m = p_{\text{on}} \cdot \tilde{w}_m$

## A.7 Per Modality Computation

Per each modality, we extract the feature as follows:

**Video:** We employ DINOv3 [43] as feature extractor, which as default takes  $448 \times 448$  input images at each timestep  $t$ . Feature extraction is applied at each timestep  $t$  in the video stream as  $\mathbf{x}_t^{(RGB)} = \phi_{RGB}(\mathbf{X}_t^{(RGB)})$ . For monochrome cameras we use low-resolution  $256 \times 256$  images to serve as an energy-efficient alternative modality. We use ViT-B version of DINOv3, which consumes approximately  $E_{total}^{(RGB)} = 9.3$  J per complete second of active sensing and computation. The low-resolution alternative reduces this to  $E_{total}^{(Mono)} = 3.09$  J per active second.

**IMU:** Accelerometer and gyroscope raw data from one IMU is directly used as a temporal input signal in PRIMUS [10]. Particularly, since PRIMUS is pre-trained in Ego-Exo4D on 5 second-length temporal signals, for each timestep  $t$ , we take a signal of bounds  $[t - 5s, t]$ . The energy consumption is very low ( $E_{total}^{(IMU)} = 4.45$  mJ per active sensor) given the simple architecture and efficient sensing. The IMU configuration (location and placement) varies depending on the acquisition set-up of each dataset. For CMU we use the right wrist IMU, for ego-exo4D we use the embedded left-IMU of the Aria headset and finally for CaptainCook4D we use the embedded IMU of the HoloLens systems.

**Audio:** We use SSAST [20] audio features as follows: for each timestep  $t$ , we take a clip segment corresponding to the last second of audio up to  $t$ . This clip is converted into a Mel spectrogram, which is the required input for the SSAST backbone.  $\mathbf{x}_t^{(Audio)} = \phi_{Audio}(\text{spectrogram}(\mathbf{X}_t^{(Audio)}))$ . We use the SSAST version where the spectrogram is splitted into a sequence of  $16 \times 16$  patches. The total energy consumption is  $E_{total}^{(Audio)} = 414$  mJ per active second. Notice that, while SSAST is typically pre-trained on 10-second clips, we reduce clip length down to just 1 second to reduce compute and memory, and allow the policy to save energy in a more granular way.

**Gaze:** For gaze modality, instead of using the 2D on the image location as a signal, we cropped a  $192 \times 192$  bounding box around the 2D gaze position. That allows us to use local visual information which could be relevant for action recognition since this crop represents the region of the image the user is interacting with or the user’s interest. This crop is also encoded with Dinov3 but only consuming 1.80 W due to its reduced size.

## A.8 Additional insights

The scenarios within each dataset differ substantially, increasing the diversity of our benchmark and introducing a range of distinct challenges. For instance, “Clean and Lubricate the Chain” is the only Ego-Exo4D scenario in which audio hurts performance. While combining modalities often improves performance, noisy or uninformative signals can instead degrade it. This effect is shown in Tab. 44: RGB alone achieves strong performance (rows 1, 7), whereas adding audio substantially reduces performance (rows 2, 3). Although it is true that using a policy that allows to disentangle audio from the rest of the modalities, like Random, allows the model to ignore it signals and get a good performance (rows 13, 15). Policies that activate all modalities at the same timesteps, such as Greedy or Frame Rate, cannot learn to ignore audio independently. Conversely, learned policies such as HCMS and AdaMML are also unsuited to this scenario (rows 4, 6). HCMS requires all lower-cost modalities to be active before enabling RGB, preventing the policy from selecting only the useful modality. AdaMML faces a different limitation: its official code implementation uses a loss that penalizes using too many modality usage only when the prediction is correct. As a result, when predictions are poor, AdaMML tends to activate all modalities and fails to disentangle their individual contributions (see Fig. 10).

## A.9 Energy-Accuracy trade-off

Figures 11 12 and 13 visualizes the Energy-Accuracy trade-offs across selected policies for Ego-Exo4D, CMU and CaptainCook4D, illustrating stark differences in routing dynamics across domains. In Ego-Exo4D (Fig. 11), we observe a clear hierarchy where dynamic policies consistently dominate. The Random routing policies ( $\tau_{tr} = 0.70$  and  $0.90$ ) maintain the highest performance frontier across nearly the entire energy spectrum, demonstrating their robustness for complex, procedural tasks. Static framerate policies exhibit distinct crossing points: ultra-lightweight modalities like IMU provide the best accuracy at extreme energy constraints ( $< 1$  mW),

Table 44: Performance on scenario “Clean and Lubricate the Chain” Ego-Exo4D [21] across energy budgets, policies and parameters.

| Budget    | #  | Policy     | Parameters                              | Video (RGB) | Audio     | IMU       | Monochrome | Gaze | Energy   | Acc   | mAP   | Edit  | F1@10 | F1@25 | F1@50 |
|-----------|----|------------|-----------------------------------------|-------------|-----------|-----------|------------|------|----------|-------|-------|-------|-------|-------|-------|
| No Budget | 1  | Frame Rate | 30.00 FPS                               | 100.0       | x         | x         | x          | x    | 09.30 W  | 32.34 | 32.98 | 14.57 | 5.52  | 4.38  | 4.00  |
|           | 2  | Frame Rate | 30.00 FPS                               | x           | 100.0     | x         | x          | x    | 00.41 W  | 0.84  | 28.93 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 3  | Frame Rate | 30.00 FPS                               | 100.0       | 100.0     | x         | x          | x    | 09.72 W  | 0.02  | 28.82 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 4  | HCMS       |                                         | 69.1        | 92.9      | 100.0     | 90.3       | 79.2 | 13.53 W  | 0.00  | 31.95 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 5  | Frame Rate | 06.00 FPS                               | 20.0        | 20.0      | 20.0      | 20.0       | 20.0 | 03.54 W  | 0.00  | 25.79 | 0.00  | 0.00  | 0.00  | 0.00  |
| 2.8 W     | 6  | AdaMML     |                                         | 1.4         | 1.6 / 100 | 1.3 / 100 | 100.0      | x    | 02.34 W  | 0.00  | 25.14 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 7  | Frame Rate | 06.00 FPS                               | 20.0        | x         | x         | x          | x    | 01.86 W  | 11.33 | 27.72 | 3.70  | 4.44  | 4.44  | 0.00  |
|           | 8  | Frame Rate | 10.00 FPS                               | x           | 33.3      | x         | x          | x    | 00.14 W  | 0.85  | 32.86 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 9  | Frame Rate | 06.00 FPS                               | 20.0        | 20.0      | x         | x          | x    | 01.94 W  | 0.00  | 28.68 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 10 | Frame Rate | 06.00 FPS                               | 20.0        | 20.0      | 20.0      | x          | x    | 01.95 W  | 0.00  | 31.22 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 11 | Frame Rate | 06.00 FPS                               | x           | 20.0      | 20.0      | 20.0       | 20.0 | 01.68 W  | 0.00  | 33.65 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 12 | Greedy     |                                         | 13.4        | 13.4      | 13.4      | 13.4       | 13.4 | 02.36 W  | 1.54  | 39.42 | 7.54  | 0.00  | 0.00  | 0.00  |
|           | 13 | Random     | $\tau_{tr}=0.70, \tau_{inj}=0.900, c=0$ | 10.3        | 10.1      | 10.3      | 10.0       | 10.0 | 01.80 W  | 53.24 | 35.95 | 24.16 | 10.06 | 7.59  | 3.14  |
| 20 mW     | 14 | Random     | $\tau_{tr}=0.97, \tau_{inj}=0.900, c=1$ | 5.2         | 11.6      | 20.7      | 8.7        | 7.6  | 01.16 W  | 46.23 | 32.75 | 25.17 | 12.97 | 9.17  | 4.72  |
|           | 15 | Frame Rate | 00.05 FPS                               | x           | 0.2       | x         | x          | x    | 00.74 mW | 1.28  | 27.53 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 16 | Frame Rate | 00.05 FPS                               | 0.2         | x         | x         | x          | x    | 16.52 mW | 12.52 | 28.56 | 2.03  | 2.91  | 2.91  | 0.00  |
|           | 17 | Random     | $\tau_{tr}=0.70, \tau_{inj}=1.000, c=0$ | 0           | 0         | 0         | 0          | 0    | 00.52 mW | 57.74 | 29.25 | 8.70  | 14.63 | 11.43 | 0.00  |
|           | 18 | Frame Rate | 00.05 FPS                               | 0.2         | 0.2       | x         | x          | x    | 17.25 mW | 0.86  | 22.68 | 0.00  | 0.00  | 0.00  | 0.00  |
|           | 19 | Random     | $\tau_{tr}=0.70, \tau_{inj}=1.000, c=1$ | 0           | 0         | 0         | 0          | 0    | 00.48 mW | 57.74 | 29.25 | 8.70  | 14.63 | 11.43 | 0.00  |

Sensors: Video (RGB), Audio, IMU, Monochrome, Gaze.

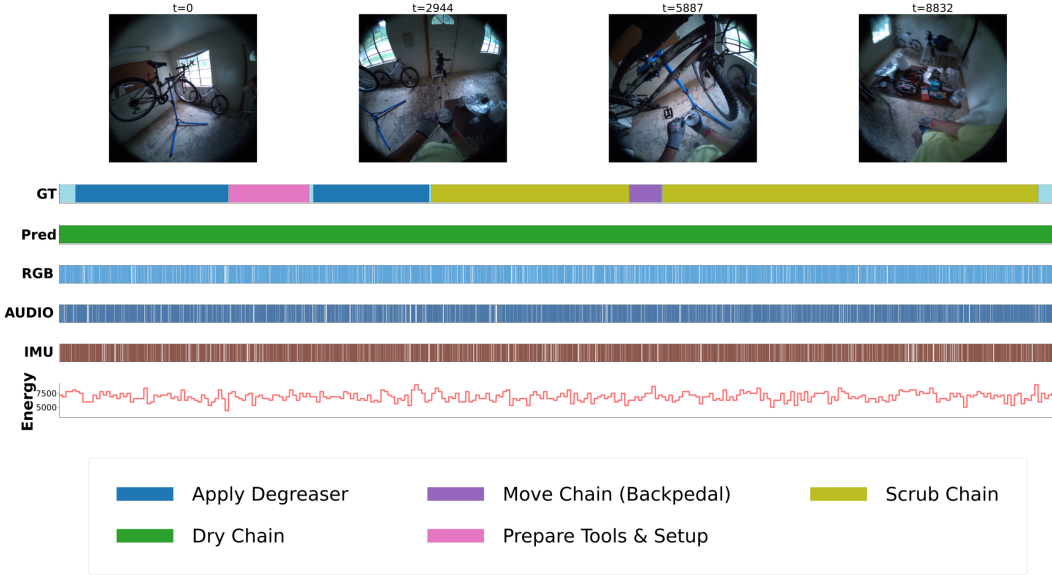


Figure 10: Qualitative example of AdaMML in scenario “Clean and Lubricate the Chain” of Ego-Exo4D [21]

while multimodal fusions (e.g., RGB + Audio or Gaze + Mono) rapidly overtake them as the energy budget relaxes toward 2.8 W.

Conversely, the trends in CMU (Fig. 12) tell a different story. Dynamic approaches like Greedy and Random ( $\tau_{tr} = 0.90$ ) fail to match this static efficiency, collapsing completely at the lowest energy scales. Furthermore, the plot highlights the severe architectural penalty of complex learned models; methods like HCMS, AdaMML, and xLSTM are clustered at the extreme right of the energy axis, fundamentally unable to scale down to the efficiency levels achieved by simpler baselines.

For Captain Cook 4D (Fig. 13) Frame Rate policies (rgb and rgb+audio) clearly overcome dynamic and learned policies across the entire energy spectrum.

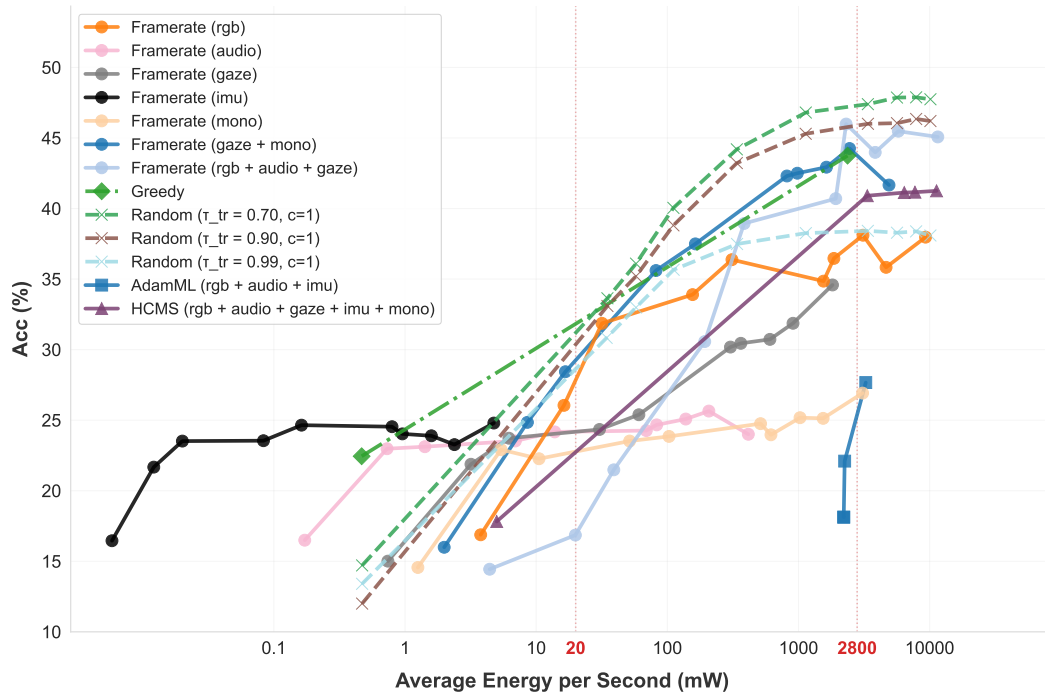


Figure 11: Energy-Accuracy trade-offs on Ego-Exo4D

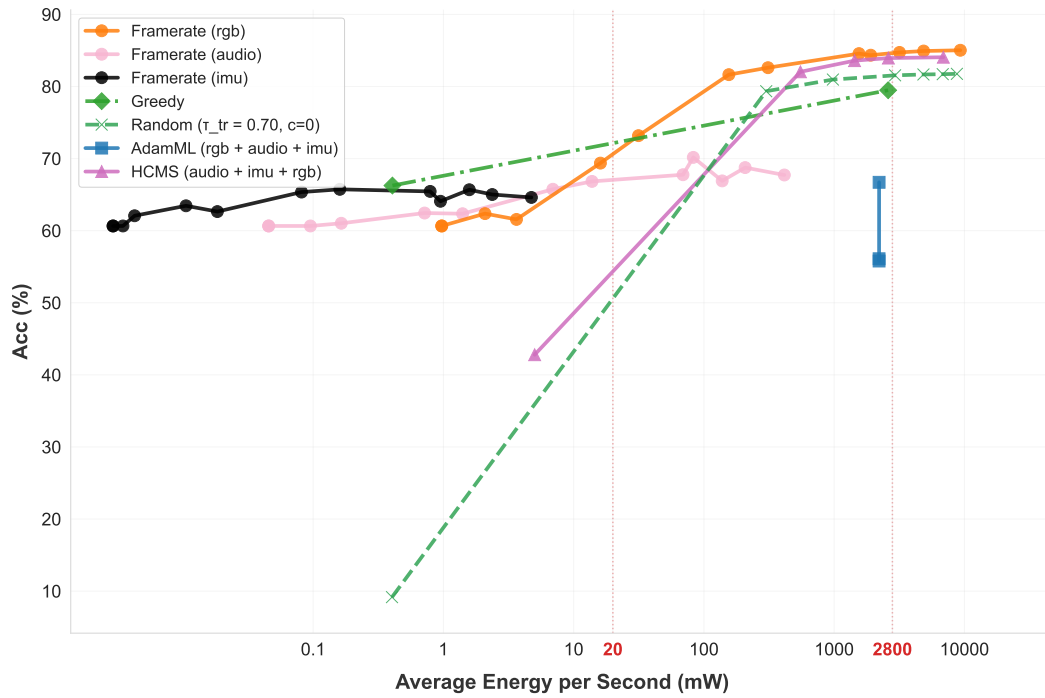


Figure 12: Energy-Accuracy trade-offs on CMU

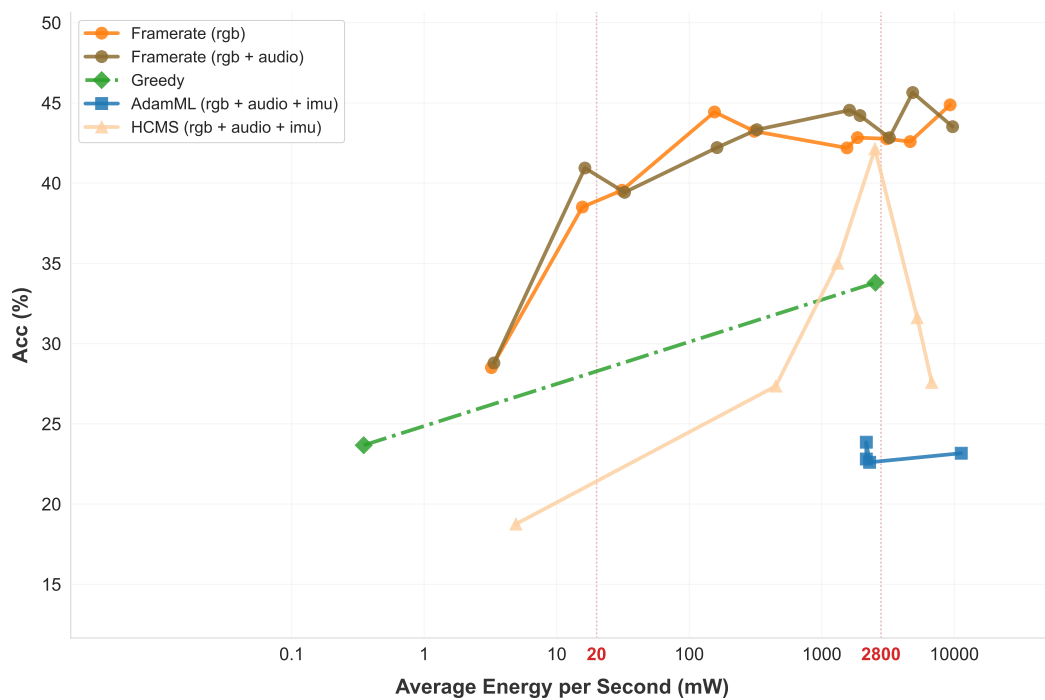


Figure 13: Energy-Accuracy trade-offs on Captain Cook 4D